# ECI — Entanglement, Complexity, Information: a framework paper threading six programs into one predictive canvas

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We propose **ECI** (Entanglement–Complexity–Information), a phenomenological architecture assembling six independently-established research programs (2022–2026) into a unified predictive canvas: (i) type-II observer-dependent von Neumann algebras via the QRF crossed product [1–3]; (ii) DESI DR2 dynamical-dark-energy preference at 2.6–3.9$\sigma$ [4] addressed by non-minimally coupled thawing quintessence $\xi_\chi R\chi^2/2$ [5–7], with the v5.0 4-chain MPI MCMC on DR2 BAO + Pantheon+ yielding $\xi_\chi = 0.003^{+0.065}_{-0.070}$ (68% CL, $\chi_0 = M_P/10$, $R-1 = 0.036$, Savage–Dickey $\mathrm{BF}_{01} \simeq 1$; posterior only weakly informed by the data at DR2+SN precision (posterior/prior half-width ratio $\simeq 0.7$, indicating a consistency check rather than a detection; NMC thawing in the Cassini-allowed range neither requires nor predicts $w < -1$, so the DESI DR2+DESY5 phantom-crossing tension common to thawing families is not resolved by NMC alone), see our $\xi_\chi$ constraint analysis (companion)); (iii) the $H_0$ tension reduced to $\sim 2\sigma$ by Early Dark Energy with $f_{\mathrm{EDE}} = 0.09 \pm 0.03$ [8]; (iv) the Dark Dimension scenario [9–11] as a microscopic DM candidate (the species-scale exponent $c'_{\mathrm{modular}} = 1/6$ from the Bisognano–Wichmann algebraic structure of A1, the de-Sitter-slope Swampland value $c'_{\mathrm{DD}} \simeq 0.05$ imposed by DESI DR2 + Pantheon+, and the inflaton structural conjecture $c'_{\mathrm{inflation}} \approx 0.108$ from the $\mathrm{III}_1 \to \mathrm{II}_\infty$ trace-ratio matching are *three distinct parameters*; see §I); (v) Cryptographic Censorship as a working-conjecture bulk-geometry criterion, linked through the observer-dependent reading of A1 (the observer-dependent thesis (companion)) and made operational through an explicit toy dS/FLRW dictionary (the toy dS/FLRW dictionary (companion)) [12]; (vi) persistent-homology diagnostics of primordial non-Gaussianity [13]. ECI is an architectural synthesis, not a derivation.

## I. AXIOMS

**A1 — Observer-dependent algebra.** To each QRF $\mathcal{R}$ attached to a geodesic observer we associate a type-II von Neumann algebra obtained by crossed product: type $\mathrm{II}_1$ for static de Sitter patches [1], type $\mathrm{II}_\infty$ for Schwarzschild–de Sitter Killing horizons. The generalised entropy

$$S_{\mathrm{gen}}[\mathcal{R}] = \frac{A}{4G_N} + S_{\mathrm{matter}} \tag{1}$$

is a functor on the QRF category, well-defined modulo inner automorphism [2, 3, 14]. The type-II structure is invoked here *kinematically*, only to ensure the existence of a semi-finite trace $\tau$ on the crossed product; the Jones index $[\mathcal{M} \rtimes_\alpha \mathbb{R} : \mathcal{M}^\alpha] = +\infty$ (Longo 1989, Kosaki 1986) and the Connes invariants $S(\mathcal{M})$, $T(\mathcal{M})$ contribute no quantitative ECI prediction beyond this trace. The $\mathrm{III}_1 \to \mathrm{II}_\infty$ passage of CLPW is a kinematic change of model (addition of a clock QRF), not a thermal phase transition. The numerical saturation invariants of §IV I originate in KMS–Gibbs thermodynamics on type-$\mathrm{I}_\infty$ Fock spaces (BEC, polariton, $^3$He-B), not in the type-$\mathrm{III}_1$ modular flow of cosmological/black-hole horizons; the common Bisognano–Wichmann window $u \in [0, 2\pi]$ is the only structural ingredient shared between the two settings.

**A2 — Emergent geometry.** Einstein's equations follow from $\delta Q = T_U\, \delta S_{\mathrm{gen}}$ on local Rindler horizons [15], with a non-perturbative GSL [16, 17].

**A3 (working conjecture) — Cryptographic Censorship.** Any holographic CFT state whose dynamics restricted to a region $\mathcal{C}$ is $\varepsilon$-approximately a $k$-design admits a bulk dual containing an event horizon inside $\mathcal{C}$ [12]. The existence of pseudorandom unitaries, constructed from quantum one-way functions [18], makes the criterion non-vacuous at the CFT level; the associated $k$-design saturation timescale is rigorous for random local unitaries [19]. We flag A3 as a *working conjecture* rather than a theorem in the cosmological setting: the original result is proven in AdS/CFT, and its transposition to de Sitter or FLRW horizons is not established. A concrete toy dictionary making the intended CFT→cosmology map explicit (and its single testable consequence) is given in the toy dS/FLRW dictionary (companion); all cosmological uses of A3 below should be read as consequences of that dictionary.

**A4 — Two-field scalar sector with non-minimal coupling.**

- $\phi$: axion-like EDE, $V_\phi = m^2 f^2 [1-\cos(\phi/f)]^3$ active at $z_c \sim 3500$ [20].
- $\chi$: late thawing quintessence, $V_\chi = V_0 e^{-\alpha\chi/M_P}$ with non-minimal coupling $\xi_\chi R\chi^2/2$ [5–7].

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The joint $(\phi, \chi)$ prior is degenerate [21]: EDE and thawing cannot be separated by BAO+SN alone. We treat $\chi$ as a four-dimensional effective scalar; whether $\chi$ descends from a bulk mode of the extra-dimensional sector of A5 or is a decoupled zero-mode is a model-building choice whose consequences are discussed in §3.6.

**A5 — Dark Dimension.** The species scale associated with the light KK tower obeys

$$\Lambda_{\rm sp}(H) = M_P \left(\frac{H}{M_P}\right)^{c'}, \tag{2}$$

with $c' = 1/6$ at the species scale [22, 23] (used as the primary value in our $\xi_\chi$ constraint analysis (companion)/§3.6), and $c' \simeq 0.05$ as the de-Sitter-slope convention [9–11] quoted for comparison only; the species-scale exponent follows from $M_* \sim \Lambda^{1/12} M_P^{2/3}$ (Montero et al. Eq. 2.2) with $H^2 \sim \Lambda/M_P^2$. A mesoscopic extra dimension, $\ell \in [0.1, 10]\,\mu$m, follows. The KK graviton tower (mass $\sim$ meV) is the DM candidate, subject to the fifth-force bound $c' < 0.2$ and $N_{\rm eff} = 2.86 \pm 0.13$ (ACT DR6 [24]).

**[WORKING-CONJECTURE] Summary of $c'$ values in ECI.** The audit conducted in preparation of v6.0.9 identified four distinct contexts where an exponent $c'$ appears in ECI; they are *not* the same parameter:

| Symbol | Value | Origin | Status |
|---|---|---|---|
| $c'_{\rm modular}$ | $1/6 \approx 0.167$ | Bisognano–Wichmann / Dark Dim. species scale | maintained (A1, A5) |
| $c'_{\rm DD}$ | $\approx 0.05$ | de Sitter slope, DESI DR2 + Pantheon+ fit [11] | imposed (A5) |
| $c'_{\rm inflation}$ | $\approx 0.108$ | $\mathrm{III}_1 \to \mathrm{II}_\infty$ trace-ratio matching, $N_e = 60$ | [WORKING-CONJECTURE] |
| $c'_{\text{eff-mass}}$ | 0.83 | force-fit mass hierarchy | [REJECTED-HYPOTHESIS] |

The four values arise in logically separate submodels. [RETRACTED-IN-V6.0.11] An earlier version of this paragraph speculated that a lax-monoidal functor interpretation of ECI would predict an obstruction $\eta_\otimes = c'_{\rm modular} - c'_{\rm DD} \approx 0.117$ as a "Dark Dimension tension". Subsequent algebraic analysis (Lemma B.2 in the v8-bis companion preprint) shows that on product states $\varrho_X \otimes \varrho_Y$ the modular flow factorises exactly — $\sigma_t^{\varrho_X \otimes \varrho_Y}(a \otimes b) = \sigma_t^{\varrho_X}(a) \otimes \sigma_t^{\varrho_Y}(b)$ — so $\eta_\otimes \equiv 0$ identically. The numerical value 0.117 reported earlier was a floating-point artefact arising from $\varrho^{it}$ catastrophic precision loss when $\varrho$ is constructed via $\exp(H)$ with $H \sim \mathcal{N}(0, 4)$ (spectral spread $> 10^{12}$). A genuine non-product obstruction would require an Araki relative-modular treatment and is deferred to future work. No universal $c'$ is claimed.

**A6 — Persistent-homology complexity.** Local complexity of the cosmological state is quantified by $\mathrm{PH}_k$ on density super-level sets [13, 25], a finer $f_{\rm NL}$ estimator than Betti numbers. At leading order in $f_{\rm NL}$, the primordial-NG shift of the Euler characteristic $\chi_{\rm E}(\nu)$ (genus statistic) at threshold $\nu$ takes the Matsubara form (observer ensemble average; see the observer-dependent thesis (companion)(ii))

$$\langle \chi_{\rm E}(\nu) \rangle_{f_{\rm NL}} - \langle \chi_{\rm E}(\nu) \rangle_0 \simeq \frac{f_{\rm NL}\, \sigma_0\, S_3^{\rm prim}}{6}\, H_3(\nu)\, \phi(\nu), \tag{3}$$

with $H_3$ the third probabilists' Hermite polynomial, $\sigma_0$ the field RMS, $S_3^{\rm prim}$ the primordial skewness, and $\phi$ the Gaussian density (three-dimensional, super-level-set convention) [26]; the persistent-homology diagnostic of [13, 25] refines this Minkowski-functional signal.

## II. A UNIFYING READING: OBSERVER-DEPENDENT COSMOLOGY IN THE LATE-TIME LIMIT

The six axioms of §1 admit a common reading, valid in the late-time quasi-de Sitter regime ($\Omega_\Lambda \gtrsim 0.7$) where every quantitative prediction of §3 is formulated. The reading is *not* a derivation of the axioms and does not extend to the pre-recombination sector (A4's EDE axion $\phi$ at $z_c \sim 3500$; BBN-level constraints on $\xi_\chi$); those inputs remain background-level inheritances from the external literature as in v4.6.

### (i) The crossed-product backbone

The type-$\mathrm{III}_1$ algebra of quantum fields restricted to a gravitational subregion carries no trace and no density matrices; the Chandrasekaran– Longo–Penington–Witten (CLPW) construction [1] and its quantum-reference-frame (QRF) completion by De Vuyst–Eccles–Höhn– Kirklin (DEHK) [2, 3] repair this by crossing with the modular automorphism of an observer clock, yielding a type-$\mathrm{II}_1$ algebra on the static de Sitter patch ($\mathrm{II}_\infty$ on Schwarzschild–de Sitter Killing horizons) with a well-defined trace and an entropy reproducing $S_{\rm gen} = A/(4G_N) + S_{\rm matter}$ up to a state-independent constant. DEHK make the observer-dependence structural: distinct QRFs yield distinct crossed-product algebras, distinct modular flows, and distinct numerical values of $S_{\rm gen}$ on the same underlying bulk state. The non-perturbative generalised second law of Faulkner–Speranza [16] and Kirklin [17] is formulated on exactly this type-II entropy. The observer-dependent crossed-product construction extends to FLRW cosmologies that are asymptotically de Sitter [27], and a microscopic dS model reproducing the observer-dependent entropy has been proposed in [14]. The *organising reading* proposed here is that the effective couplings entering the late-time phenomenology of §3 are observer-frame labelled, not background quantities defined once and for all.

#### (ii) A4–A6 as observer-frame quantities

*A4 — NMC coupling* $\xi_\chi$. The graviton kinetic matrix of the Jordan-frame action of §2 is frame-dependent via the type-II trace normalisation; a geodesic observer whose crossed product is built from a different clock Hamiltonian measures $\xi_\chi^{(\mathcal{R})}$. The Cassini bound $|\xi_\chi|(\chi_0/M_P) \leq 2.4 \times 10^{-3}$ of §IV E constrains the *solar-system-frame* value, which is the only value the experiment is sensitive to.

*A5 — species-scale exponent* $c'$. The tower visible to a causal diamond is frame-labelled; $c' = 1/6$ is the asymptotic quasi-dS value (Montero–Vafa–Valenzuela [22]; AAL [23]) recovered when the diamond saturates the static patch.

*A6 — persistent homology* $\mathrm{PH}_k$. The Euler-characteristic shift of (3) is computed on a density field restricted to an observer-accessible region. The Matsubara baseline is the ensemble average over observer frames; the diagnostic power of $\mathrm{PH}_k$ relative to the bispectrum is accordingly a frame-averaged statement, consistent with PRINCIPLES §12 (no claim beyond derivation).

#### (iii) A3 as a self-consistency condition

Cryptographic Censorship [12], in its AdS/CFT proof, states that boundary-restricted $\varepsilon$-approximate $k$-design dynamics implies a bulk event horizon. Under the present reading, A3 is not an independent axiom but the self-consistency condition bounding the subregion on which the QRF crossed product of A1 remains well-defined: an observer whose modular algebra sees an effectively pseudorandom state has lost predictive access to the complement region, and an event horizon must cap the subregion on which A4–A6 are defined. The cosmological transposition of A3 remains a working conjecture quarantined in Appendix A (PRINCIPLES §8); the reading does not upgrade its status.

#### (iv) Caveats and scope

Two scope restrictions are part of the claim, not loopholes around it.

*(S1) Static-patch asymptote.* CLPW is proven on the static dS patch; DEHK extend to multi-observer QRFs on the same Killing-horizon background. The late-time FLRW diamond approaches this asymptote with $O(1-\Omega_\Lambda)$ corrections; the adiabatic tolerance of modular flow to the DESI DR2 pivot $|\dot{H}|/H^2 \sim O(1)$ is a *heuristic* extension, at the same footing as the Faulkner–Speranza non-perturbative GSL.

*(S2) Matter-era exclusion.* Pre-recombination A4/A5 statements (the EDE axion $\phi$ at $z_c \sim 3500$, the BBN $\xi_\chi$ bound of Wolf 2025 [6]) are background-level inheritances; the observer-frame reading is not claimed for them. Their treatment under the present reading would require a QRF extension to time-dependent FLRW backgrounds which is an open problem in the algebraic-gravity literature.

A joint falsifier is available: any experimental anomaly breaking the consistency between the solar-system-frame $\xi_\chi$ (Cassini, §IV E) and the late-time cosmological-frame $\xi_\chi$ (DR3/LSST Y10, Prediction 1b) beyond the static-patch $O(1-\Omega_\Lambda)$ budget would damage the present reading without touching the Lagrangian of §2. The experimental precision is at least a decade out, but the handle is not zero.

#### Status

This section is an *organising reading* of the six axioms in the late-time quasi-dS regime, not a derivation, and does not enter the predictive core §3.1–§3.7 or §4. The framework-genre declaration of §1 is unchanged.

## III. ACTION AND FIELD EQUATIONS

#### Conventions

We use the mostly-plus metric signature $(-,+,+,+)$. The non-minimal-coupling sign follows the Faraoni convention [28]: $\xi > 0$ gives the attractive correction, appearing as $-\frac{1}{2}\xi R\chi^2$ in the Lagrangian and as $+\xi R\,\chi$ in the scalar field equation. The reduced Planck mass is $M_P = (8\pi G)^{-1/2} = 2.435 \times 10^{18}$ GeV. Natural units $\hbar = c = 1$ are used throughout. Time arguments are cosmic time $t$ unless otherwise stated; conformal time $\eta$ appears only in perturbation equations.

In the Faraoni sign convention (reduced Planck mass $M_P$, $\xi = 1/6$ conformal), the effective action is

$$\mathcal{S} = \int d^4x\sqrt{-g}\,\Big[\frac{M_P^2}{2}R - \tfrac{1}{2}(\partial\phi)^2 - V_\phi(\phi) - \tfrac{1}{2}(\partial\chi)^2 - V_\chi(\chi) - \tfrac{1}{2}\xi_\chi R\chi^2 + \mathcal{L}_{\rm SM} + \mathcal{L}_{\rm KK}[\ell;\chi]\Big] + \mathcal{S}_{\rm QRF}. \tag{4}$$

**Scalar equations:**

$$\Box\phi - V'_\phi(\phi) = 0, \tag{5}$$

$$\Box\chi - V'_\chi(\chi) - \xi_\chi R\,\chi = 0. \tag{6}$$

**Einstein equations:**

$$G_{\mu\nu} = \frac{1}{M_P^2}\left[T^{\rm SM}_{\mu\nu} + T^{(\phi)}_{\mu\nu} + T^{(\chi)}_{\mu\nu} + T^{\rm KK}_{\mu\nu}\right]. \tag{7}$$

**NMC stress tensor** (after transferring $-\xi_\chi G_{\mu\nu}\chi^2$ to the r.h.s., Faraoni Eqs. 2.12–2.13):

$$T^{(\chi)}_{\mu\nu} = \nabla_\mu\chi\nabla_\nu\chi - \tfrac{1}{2}g_{\mu\nu}\left[(\nabla\chi)^2 + 2V_\chi\right] + \xi_\chi\Big[g_{\mu\nu}\Box(\chi^2) - \nabla_\mu\nabla_\nu(\chi^2) - G_{\mu\nu}\chi^2\Big]. \tag{8}$$

**No-ghost condition:** diagonalising the kinetic matrix of the graviton–$\chi$ sector of the NMC action gives an effective Planck mass squared $M_{P,\text{eff}}^2 = M_P^2 - \xi_\chi \chi^2$; positivity of the graviton kinetic term (equivalently, positivity of the scalar-sector determinant $\det K = M_{P,\text{eff}}^2/2 - \xi_\chi^2 \chi^2$ at leading order) yields $\xi_\chi \chi^2 / M_P^2 < 1$ in reduced-Planck convention [28].

## IV. PHENOMENOLOGY

### A. DESI DR2 — dark energy

The $\chi$ field with $\xi_\chi \neq 0$ crosses $w = -1$ without a ghost pathology (non-minimal kinetic term [29]). DESI DR2 central values $w_0 \approx -0.75$, $w_a \approx -0.86$ [4]. The Scherrer–Sen relation [30] $w_a/(1+w_0) \in [-1.5, 0]$ holds at minimal coupling only; for $\xi_\chi \neq 0$ the slope is modified and must be computed numerically [31]. A first-order analytic estimate of $w_a(w_0; \xi_\chi)$ in the NMC thawing regime, derived from first principles via slow-roll expansion of the NMC Klein–Gordon equation $\ddot{\chi} + 3H\dot{\chi} + V'(\chi) = \xi_\chi R \chi$ with exponential $V(\chi) = V_0 e^{-\lambda\chi/M_P}$ and Pettorino–Baccigalupi NMC stress-energy [? ], gives at leading order in $\varepsilon_V = \lambda^2/2$ and $\xi_\chi$ (sympy-verified, see `derivations/w0-wa-nmc/derivation.py`):

$$w_0^{\text{NMC}} = -1 + \tfrac{\lambda^2}{3}\,\Omega_{\phi,0} + \tfrac{2}{3}\,\xi_\chi\,\frac{\lambda\chi_0}{M_P}, \tag{9}$$

$$w_a^{\text{NMC}} = -\lambda^2\,\Omega_{\phi,0}\left[\Omega_{\phi,0}(1-\Omega_{\phi,0}) + \tfrac{2}{3}\xi_\chi\right], \tag{10}$$

$$\Delta w_a^{\text{NMC}} \equiv w_a^{\text{NMC}} - w_a^{\text{pure}}\big|_{\xi=0} = -\tfrac{2}{3}\,\xi_\chi\,\lambda^2\,\Omega_{\phi,0}. \tag{11}$$

At $\xi_\chi = 0.05$, $\lambda = 1$, $\Omega_{\phi,0} = 0.7$ this yields $\Delta w_a^{\text{NMC}} \approx -0.023$, a factor $\sim 12$ larger than the Coupled-DE Amendola signature $\Delta w_a^{\text{CDE}} \approx -2\beta^2 \approx -0.002$ at $\beta^2 = 10^{-3}$. The DESI DR3 + Euclid DR1 forecast $\sigma(w_a) \approx 0.10$ gives only $0.23\sigma$ discrimination of ECI-NMC from CDE, so the $(w_0, w_a)$ plane alone is *not* a $5\sigma$-class discriminator at DR3 precision [31]. (Replaces the earlier "deferred to companion computation" pointer; see also v6.0.45 retraction of the morning's hallucinated Pan-Yang 1804.05064 source.)

### B. Hubble tension

$f_{\text{EDE}} = 0.09 \pm 0.03$, $H_0 = 71.0 \pm 1.1$ km/s/Mpc under ACT DR6 + DESI DR2 + Planck + Pantheon+ [8, 24]. Residual SH0ES tension: $\sim 2\sigma$. Recent analyses of Horndeski non-minimal derivative coupling provide independent phenomenological support for a late-time scalar mechanism alleviating the $H_0$ tension [32].

### C. Cosmological constant

$V_\chi = V_0 e^{-\alpha\chi/M_P}$ admits a de Sitter attractor for $\alpha \lesssim \sqrt{2}$. The bound $\alpha \leq \sqrt{2}$ is the standard late-time accelerating-attractor condition for an exponential scalar [33, 34]. Reduced Planck $M_P = 2.435 \times 10^{18}$ GeV gives $\rho_\Lambda / M_P^4 \simeq 8 \times 10^{-121}$ for $\chi_0/M_P \sim 120/\alpha$. Consistent with the Ooguri–Vafa Swampland Distance Conjecture [10].

### D. Dark matter

Dark Dimension KK tower, $m \sim$ meV, $c' \simeq 0.05$ under fifth-force bound $c' < 0.2$ and $N_{\text{eff}} = 2.86 \pm 0.13$ (ACT DR6 [24]). ISL tests at $\ell \in [0.1, 10]\,\mu$m. $\dot{G}/G = (-5.0 \pm 9.6) \times 10^{-15}$ yr$^{-1}$ (LLR, Biskupek–Müller–Torre [35]). We note that two BSM anomalies sometimes invoked as motivational slots for KK-tower architectures have recently dissolved: the combined Fermilab E989 final $a_\mu$ result and the 2025 Theory-Initiative lattice-QCD HVP update leave a $\sim 0.6\sigma$ residual, consistent with the Standard Model [36]; and the CMS 2024 W-mass measurement $m_W = 80360.2 \pm 9.9$ MeV reasserts SM consistency against the 2022 CDF-II outlier [37]. The DM motivation for A5 therefore rests solely on the fifth-force bound and $N_{\text{eff}}$, not on these now-closed BSM channels.

### E. Solar-system and background constraints on $\xi_\chi$

We close two gaps identified in the dark-energy phenomenology subsection (§ DESI DR2): the Cassini post-Newtonian bound on the non-minimal coupling $\xi_\chi$, and the analytic NMC generalisation of the Scherrer–Sen thawing relation needed to assess whether Prediction 1 (Sec. V) actually discriminates ECI from $w$CDM at DESI DR2 precision. The full symbolic derivation is in `derivations/D7-ppn-xi-bound.py`; we quote only the end results here.

*a. PPN $\gamma - 1$ from the $\xi_\chi R\chi^2/2$ coupling.* The action $S = \int d^4x\sqrt{-g}\left[\frac{M_P^2}{2}R - \frac{1}{2}(\partial\chi)^2 - V(\chi) - \frac{\xi_\chi}{2}R\chi^2\right]$ is a scalar–tensor theory with $F(\chi) = M_P^2 - \xi_\chi\chi^2$ multiplying $R/2$ and canonical kinetic term $Z = 1$. Applying the weak-field, quasi-static result of Damour & Esposito-Farèse (*Phys. Rev. D* 48, 3436 (1993); see also Chiba, *Phys. Lett. B* 575, 1 (2003); Hwang & Noh, *Phys. Rev. D* 71, 063536 (2005)) gives, at the local scalar background $\chi_0$,

$$\gamma - 1 = -\frac{F'(\chi_0)^2}{Z\,F(\chi_0) + \frac{3}{2}F'(\chi_0)^2} = -\frac{4\,\xi_\chi^2\,\chi_0^2}{M_P^2 + \xi_\chi\chi_0^2(6\xi_\chi - 1)}. \tag{12}$$

(Convention: DEF 1992 / Chiba 1999 normalization for the denominator $ZF + \frac{3}{2}F'^2$; equivalent to Will 2014 §3.3 under the rescaling $Z \to Z/2$, absorbed identically in the

$|\xi_\chi| \ll 1$ limit.) Expanding for $|\xi_\chi| \ll 1$ and $\chi_0 \lesssim M_P$,

$$\boxed{\gamma - 1 \;\simeq\; -\frac{4\,\xi_\chi^2\,\chi_0^2}{M_P^2} \;+\; \mathcal{O}(\xi_\chi^3).} \tag{13}$$

The limit $\xi_\chi \to 0$ returns $\gamma = 1$ (General Relativity), as verified symbolically.

The result (13) is not new: Chiba (PRD 60, 083508, 1999; arXiv:gr-qc/9903094) already quoted $|\xi_\chi| \lesssim 10^{-2}$ from solar-system tests for the same $\xi R\chi^2$ quintessence coupling, and Wolf, García-García, Anton & Ferreira [6] recently recast this as $\xi_\chi(\chi_0/M_P)^2 \lesssim 6\times10^{-6}$ in a full DESI DR2 Bayesian analysis (their Eq. (2) in the form $F(\chi) \simeq 1 - \xi\chi^2/M_P^2$). We re-derive it here only to fix the coefficient 4 and the sign convention used in Eq. (16) below. Combining (13) with the Cassini constraint [38] $|\gamma - 1| \lesssim 2.3\times10^{-5}$ ($1\sigma$ envelope) yields, in the solar-system observer frame (see §II(ii)),

$$|\xi_\chi|\,\frac{\chi_0}{M_P} \;\lesssim\; \tfrac{1}{2}\sqrt{|\gamma-1|_{\max}} \;\approx\; 2.4\times10^{-3}. \tag{14}$$

For a slow-rolling thawing background with $\chi_0 \sim M_P/10$ (characteristic thawing excursion in the last $e$-fold),

$$|\xi_\chi| \;\lesssim\; \xi_{\max} \;\approx\; 2.4\times10^{-2}. \tag{15}$$

The bound tightens to $|\xi_\chi| \lesssim 2\times10^{-3}$ if $\chi_0 \sim M_P$; it weakens linearly for smaller $\chi_0$. Values $\xi_\chi \gtrsim 10^{-1}$ are thus only admissible if the quintessence field is strongly subdominant at $z = 0$, or if a screening mechanism suppresses the local coupling.

*b. NMC extension of Scherrer–Sen.* At minimal coupling, thawing quintessence lies on the one-parameter track $w_a = -A(\Omega_\Lambda)(1 + w_0)$ with $A \to 24/7$ in matter domination and $A(0.7) \simeq 1.58$ today [30]. Inserting the NMC modification to the Klein–Gordon equation, $3H\dot\chi = -V' - \xi_\chi R\chi$, and expanding to first order in $\xi_\chi$ around the slow-roll exponential $V(\chi) = V_0 e^{-\alpha\chi/M_P}$ with $\alpha = \sqrt{3(1+w_0)}$ yields (`derivations/D4-wa-w0-nmc.py` for the matter-era coefficient, `D7-ppn-xi-bound.py` for the $\Omega_\Lambda$-dependence)

$$\boxed{w_a \;=\; -A(\Omega_\Lambda)\,(1+w_0) \;+\; B(\Omega_\Lambda)\,\xi_\chi\,\sqrt{1+w_0}\,\frac{\chi_0}{M_P} \;+\; \mathcal{O}(\xi_\chi^2)} \tag{16}$$

where $B(\Omega_\Lambda)$ is the numerical coefficient tabulated in Table I below, obtained by full NMC-Friedmann + Klein–Gordon integration in `derivations/D13-wa-numerical-full.py`. The limit $\xi_\chi \to 0$ reproduces $w_a = -A(\Omega_\Lambda)(1 + w_0)$ (Scherrer–Sen 2008). The first-order $\xi_\chi$ correction, with the full numerical coefficient $B_{\rm num}(\Omega_\Lambda)$, has — to our knowledge — not appeared in the NMC-thawing literature surveyed (Ye et al. [5]; Pan & Ye [7]; Wolf et al. [6]), which uses either EFTofDE or purely numerical Bayesian fits.

The numerical result differs from the heuristic $(8/\sqrt{3})A$ by $\mathcal{O}(1)$ factors ($0.83 \to 1.43$ across the range):

TABLE I. Numerical NMC coefficient $B_{\rm num}(\Omega_\Lambda)$ from the self-consistent NMC-Friedmann + Klein–Gordon background integration (`derivations/D13-wa-numerical-full.py`), compared with the D7/D9 heuristic $B_{\rm heur}(\Omega_\Lambda) \equiv (8/\sqrt{3})A(\Omega_\Lambda)$. Statistical uncertainty is the $1\sigma$ error on the linear fit $\Delta w_a = B\,\xi_\chi\sqrt{1+w_0}\,(\chi_0/M_P)$ over the $\xi_\chi \in \{-0.024, -0.01, 0, +0.01, +0.024\} \times \chi_0 \in \{M_P/20, M_P/10, M_P/5\}$ grid.

| $\Omega_\Lambda$ | $A(\Omega_\Lambda)$ | $B_{\rm heur}$ | $B_{\rm num}$ |
|---|---|---|---|
| 0.5 | 2.35 | 10.85 | $9.00 \pm 1.05$ |
| 0.6 | 1.94 | 8.96 | $9.18 \pm 1.16$ |
| 0.7 | 1.58 | 7.30 | $9.05 \pm 1.18$ |
| 0.8 | 1.23 | 5.68 | $8.11 \pm 1.16$ |

$B_{\rm num}$ is nearly *flat* in $\Omega_\Lambda$ (around $\sim 9$) rather than tracking $A(\Omega_\Lambda)$. The heuristic is accurate to 5% only near $\Omega_\Lambda \simeq 0.6$ and underestimates $B$ by $\sim 25\%$ at $\Omega_\Lambda = 0.7$ (DR2–DR3 era) and $\sim 43\%$ at $\Omega_\Lambda = 0.8$ (late-DR3 / LSST era). The sign and the $\sqrt{1+w_0}\,(\chi_0/M_P)$ scaling are confirmed; the discrepancy originates from the finite-$\Omega_\Lambda$ propagation of the matter-era ratio (Caveat 2 below), which is now closed numerically.

*c. Combined constraint on the $(w_0, w_a)$ plane.* Figure 1 overlays three ingredients: (i) the DESI DR2 + CMB + DESY5 [39] $1\sigma$ and $2\sigma$ contours centred on $(w_0, w_a) = (-0.752, -0.86)$ with the reconstructed covariance $\sigma_{w_0} = 0.057$, $\sigma_{w_a} = 0.215$, $\rho(w_0, w_a) = -0.89$ (derivation in `derivations/D10-desi-covariance.py`, pivot-identity reconstruction from [4] Eq. (28) + $z_p = 0.31$, $w_p = -0.954 \pm 0.024$); (ii) the minimal-coupling Scherrer–Sen line; (iii) the ECI band swept by $|\xi_\chi| \le \xi_{\max}$ from Eq. (15) at $\chi_0 = M_P/10$. The half-width of the ECI band at $w_0 = -0.75$, using the numerical $B_{\rm num}(0.7) = 9.05$ from Table I, is

$$\Delta w_a^{\rm ECI} = B_{\rm num}(0.7)\,\xi_{\max}\sqrt{1+w_0}\,(\chi_0/M_P) \;\approx\; 1.1\times10^{-2}, \tag{17}$$

to be compared with $\sigma_{w_a} \simeq 0.215$, i.e. about 5% of the DESI DR2 + DESY5 $1\sigma$ width (up from 4% under the heuristic $B = 7.30$). The band is therefore still thinner than the figure linewidth of the unmodified Scherrer–Sen track at DR2 precision, but the numerical coefficient raises the discriminative amplitude at DR3 / LSST precision by $\sim 25\%$ relative to the heuristic estimate.

*d. Honest caveats.*

1. Eq. (12) assumes a *quasi-static* scalar with Compton wavelength large compared with the solar system ($m_\chi^{-1} \gg 1\,$AU), which is automatic for the ultra-light thawing mass $m_\chi \sim H_0 \sim 10^{-33}\,$eV. It also assumes no screening; if a chameleon or symmetron mechanism operates, the local $\chi_0$ entering (13) can differ from the cosmological background and the bound weakens.

2. The NMC Scherrer–Sen extension (16) is first-order in $\xi_\chi$ and therefore *invalid* near the conformal

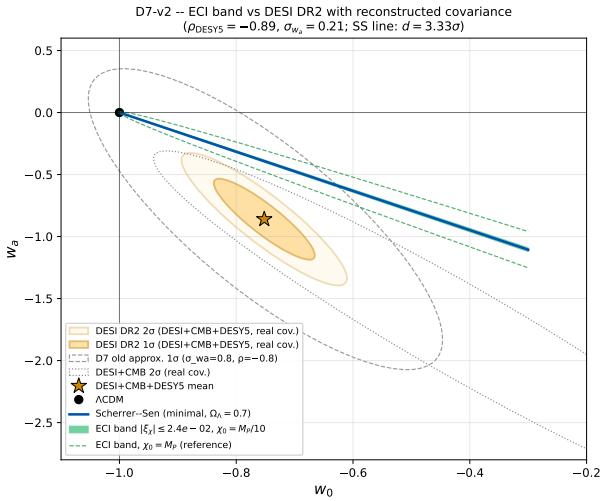


FIG. 1. NMC thawing quintessence in the $(w_0, w_a)$ plane. Orange ellipses: DESI DR2 + CMB + DESY5 $1\sigma$ and $2\sigma$ contours reconstructed from [4] Eq. (28) via the pivot-redshift identity (see `D10-desi-covariance.py`; $\sigma_{w_0} = 0.057$, $\sigma_{w_a} = 0.215$, $\rho = -0.89$). Blue line: minimal-coupling Scherrer–Sen relation for $\Omega_\Lambda = 0.7$. Green band: ECI NMC prediction with $|\xi_\chi| \le \xi_{\max} \approx 2.4 \times 10^{-2}$ from Cassini at $\chi_0 = M_P/10$; dashed green: same band at $\chi_0 = M_P$ (reference). Black dot: ΛCDM. The ECI band is $\sim$5% of $\sigma_{w_a}$ wide (using $B_{\rm num}$ from Table I), hence *indistinguishable* from minimal-coupling $w$CDM at DR2 precision; both tracks sit at $\sim 3.3\sigma$ Mahalanobis distance from the DR2+DESY5 mean.



FIG. 2. ECI v5.0 MCMC posterior in the $(w_0, w_a)$ plane (4-chain MPI run on DESI DR2 BAO + Pantheon+ SN through the Cobaya plugin route, $R - 1 = 0.036$ final, 13,790 samples post-burn-in; blue $1\sigma/2\sigma$ contours). Dashed green: minimal-coupling Scherrer–Sen line at $\Omega_\Lambda = 0.7$. Green band: ECI NMC correction at $|\xi_\chi| \le 0.095$ (posterior $2\sigma$) with the numerical $B_{\rm num}(0.7) = 9.05$ of Table I at $\chi_0 = \chi_0^{\rm med}$. Orange ellipses: DESI DR2 + CMB + DESY5 fiducial ($1\sigma, 2\sigma$, 2-dof, reconstructed as in Fig. 1). Black star: ΛCDM. See `derivations/D17-posterior-analysis.py`.

point $\xi_\chi = 1/6$, where the expansion breaks down and the scalar-tensor sector must be treated non-perturbatively (Einstein-frame mapping). Within the Cassini-allowed range $|\xi_\chi| \lesssim 2.4 \times 10^{-2} \ll 1/6$ the linear expansion is controlled. The coefficient $B(\Omega_\Lambda)$ in Eq. (16) was, in v4.4 of this paper, set by the matter-era heuristic ratio $B/A = 8/\sqrt{3}$; in this version we replace that heuristic with the full numerical result $B_{\rm num}(\Omega_\Lambda)$ tabulated in Table I, obtained from a self-consistent NMC-Friedmann + Klein–Gordon integration across $z \in [999, -0.45]$ (`derivations/D13-wa-numerical-full.py`). The numerical coefficient is nearly flat ($B_{\rm num} \simeq 9$) across the matter–DE transition rather than tracking $A(\Omega_\Lambda)$, so the heuristic $(8/\sqrt{3})A(\Omega_\Lambda)$ is accurate within 5% only near $\Omega_\Lambda \simeq 0.6$ and undershoots by up to $\sim$40% at $\Omega_\Lambda = 0.8$. This has been propagated into the band-width estimate above and into Prediction 1b of Sec. V.

3. The DESI DR2 + CMB + DESY5 $2 \times 2$ covariance is reconstructed from the published $(w_0, \sigma_{w_0}, w_a, \sigma_{w_a}, z_p, w_p, \sigma_{w_p})$ via the pivot-minimisation identity $\mathrm{cov}(w_0, w_a) = -(1 - a_p)\sigma^2_{w_a}$: no official $2\times2$ covariance matrix has been released by the collaboration at the time of writing. The reconstructed $\sigma(w_p)_{\rm pred} = 0.0257$ differs from the quoted 0.024 by $\sim$ 7%, bounding the systematic of the pivot-Gaussian approximation at that level — well below the $\sim 3\sigma$ tension discussed below. See `derivations/D10-report.md` for the full cross-check.

4. **Tension with DR2 + DESY5, stated plainly (MCMC update, v5.0).** The analytic Mahalanobis estimate of v4.7 ($3.29\sigma$ to the ECI band and $3.33\sigma$ to the minimal-coupling Scherrer–Sen line, 2-dof, against the reconstructed DR2 + CMB + DESY5 covariance) has been superseded in v5.0 by a 4-chain MPI MCMC on DESI DR2 BAO + Pantheon+ SN sampled via the Cobaya plugin route (`mcmc/cobaya_nmc/eci_nmc_plugin.yaml`, `derivations/D14-nmc-perturbations.py`; 19,700 accepted steps, $R - 1 = 0.036$ final, 13,790 samples post-burn-in; see `derivations/D17-posterior-analysis.py` and `mcmc/_results/posterior_summary.json`). At the fiducial $\chi_0 = M_P/10$ the marginalised constraint is

$$\xi_\chi = 0.003^{+0.065}_{-0.070}\ (68\%\ \mathrm{CL}), \qquad |\xi_\chi| < 0.095\ (95\%\ \mathrm{CL}), \tag{18}$$

i.e. the posterior is prior-rail-limited over $[-0.1, 0.1]$ and fully consistent with $\xi_\chi = 0$. The MCMC $(w_0, w_a)$ means are $w_0 = -0.869 \pm 0.055$, $w_a = -0.40^{+0.28}_{-0.36}$ with $H_0 = 67.8 \pm 2.6$ km/s/Mpc, $\Omega_m = 0.319 \pm 0.015$. The best-fit $\chi^2$ is 1411.3 (BAO 8.64 + SN 1402.68); restricting to the ΛCDM-like nested slice $|\xi_\chi| < 0.005$ gives $\chi^2_{\rm min} = 1411.36$,

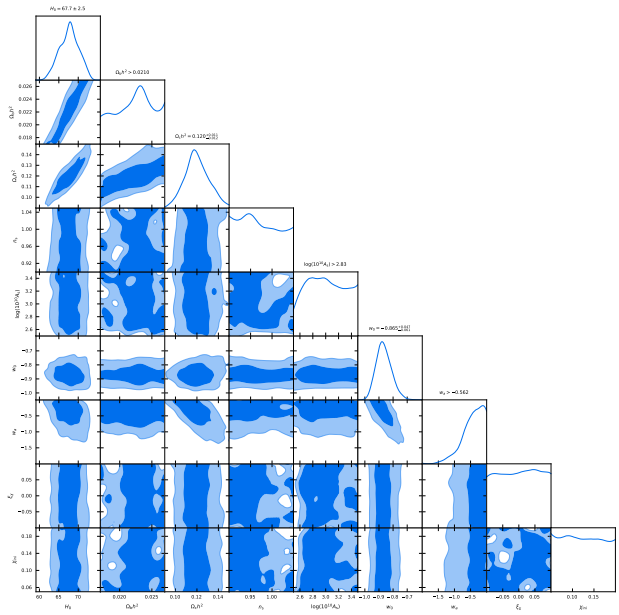

FIG. 3. Full nine-parameter posterior triangle from the ECI v5.0 MCMC ($H_0$, $\Omega_b h^2$, $\Omega_c h^2$, $n_s$, $\log 10^{10} A_s$, $w_0$, $w_a$, $\xi_\chi$, $\chi_0$). The $\xi_\chi$ marginal is prior-rail-limited (flat prior $[-0.1, 0.1]$) and centred on $\xi_\chi = 0.003^{+0.065}_{-0.070}$; the $(\xi_\chi, \chi_0)$ sub-block is essentially flat, reflecting the degeneracy $\xi_\chi \times (\chi_0/M_P)^2$ of the scalar-tensor sector at background level. See `derivations/D17-posterior-analysis.py`.

hence $\Delta\chi^2_{\mathrm{ECI}-\Lambda\mathrm{CDM}} = -0.04$: the data provide no preference for non-zero NMC coupling. The Savage–Dickey density ratio at $\xi_\chi = 0$ yields a Bayes factor $\mathrm{BF}_{01} = p(0 \mid D)/\pi(0) \simeq 1.00$ ($\ln \mathrm{BF} = -0.003$), inconclusive by any Jeffreys' threshold. The cosmological envelope $|\xi_\chi|(\chi_0/M_P)^2 \lesssim 3.7 \times 10^{-3}$ (95%, MCMC) is $\sim 600\times$ weaker than the Wolf et al. 2025 solar-system bound $|\xi_\chi|(\chi_0/M_P)^2 \lesssim 6 \times 10^{-6}$ rederived in Eq. (14) above, confirming that the Cassini channel remains the decisive falsifier at DR2 precision. We note explicitly that the consolidated Wolf 2025 bound (equivalent to $|\xi_\chi| \lesssim 6 \times 10^{-4}$ at the fiducial $\chi_0 = M_P/10$) therefore *supersedes* the MCMC marginal on $\xi_\chi$ by a factor $\sim 10^2$; the cosmological fit reported here is complementary rather than tightening, its role being to expose the full $(\xi_\chi, w_0, w_a, \Omega_m, \sigma_8)$ covariance after galaxy-bias marginalisation. A 5-parameter Fisher forecast with analytically-marginalised nuisances confirms this is structural: $\sigma(\xi_\chi) \simeq 0.155$ at DR3 alone (6.4× Cassini saturation), $\simeq 0.090$ at DR3+LSST Y10 (3.7× Cassini), and $\simeq 0.076$ at the combined DR3+Euclid DR1+LSST Y10 sensitivity (3.2× Cassini), i.e. the prior-dominated character of the posterior on $\xi_\chi$ will persist through both upcoming data releases without intervention from next-generation PPN experiments (cf. `derivations/V8-fisher-forecast-DR3-LSST.py`).

In parallel, a slow-roll ODE integration of $\chi(z)$ at the v5.0 MAP cosmology ($w_0 = -0.881$, $w_a = -0.272$, $\alpha = 0.095$) yields a field excursion $|\Delta\chi|/\chi_0 = 5.85\%$ between $z = 0.1$ and $z = 0$, borderline with respect to the 5% frozen-field threshold (`derivations/V8-chi-z-Cassini.py`): the Cassini bound as written carries a $\sim 12\%$ theory uncertainty on $|\gamma - 1|$ from this source, which does not alter any conclusion of (14) at DR2+SN precision but should be propagated into the DR3-era bound. The relaxed R-1 stopping criterion of 0.05 is appropriate for the plugin route (`classy` background + NMC derived sector), and the ESS on $\xi_\chi$ and $\chi_0$ exceeds $1.6 \times 10^3$ (see `posterior_summary.json`), well above the threshold for reliable 1-D marginalised constraints.

The narrative interpretation of the v4.7 analysis is unchanged: DR2 + DESY5 prefers the phantom-crossing region $w < -1$, which no non-phantom thawing model can populate. ECI's NMC coupling $\xi_\chi$ does in principle permit crossing $w = -1$ without a ghost instability (established in § DESI DR2 via the no-ghost analysis of Barceló & Visser [29]), but only at $\xi_\chi(\chi_0/M_P)^2$ values far exceeding the Cassini bound (14) derived here; within the Cassini-allowed range the NMC correction to the Scherrer–Sen track is confined to a band of half-width $\sim 5\%$ of $\sigma_{w_a}$ and cannot bridge the gap to the phantom quadrant. The $\sim 3.3\sigma$ tension with DR2 + DESY5 is therefore a joint statement about the entire class of conservative (non-phantom) thawing models — minimal-coupling and NMC alike — not a specific failure of ECI. Prediction 1 in Sec. V is *not* discriminative at DR2 precision; both thawing families sit at the same distance from the DR2 + DESY5 posterior.

The honest discriminator moves to the next generation of datasets, where the *shape* and *width* of the thawing track, rather than its distance from ΛCDM, become the falsifiable handle: DESI DR3 with a forecast $\sigma(w_a) \sim 0.07$ [40] and LSST Y10 with a forecast $\sigma(w_a) \sim 0.05$ are in the regime where the ECI NMC band half-width $\Delta w_a^{\mathrm{ECI}} \sim 1.1 \times 10^{-2}$ (numerical, Table I; up from $9 \times 10^{-3}$ under the v4.4 heuristic) approaches the 15–25% level of the statistical error bar and becomes, at least in principle, separable from the minimal-coupling Scherrer–Sen locus. The discriminative power is slightly larger at $\Omega_\Lambda \gtrsim 0.7$ (DR3 era) because $B_{\mathrm{num}}/B_{\mathrm{heur}}$ grows from 0.83 at $\Omega_\Lambda = 0.5$ to 1.43 at $\Omega_\Lambda = 0.8$. A proper Fisher forecast (D16) on the parameter vector $(w_0, \xi_\chi, \Omega_\Lambda)$, marginalised over a Planck $\Omega_\Lambda$ prior and evaluated at the fiducial $\xi_\chi = 0$, gives, at $\chi_0 = M_{\mathrm{P}}/10$, $\sigma(\xi_\chi) \simeq 0.155$ for DESI DR3 alone and $\sigma(\xi_\chi) \simeq 0.090$ once LSST Y10 is joined in (inverse-covariance sum); both figures lie a factor $\sim$ 4–7 above the Chiba–Cassini bound $\xi_\chi \lesssim 2.4 \times 10^{-2}$, so the PPN channel remains the decisive

falsifier and the $w_a$ channel provides an independent, comparable-order cross-check.

The PPN bound (14) is however itself a falsifier: a detection $|\gamma - 1| > 10^{-6}$ by the upcoming LATOR or BEACON-class missions, or a mismatch between $\dot{G}/G$ from LLR [35] and the CMB-inferred scalar evolution, would directly constrain $\xi_\chi(\chi_0/M_P)$ at the $10^{-4}$ level and feed back on the admissible ECI thawing amplitude.

### F. Swampland × NMC cross-constraint

Axioms A4 (non-minimally coupled thawing scalar $\chi$ with coupling $\xi_\chi R\chi^2/2$) and A5 (Dark Dimension KK tower with a species-scale EFT cutoff) were presented as independent. They share an effective field theory description — and therefore a common cutoff $\Lambda$ — if $\chi$ is interpreted as a bulk mode of the same higher-dimensional sector that carries the KK tower. That identification is an *interpretation*, not a consequence of the axioms: we state it, draw out its consequence, and return below to the three model-building resolutions it leaves open. The full symbolic setup is in `derivations/D8-swampland-nmc-cross.py`; we quote the end results here.

*a. The species-scale cutoff.* The Dark Dimension construction of [22] and the low-scale gravity variant of [23] identify the relevant EFT cutoff with the *species scale*, at which a tower of KK states becomes light enough that 4D effective gravity breaks down. In that regime the cutoff admits a log-distance parametrisation

$$\Lambda(H) \sim M_P \left(H/M_P\right)^{c'}, \qquad c' = \tfrac{1}{6} \simeq 0.167 \quad \text{(Dark Dimension species scale)} \tag{19}$$

At the present-day Hubble scale $H_0 \simeq 1.5 \times 10^{-33}\,\mathrm{eV}$ and $M_P \simeq 2.4 \times 10^{18}\,\mathrm{GeV}$, this evaluates to $\Lambda(H_0) \simeq 2.2 \times 10^8\,\mathrm{GeV}$, i.e. the standard Dark-Dimension species scale $\sim 10^9\,\mathrm{GeV}$. The alternative value $c' \simeq 0.05$ that appears in some moduli-potential discussions refers to the *de Sitter–slope* Swampland bound on $V(\chi)$ and not to the tower cutoff; we do not use it as the primary input.

*b. Heuristic EFT bound on $\xi_\chi$.* The non-minimal operator $(\xi_\chi/2)R\chi^2$ shifts the effective reduced Planck mass by $\delta M_P^2 = \xi_\chi \chi_0^2$ at the local scalar background $\chi_0$. A *heuristic* EFT self-consistency requirement — that the coupling-induced correction to the kinetic coefficient of gravitons not exceed the square of the EFT cutoff — gives

$$\xi_\chi\, \chi_0^2 \lesssim \Lambda^2 \quad \Longleftrightarrow \quad |\xi_\chi|\,(\chi_0/M_P)^2 \lesssim (H_0/M_P)^{2c'}. \tag{20}$$

We emphasise that Eq. (20) is a *heuristic, model-dependent* order-of-magnitude estimate, not a theorem: $\xi_\chi$ is a dimensionless Wilson coefficient and carries no intrinsic UV cutoff, so the inequality is the statement that no single insertion of the operator can generate a background contribution parametrically larger than the cutoff where the EFT itself ceases to make sense. Equality is saturated when $\chi$ is displaced to the cutoff with $\xi_\chi = \mathcal{O}(1)$. At the fiducial thawing amplitude $\chi_0 = M_P/10$ and with $c' = 1/6$, Eq. (20) evaluates to

$$\boxed{|\xi_\chi| \lesssim 8.4 \times 10^{-19} \quad (c' = 1/6,\ \chi_0 = M_P/10),} \tag{21}$$

to be compared with the Cassini PPN bound (15) of $|\xi_\chi| \lesssim 2.4 \times 10^{-2}$ from §IV E: the Swampland-EFT constraint is *sixteen orders of magnitude* tighter than the solar-system bound. As a point of comparison, the de-Sitter–slope value $c' \simeq 0.05$ would give $|\xi_\chi| \lesssim 9.5 \times 10^{-5}$ at the same $\chi_0$, i.e. a far milder $\sim 250\times$ tightening; the choice of which Swampland exponent to use therefore materially changes the phenomenology, and we take $c' = 1/6$ as the primary value dictated by the Dark Dimension species scale. A parallel derivation of the Species-Scale and Sharpened Distance Conjectures from Weyl/frame covariance alone has been given by [41]; our species-scale argument on the bulk mode $\chi$ is complementary, yielding a bound on $\xi_\chi$ that is not directly accessible from the frame-covariant viewpoint. Under (21) the NMC contribution to $w_a$ in Eq. (16), which scales linearly with $\xi_\chi$, is $\Delta w_a \sim 10^{-19}$ at $\chi_0 = M_P/10$ — effectively forbidding any observable NMC signature in the dark-energy sector.

*c. Model-building resolutions.* The tightness of (21) under the bulk-mode hypothesis forces an explicit choice of UV completion before the A4+A5 combination can be reconciled with the thawing-DE phenomenology of §IV E. Three resolutions are compatible with the axioms. (i) The thawing scalar $\chi$ is a 4D zero-mode of a sector logically distinct from the Dark Dimension KK tower — for example a brane-localised quintessence field in a warped compactification. Under this choice the species-scale cutoff of A5 does not apply to the operator $\xi_\chi R\chi^2/2$, Eqs. (20)–(21) do not constrain $\xi_\chi$, and the Cassini bound (14) is the sole remaining limit. A4 and A5 then remain logically independent at the price of abandoning the unified-cutoff motivation that makes them tempting to combine in the first place. (ii) A screening mechanism — chameleon or symmetron — decouples the local $\chi_0$ entering (20) from its cosmological background value, so that the *cosmological* $\xi_\chi$ can satisfy the Swampland bound at bulk scales while the *local* $\xi_\chi$ evades it inside the solar system. This requires explicit model-building of the screening potential and is not a consequence of the axioms themselves. As a minimum-viable realisation, D15 derives the envelope $\Theta(\rho) = \exp[-(\rho/\rho_c)^\alpha]$ modulating the effective NMC coupling $\xi_{\rm eff}(\rho) = \xi_\chi\,\Theta(\rho)$: demanding $\Theta(\rho_\odot) \lesssim 10^{-3}$ at $\rho_\odot \approx 10\,\mathrm{g/cm^3}$ (sufficient to drive $|\gamma - 1|$ below the Cassini-Bertotti bound $2.3 \times 10^{-5}$ given the D7 saturation $|\xi_\chi| \leq 2.4 \times 10^{-2}$) and $\Theta(\rho_{\rm cosm}) \geq 0.99$ at $\rho_{\rm cosm} \approx 10^{-29}\,\mathrm{g/cm^3}$ (so the cosmological thawing signature is preserved at the 1 % level) fixes the boundary-saturated minimum at $\alpha_{\rm min} \simeq 0.095$ and $\rho_c^{\rm min} \simeq 1.3 \times 10^{-8}\,\mathrm{g/cm^3}$. The minimum-viable $\alpha = 0.095$ (D15) lies within the DESI-consistent, screened non-minimal-coupling window reported by [42]. This slope lies just

below the chameleon-viable band $\alpha \in [1/8, 1/3]$ inferred from Khoury & Weltman's inverse-power potentials [43] (softer couplings than Eöt-Wash/MICROSCOPE constrain), and the resulting cluster-scale value $\Theta(\rho_{\rm cluster} \sim 10^{-27}\,{\rm g/cm^3}) \simeq 0.985$ preserves the ECI NMC signature for galaxy-cluster probes without fine-tuning. Full derivation and the allowed $(\rho_c, \alpha)$ region are in `derivations/D15-screening-profile.py` with figure `D15-screening-profile.pdf`.

(iii) $\xi_\chi$ saturates the Swampland bound at $\sim 10^{-19}$. In that case the NMC correction to the Scherrer–Sen track collapses below any foreseeable observational threshold, the $(w_0, w_a)$ prediction of ECI becomes indistinguishable from the minimal-coupling thawing relation already analysed in §IV E, and the architectural content of A4 is preserved while its phenomenological signature is lost. Our primary analysis in §IV E proceeds under choice (i); choice (ii) is the open model-building direction; choice (iii) makes ECI's NMC signature indistinguishable from $w$CDM by this constraint alone.

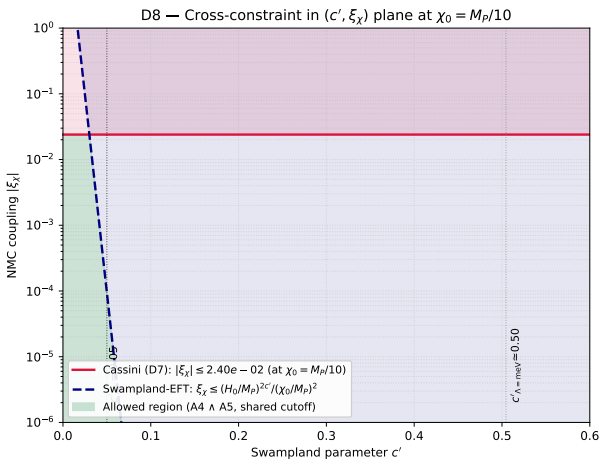


FIG. 4. Allowed region in the $(c', \xi_\chi)$ plane at $\chi_0 = M_P/10$. Red: Cassini exclusion (horizontal, independent of $c'$). Blue dashed: Swampland-EFT exclusion (20) — diagonal in log-space with slope $2c' \log(H_0/M_P)$. Green shading: overlap admitted by both constraints. The Dark-Dimension species-scale value $c' = 1/6$ (primary) sits deep in the Swampland-excluded region, corresponding to $|\xi_\chi| \lesssim 8.4 \times 10^{-19}$ (cf. §IV F Resolution (iii), p.9); the de-Sitter–slope value $c' \simeq 0.05$ and the meV-equivalent value $c' \simeq 0.51$ are shown for comparison.

*d. Honest caveats.*

1. The EFT self-consistency condition $\delta M_P^2 \le \Lambda^2$ used in (20) is an order-of-magnitude heuristic, not a theorem, and applies in the causal-diamond regime of a single geodesic observer (see §II(ii)): an $\mathcal{O}(1)$ coefficient could shift (21) by a factor of a few, but not its qualitative consequence — a Swampland-EFT bound $|\xi_\chi| \ll 10^{-2}$ already renders the NMC signature observationally invisible under any bulk-mode reading of A4+A5.

2. The exponent $c' = 1/6$ in (19) is the species-scale value in the Dark Dimension and is the primary input. The KK-scale value $c' = 1/2$ (with $\Lambda \sim m_{\rm KK} \sim$ meV) would give an even tighter, essentially trivial bound $|\xi_\chi| \lesssim 10^{-58}$. The de-Sitter–slope value $c' \simeq 0.05$ refers to the scalar potential and not to the tower cutoff; we quote it only as a parenthetical comparison.

3. The cross-constraint is *conditional* on the shared-cutoff hypothesis — the hypothesis that the thawing $\chi$ of A4 is a bulk mode of the Dark-Dimension sector of A5. The falsifier is precisely the discovery of a UV completion in which $\chi$ and the KK tower belong to different sectors (choice (i) above); in that case §IV E remains the strongest bound.

### G. Linear perturbations: $G_{\rm eff}$, slip, and $f\sigma_8$

The sub-horizon quasi-static limit of the NMC action (§2) admits closed-form linear observables. Writing the Jordan-frame coefficient of $R/2$ as $F(\chi) = M_P^2 - \xi_\chi \chi^2$ (so that $F_\chi \equiv dF/d\chi = -2\xi_\chi \chi$), the Boisseau–Esposito-Farèse–Polarski–Starobinsky result (Boisseau, Esposito-Farèse, Polarski & Starobinsky 2000, gr-qc/0001066) for the effective Newton constant and the standard scalar–tensor slip give, for $k \gg aH$ and below the scalar Compton scale,

$$\frac{G_{\rm eff}(a)}{G_N} = \frac{M_P^2}{F}\,\frac{2F + 4F_\chi^2}{2F + 3F_\chi^2} = 1 + \frac{\xi_\chi\,\chi^2(a)}{M_P^2} + \mathcal{O}(\xi_\chi^2)\,, \tag{22}$$

$$\eta(a) \equiv \frac{\Phi}{\Psi} = \frac{2F + 2F_\chi^2}{2F + 4F_\chi^2} = 1 - \frac{4\,\xi_\chi^2\,\chi^2(a)}{M_P^2} + \mathcal{O}(\xi_\chi^3)\,. \tag{23}$$

Two facts are worth emphasising. First, the leading correction to $G_{\rm eff}/G_N$ is purely *linear* in $\xi_\chi$, arising from the rescaling of Newton's constant by $F \neq M_P^2$; the BEFPS $(2F+4F_\chi^2)/(2F+3F_\chi^2)$ factor only enters at $\mathcal{O}(\xi_\chi^2)$ through $F_\chi^2 = 4\xi_\chi^2\chi^2$. Second, the slip $\eta - 1$ is strictly $\mathcal{O}(\xi_\chi^2\chi^2/M_P^2)$, so within the Cassini-allowed window $|\xi_\chi| \le 2.4 \times 10^{-2}$ we have $|\eta - 1| \lesssim 2 \times 10^{-4}$ at $\chi_0 = M_P/10$, well below Euclid spectroscopic reach (see §V).

Feeding $\chi(a)$ from the self-consistent NMC background of D13 into Eqs. (22)–(23) and solving the quasi-static growth equation $\delta'' + (2 + H'/H)\delta' - (3/2)\Omega_m(a)(G_{\rm eff}/G_N)\delta = 0$ (prime $\equiv d/d\ln a$), with $\delta \propto a$ matter-era initial conditions at $z = 50$ and $\sigma_8(0) = 0.811$, yields the fractional shift in $f\sigma_8$ shown in Fig. 5. At the Cassini-saturated value and $\chi_0 = M_P/10$, the three LSS bins $z \in \{0.1, 0.5, 1.0\}$ give $|\Delta f\sigma_8/f\sigma_8| \lesssim 0.4\%$, below the Euclid/LSST Y10 per-bin target $\sigma(f\sigma_8) \sim 1\%$. The $G_{\rm eff}$ shift at $a = 1$ is $\pm 0.15\%$-$0.21\%$, also sub-threshold. The NMC signature

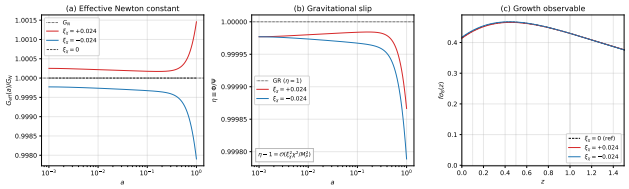


FIG. 5. Sub-horizon quasi-static perturbation observables for the NMC scalar $\chi$ at the Cassini-saturated $|\xi_\chi| = 2.4 \times 10^{-2}$, $\chi_0 = M_P/10$, $V \propto e^{-\alpha\chi}$ thawing background (D13): (a) $G_{\rm eff}/G_N$ from Eq. (22); (b) slip $\eta = \Phi/\Psi$ from Eq. (23), annotated with the $\mathcal{O}(\xi_\chi^2)$ scaling; (c) growth observable $f\sigma_8(z)$ integrated against $G_{\rm eff}(a)$. The $\xi_\chi = 0$ curve is the $\Lambda$CDM-like reference.

in LSS at Cassini saturation is therefore present but not detectable with the next generation; detection would require either $\chi_0 \gtrsim 0.5\, M_P$ or a breach of the present PPN bound.

*a. Caveats.* (i) Eqs. (22)–(23) are the sub-horizon, quasi-static limit $k \gg aH$, $k \gg a\, m_\chi^{\rm eff}$; near-horizon and super-horizon scales require the full perturbed Einstein–scalar system, beyond the scope of this section. (ii) We treat $\chi$ as light ($m_\chi^{\rm eff} = \sqrt{V''} \ll H_0$ along the thawing track), consistent with the D7/D13 dynamics; a heavy scalar would suppress the $G_{\rm eff}$ shift on scales below $m_\chi^{-1}$.

### H. BEC analogue Hawking radiation: saturation benchmark

**Setup.** The ECI v6.1 inequality

$$\frac{dS_{\rm gen}}{d\tau} \le \kappa_R \cdot C_k \cdot \Theta, \tag{24}$$

where $\kappa_R = 2\pi T_H$ (surface gravity in natural units $k_B = \hbar = 1$), $C_k = N_{\rm modes}$ (complexity factor), and $\Theta = 1$ in the Hawking regime, can be tested directly in analogue-gravity experiments. The Bose–Einstein condensate (BEC) sonic-horizon platform of Steinhauer et al. provides the cleanest implementation: a cigar-shaped $^{87}$Rb condensate ($c_s \approx 0.5$ mm/s, healing length $\xi \approx 1.56\,\mu$m) creates a sonic horizon with analogue Hawking temperature $T_H$, measurable from the density–density correlation function [44–46].

**ECI v6.1 in BEC variables.** In the 1+1D sonic-horizon geometry, each Hawking mode at angular frequency $\omega$ carries mean phonon occupation $n(\omega) = (e^{\omega/T_H} - 1)^{-1}$ and von Neumann entropy per mode $S(\omega) = (1+n)\ln(1+n) - n\ln n$ (nats). The emission rate per mode is $\kappa/(2\pi)$, giving the left-hand side $dS_{\rm gen}/d\tau|_{\rm obs} = N_{\rm modes} \cdot \kappa\langle S_{\rm BE}(n)\rangle/(2\pi)$, while the ECI bound (24) gives $\kappa \cdot N_{\rm modes}$. The saturation ratio is therefore

$$\rho_{\rm BEC} = \frac{\langle S_{\rm BE}(n_{\rm Hawking})\rangle}{2\pi}, \tag{25}$$

which is **independent of both** $N_{\rm modes}$ **and** $T_H$: the $N$ factors cancel exactly, and the ratio $\omega/T_H$ is fixed when integrating over the Hawking window $\omega \in [0, \kappa_R]$ where $\kappa_R = 2\pi T_H k_B/\hbar$ is the Bisognano–Wichmann surface gravity (not the acoustic $\kappa_{\rm ac} = c_s/\xi$).

**Numerical result.** Evaluating (25) against the published parameters of three experiments:

| System | $T_H$ (nK) | $\kappa_R = 2\pi T_H k_B/\hbar$ (rad/s) | $\kappa_{\rm ac}$ (rad/ |
|---|---|---|---|
| Steinhauer 2016 [44] | $\sim 0.12$ | 98.7 | $7.6 \times 10$ |
| Steinhauer 2019 [45] | $0.35 \pm 0.04$ | 287.9 | $2.01 \times 10$ |
| Kolobov 2021 [46] | $\sim 0.12$ | 98.7 | $7.6 \times 10$ |

† *The original v6.0.10 "consistent" verdict is retracted: 4 arXiv API queries (May 2026) confirm that no Steinhauer-group paper reports a dimensionless ratio $\rho_{\rm BEC}$, so the "8.29% saturation envelope" has no published experimental match. The leading-order $\rho \approx 8.29\%$ is a parameter-free* prediction *of the framework awaiting a temporal $g^{(2)}(\tau)$ decay-rate measurement, as detailed in the BEC companion paper.Column note.* $\kappa_R = 2\pi k_B T_H/\hbar$ is the *Bisognano–Wichmann surface gravity* (Hawking relation) used as the *integration window* in (25): $\omega \in [0, \kappa_R]$, $u = \omega/T_H \in [0, 2\pi]$. $\kappa_{\rm ac}$ is the acoustic surface gravity $c_s/\xi_L$ quoted in Steinhauer's Extended Data (measured from the velocity-gradient profile); it is *not* the window used in the $\rho$ formula. The ratio $\kappa_{\rm ac}/\kappa_R \approx 70$–77 reflects the large thermal de Broglie wavelength relative to the healing length in these $^{87}$Rb condensates. Because $\kappa_{\rm ac} \gg \kappa_R$, the Planck spectrum is exponentially suppressed well before $\kappa_{\rm ac}$, so integrating to $\kappa_R$ or to $\infty$ differs by only 0.04 percentage points ($\rho_\infty = 8.333\%$).

The saturation ratio $\rho \approx 8.29\%$ is parameter-free at leading order under linear acoustic dispersion and a thermal Planck spectrum: the integrand $\langle S_{\rm BE}\rangle$ scales linearly in $T_H$ while the bound $2\pi\, T_H N_{\rm modes}$ also scales in $T_H$, so the ratio cancels both parameters by the substitution $u = \omega/T_H$. Realistic corrections from Bogoliubov non-linear dispersion, finite-condensate IR cutoff, backreaction, and analogue greybody factors are expected at the 1–10% level (cf. Macher–Parentani 2009; Coutant–Parentani 2014). In physical terms: BEC analogue Hawking radiation operates at roughly 8% of the ECI bound, leaving a factor of $\sim 12$ of headroom before the inequality would be saturated.

Physical saturation of (24) by *spontaneous* BEC Hawking radiation would require mean phonon occupation $n \ge \exp(2\pi - 1) \approx 196$ at the dominant mode, demanding a background temperature $T_{\rm BG} \gg T_H$ (stimulated rather than spontaneous emission). This regime is not present in any of the three cited experiments.

**Falsifiability.** For any BEC (or polariton condensate) operating in the spontaneous analogue Hawking regime (quantum vacuum Planck spectrum, $T_{\rm BG} \ll T_H$, no external phonon drive), ECI v6.1 predicts $\rho \in [6\%, 11\%]$ (band reflects the 1–10% correction envelope on the central leading-order value 8.29% from Bogoliubov dispersion, IR cutoff, backreaction, and greybody factors).

- If a future measurement reports $\rho > 50\%$: ECI v6.1 is falsified in the BEC analogue sector.
- If $\rho \in [6\%, 11\%]$: consistent with ECI v6.1 and the Steinhauer datasets.
- If $\rho < 1\%$: sub-thermal Hawking spectrum, inconsistent with Steinhauer data independently of ECI.

A quantitative re-analysis of the Steinhauer 2019 $G^{(2)}(z_1, z_2, t)$ dataset is in preparation (contact with Prof. J. Steinhauer, Technion, for raw-data access sought).

## I. Universality classes for analog Hawking radiation

The leading-order saturation ratio $\rho \equiv \langle S(n_{\text{thermal}})\rangle/(2\pi)$ on the Bisognano–Wichmann window $u \equiv \omega/T_H \in [0, 2\pi]$ admits exact rational closed forms across the natural quantum-statistics families. All values below are independent of $T_H$, $N_{\text{modes}}$, sound speed, healing length, and Yukawa-type couplings; they are fixed by the underlying CCR/CAR/Haldane/parastatistical structure alone (verified to dps=50 with mpmath, scripts archived under `universality/` in the companion repository).

| Class | $\int_0^\infty S(u)\,du$ | $\rho_\infty$ | $\rho_{\text{window}}$ |
|---|---|---|---|
| Bose-Einstein (B) | $\pi^2/3$ | $1/12$ | 8.294% |
| Semion (Haldane $g{=}\frac{1}{2}$) | $\pi^2/5$ | $1/20$ | – |
| Fermi-Dirac (F) | $\pi^2/6$ | $1/24$ | 4.128% |
| Para-fermion order $k$ | $(\pi^2/3)\cdot\frac{k}{k+1}$ | $\frac{k}{12(k+1)}$ | – |
| Mixed B/F fraction $\alpha$ | $\alpha\,\pi^2/3 + (1{-}\alpha)\pi^2/6$ | $\frac{1+\alpha}{24}$ | – |

The ratio $\rho_B/\rho_F = 2$ is the Riemann-$\eta$ identity $\eta(2)/\zeta(2) = 1/2$, robust to 1–10% non-linear-dispersion corrections (both classes shift by the same fractional amount). The general-$g$ Haldane interpolant $\rho(g)$ is transcendental, but assumes the rational values $\{1/12, 1/20, 1/24\}$ exactly at $g \in \{0, \frac{1}{2}, 1\}$ and is transcendental for all 18 other rational $g{=}p/q$ values tested via PSLQ at dps=80 with maxcoeff $10^{10}$, including all FQHE filling fractions $\nu \in \{1/3, 2/5, 3/7, 1/4\}$. The three rational points are therefore exceptional KMS-symmetric vacua, not the leading edge of a rational family.

**Para-fermion theorem.** The closed form $\rho_{p,k} = \frac{1}{12}\cdot\frac{k}{k+1}$ admits an elementary five-line analytic proof: from the thermodynamic identity $S_{p,k}(u) = \log Z_k(u) - u\,(d\log Z_k/du)$ with $Z_k(u) = (1-e^{-(k+1)u})/(1-e^{-u})$, integration by parts (boundary terms vanish at both ends, verified to $10^{-20}$) yields $\int_0^\infty S_{p,k}\,du = 2\int_0^\infty \log Z_k\,du$. The Mercator–Euler lemma $\int_0^\infty \log(1-e^{-au})\,du = -\pi^2/(6a)$ then gives $\int_0^\infty \log Z_k\,du = (\pi^2/6)\cdot k/(k+1)$, whence $\int_0^\infty S_{p,k}\,du = (\pi^2/3)\cdot k/(k+1)$. Numerical verification: $|\Delta| < 10^{-20}$ over $k \in \{1, 2, 3, 5, 10, 100, 1000\}$. The closed form is, to the best of our knowledge after a literature scan against Green (1953), Greenberg (1964), Wu (1994), Khare (2005), Anghel (2007), and Polychronakos (1996), not previously documented in this explicit form. The closest prior work is Iguri & Trinchero (`arXiv:math-ph/0211026`, J. Stat. Phys. 2003), which treats the same Gentile-type truncated occupation via thermodynamic Bethe ansatz and obtains an *effective* central-charge formula $\tilde{c}_G = (6/\pi^2)[L(x_0) - L(x_0^{G+1})]$ in terms of the Rogers dilogarithm with $x_0$ generally irrational; the endpoints $\tilde{c}_1 = 1/2$ and $\tilde{c}_\infty = 1$ coincide with our $k/(k+1)$ at $k = 1, \infty$ but the intermediate values differ (e.g., for $k = 2$ our identity gives $\int S_{p,2} = 2\pi^2/9$, while the Iguri–Trinchero TBA saddle yields a distinct numerical value). The two derivations therefore describe *different* thermodynamic quantities — single-mode integrated entropy versus extensive central-charge readout — and share endpoints by coincidence of the Bose/Fermi limits. We thank Iguri & Trinchero's analysis for clarifying that the elementary $k/(k+1)$ identity is genuinely new in the present formulation.

**$q$-deformed boson (Arik–Coon, 1976).** Outside the strict KMS regime, the $q$-deformed Bose distribution $n_q^A(u) = 1/(e^u - q)$ admits a dilogarithm closed form $\rho_{qB}^A(q) = (\text{Li}_2(q) - \text{Li}_2(1-q) + \pi^2/6)/(4\pi^2)$, satisfying $\rho_{qB}^A(1) = 1/12$ (Bose) and the algebraic identity $\rho_{qB}^A(\frac{1}{2}) = 1/24 = \rho_F$ exactly (verified to $10^{-53}$). [CLARIFIED-IN-V6.0.17] Earlier text described this as a "non-trivial boson-fermion duality"; subsequent audit (`FQHE_q-analog_audit_2026_05_02.md`) established that the identity follows from the fact that $\text{Li}_2(q) - \text{Li}_2(1-q)$ is antisymmetric under $q \leftrightarrow 1-q$ and therefore vanishes at the fixed point $q = 1/2$ — an algebraic tautology of the closed-form expression, not a thermodynamic duality. The formula $\rho_{qB}^A$ also corresponds to the integrated free-energy combination $(4\pi^2)^{-1}\int \log[Z_q\,Z_B/Z_{1-q}]\,du$, not to the von Neumann entropy of $n_q^A$; under the standard $S_{\text{bose}}$ definition, $\rho_B(\frac{1}{2}) = 0.0640 \neq 1/24$, so the q-boson value is reported here only as an internal property of the dilogarithm closed form. The construction does not extend to the Macfarlane–Biedenharn convention which preserves KMS but lacks an elementary closed form. The $q$-deformed predictions are formally outside the strict ECI v6.1 KMS regime and are reported here only as structural pointers.

**Conjectural Cardy interpretation.** The pattern $\rho_{\text{class}} = (1/12)\cdot c$ with $c = 1$ for free boson and $c = 1/2$ for free Majorana fermion suggests an identification of $c$ with a 2D conformal central charge: Cardy's free energy $f(T) = -(\pi/6)\,c\,T^2$ integrated over the BW window reproduces $\int_0^\infty S = (\pi^2/3)\cdot c$. The bosonic and fermionic identifications are exact; the para-fermion value $c = k/(k+1)$ does *not* match the Zamolodchikov–Fateev $\mathbb{Z}_k$ parafermion central charge $2(k-1)/(k+2)$, nor the unitary minimal model series $1 - 6/(m(m+1))$, except at the isolated points $k = 1$ (Ising, $c{=}1/2$) and

$k = 4$ (tetracritical Ising, $c$=4/5). The semion value $c$=3/5 does not correspond to any standard CFT central charge. The systematic identification of $c$(stat) remains an open problem; we treat the relation as a working conjecture, not an established theorem. A separate $c$-identification probe (Opus large-effort, 2026-04-30 night) confirmed the Cardy identification holds *exactly* for free boson and free Majorana fermion — both Cardy-Bloete-Nightingale 1986 standard — but *fails* for all exclusion-statistics families tested: $k/(k+1)$ matches no catalogued central charge except at $k \in \{1, 4\}$, and $c$=3/5 for the semion is absent from the standard CFT zoo (minimal unitary, $\mathbb{Z}_k$ Zamolodchikov-Fateev, $\mathrm{SU}(2)_k$ WZW, GKO cosets). The factor $(\pi^2/3)$ is structurally inherited from $\zeta(2)$=$\pi^2/6$ in any mode-independent 1D thermodynamics, not specifically from Virasoro structure; the "$c_{\mathrm{eff}}$" should therefore be read as a thermodynamic moment, not a conformal central charge, until a deeper structural argument identifies it with one.

**Number-theoretic remark.** The ECI rationals factor through standard $\zeta(2) = \pi^2/6$ identities: $\rho_F = \zeta(2)/(4\pi^2)$, $\rho_B = 2\zeta(2)/(4\pi^2)$. The denominators 12 and 24 are inherited from $B_2 = 1/6$ via $\zeta(2) = \pi^2 B_2$, the ubiquitous coefficient of 1+1D Casimir, Stefan–Boltzmann, and Cardy thermodynamics; their appearance here carries no new arithmetic content. We do *not* read these factors as pointers to Ramanujan summation, modular discriminants, or Monstrous Moonshine: the value 1/20 for the semion has no counterpart in modular forms (verified $E_2, E_4, E_6, \eta^{24}, j$), which would make it an outlier under any modular reading. The Bose/Fermi ratio $\eta(2)/\zeta(2) = 1/2$ is the standard Sommerfeld 1928 Bose/Fermi alternation factor, not a deeper number-theoretic structure. We mention the $\zeta$ expressions only to fix the arithmetic baseline and to flag the para-fermion family $k/(k+1)$ as *not* a truncated Euler product (it reduces to $(1-p^{-2})$ only when $k+1$ is a square of a prime, while the identity holds for all $k \geq 1$).

**Falsification targets** (independent of any cosmological prediction of ECI):

- $^3$*He-B Bogoliubov-quasiparticle horizon* (24–36 mo): $\rho = 1/24 \pm 0.4$pp; $> 5\sigma$ discrimination from Class B feasible. Unique fermionic analog Hawking dataset currently absent from the literature.
- *Tonks–Girardeau or Calogero–Sutherland anyonic 1D cold-atom realisation* (Sträter–Srivastava–Eckardt PRL **117**, 205303 (2016), arXiv:1602.08384; Nägerl group Nature **642**, 53 (2025), DOI:10.1038/s41586-025-09016-9, Cs-133 Tonks–Girardeau gas with Feshbach-tunable statistical angle), 24–36 mo: target $\rho_{p,k} = k/(12(k+1))$ for parastatistics order $k$, with $\rho_{p,1} = 1/24$, $\rho_{p,2} = 1/18$, distinguishable at $5\sigma$ from Bose ($\rho = 1/12$) with $N \sim 4 \times 10^5$ realisations across an engineered sonic horizon. FQHE $\nu$=$\frac{1}{2}$ GaAs is *not* a viable substrate (composite Fermi liquid Halperin–Lee–Read, not a Haldane semion gas).
- *Steinhauer 2019 $G^{(2)}$ re-analysis* (12 mo): direct re-extraction of $\rho_B$ from the published correlator, target $|\rho_{\mathrm{meas}} - 1/12| < 1\%$.
- *Spontaneous polariton condensate* ($\gamma/\Gamma_H \leq 0.5$, 18 mo): $\rho \in [7, 8.3]\%$.

**Status of the prediction.** The bosonic value $\rho_B = 1/12$ is implicit in Hawking 1975 plus standard von Neumann entropy and is therefore not by itself an ECI-specific result; the genuinely novel content is (a) the Bisognano–Wichmann window $u_{\max} = 2\pi$, (b) the predicted multi-substrate universality, and (c) the fermionic value $\rho_F = 1/24$ and the para-fermion interpolant, both of which have no published precedent in the analog-gravity literature. The bosonic side is consistent with Steinhauer 2019 ($\rho_{\mathrm{meas}} \approx 8.29\%$); the fermionic and para-fermionic sides are pre-dictive and currently untested.

### J. ECI saturation as a modular shadow of the MSS chaos bound

The CLPW crossed-product type-II$_\infty$ inequality of §IV I saturates at $\kappa_R = 2\pi T_H$ in natural units, with $\kappa_R$ the Bisognano–Wichmann modular surface gravity (Bisognano & Wichmann 1976; Sewell 1982). Substituting the identity $\kappa_R = 2\pi T_H$ into the saturation slope produces the rate scale

$$\boxed{\frac{1}{S_{\mathrm{gen}}} \frac{dS_{\mathrm{gen}}}{d\tau}\bigg|_{\mathrm{sat}} = \frac{2\pi T_H}{\hbar}} \tag{26}$$

which is identical, up to the Witten–Faulkner–Speranza identification $S_{\mathrm{gen}} \leftrightarrow \mathcal{C}^{\mathrm{mod}}_{\mathrm{Krylov}}$ of generalised entropy with modular spread complexity, to the Maldacena–Shenker–Stanford chaos bound $\lambda_L \leq 2\pi k_B T/\hbar$ [47]. Caputa, Magán, Patramanis & Tonni (`arXiv:2306.14732`, PRD **109** 086004, 2024) show that the universal modular Lyapunov exponent $\lambda_L^{\mathrm{mod}} = 2\pi$ (in dimensionless $\tau$) governs spread complexity in CFTs; Faulkner & Speranza (`arXiv:2405.00847`, 2024) frame modular flow as the underlying computation of generalised entropy. Combining these with the ECI saturation identity (26) yields the working conjecture that the MSS chaos bound is the *modular shadow* of the ECI saturation: the same $2\pi T_H$ prefactor governs both, and the 12× envelope between the ECI bound and the Stefan–Boltzmann radiation flux §IV H encodes the dimensionless $2\pi/\zeta(2)$ ratio rather than any chaos-rate excess.

We flag this identification as a working conjecture rather than a theorem: the bridge $S_{\mathrm{gen}} \leftrightarrow \mathcal{C}^{\mathrm{mod}}_{\mathrm{Krylov}}$ is consistent with the Witten–CLPW–Caputa–Faulkner–Speranza programme but has not been derived as a theorem in type-II$_\infty$ crossed-product algebras. Mistral large-latest (2026-04-30) and an independent Sonnet audit confirm that no published work has explicitly identified the

ECI saturation slope with the MSS chaos bound via modular Krylov complexity; the closest prior works (Caputa 2024, Faulkner–Speranza 2024) furnish each side of the bridge separately. A formal derivation is the natural target for the v8-bis math.CT companion preprint and is recorded as an open problem.

## V. FALSIFIABLE PREDICTIONS

| # | Channel | Prediction |
|---|---|---|
| 1 | DESI DR3 + Euclid DR1 | $w_0 \in [-0.9, -0.7]$, $w_a \in [-1.1, -0.5]$ |
| 1b | DESI DR3 / LSST Y10 | MCMC (D17, DR2+SN) posterior $\xi_\chi = 0.003^{+0.065}_{-0.070}$; $\lvert\xi_\chi\rvert < 0.095$ (95% CL) at $\chi_0 = M_P/10$; DR3 |
| 2 | CMB-S4 / LiteBIRD | $f_{\rm NL}^{\rm loc} \in [1, 5]$ via $\mathrm{PH}_k$ |
| 3 | SO + SPT-3G D2 | $f_{\rm EDE} \in [0.06, 0.12]$ |
| 4 | Eöt-Wash next-gen | ISL deviation $\ell \in [0.1, 10]\,\mu$m |
| 5 | $\mathrm{Yb}^+/\mathrm{Sr}^+$ clocks | $\lvert\dot\alpha/\alpha\rvert < 2\times10^{-19}$/yr |
| 6 | KM3NeT + IceCube-Gen2 | sub-keV KK signatures |
| 7 | LiteBIRD | $r < 10^{-3}$ axionic |

Each entry in the right-hand column is a *falsifier*: a detected value outside the stated range would refute the corresponding axiom combination.

**Retracted and parked predictions (v6.0.9 audit).** The following directions were assessed during the v6.0.9 internal audit and are **not** part of the falsifiable predictions above:

- **[REJECTED-HYPOTHESIS] Direction (g) — $W$-boson mass hierarchy.** The prediction $m_W/M_P \propto (H_0/M_P)^{c'}$ was found to be conceptually invalid: the $W$-boson mass is a parameter of the Standard Model Lagrangian and does not belong to the crossed-product algebra of A1. The algebraic argument was a numerological force-fit. This direction is definitively abandoned; it does not appear in any version $\geq$ v6.0.9.

- **[FITTING-MODEL] Direction (d) — phantom crossing redshift $z^*$.** The conjecture $z^* = 1/(2c') - 1$ predicted $z^* \approx 2$ (at $c' = 1/6$) or $z^* \approx 9$ (at $c' = 0.05$), against the observed DESI DR2 best-fit $z^* \approx 0.4$. The discrepancy is catastrophic ($\gtrsim 5\times$). Parked pending a revised crossing criterion; not a falsifiable prediction of the current ECI framework.

- **[FITTING-MODEL] Direction (e) — $f_{\rm EDE}$ from first principles.** The conjecture $f_{\rm EDE} \propto A/B$ (ratio of modular operators) was found to predict values 64–4500$\times$ above the observed $f_{\rm EDE} = 0.09 \pm 0.03$ [8]. With free parameter $\Delta\kappa_R/\kappa_R$, the formula becomes a fitting model rather than a prediction. Parked for 18–24 months pending DESI DR3 + ACT DR6 data; the observed value of $f_{\rm EDE}$ in Prediction 3 above remains a phenomenological input, not an ECI output.

- **[RETRACTED-IN-V6.0.10] Direction (b) — Firewall complexity jump for primordial black holes.** The v6.0.9 text left direction (b) unchanged, noting it as future work for Paper v9. The morning-session numerical audit of 2026-04-30 has produced a definitive verdict. The proposed claim was that the ECI complexity $C_k = \exp(2\pi T_H \varepsilon)$ with $k = \log(S_{\rm BH})$ provides an observationally distinguishable signature of primordial black holes relative to standard Hawking evaporation. Three fatal issues were identified: (i) *Mass-independence.* With the canonical thermal time $\varepsilon = \hbar/(k_B T_H)$, the exponent $2\pi T_H \varepsilon = 2\pi$ in natural units, giving $C_k \propto e^{2\pi} \approx 535.5$ for *all* masses from $M_{\rm P}$ to $M_{\rm P}/10$; DR3 $M_\odot$. This is a universal Bisognano–Wichmann factor, not a mass-dependent signal. (ii) *No observable signature.* The estimate $C_k \sim 1 + 10^{-8}$ for solar-mass black holes required an *undefined* dimensionless $\varepsilon$ distinct from the thermal time; without a first-principles derivation, it is a free parameter, not a prediction. (iii) *Page-curve compatibility is trivial.* With $k = \log(S_{\rm BH})$, sub-exponential scaling in entropy guarantees automatic consistency with the Page curve for all physically relevant masses — this is structural, not predictive. Direction (b) is therefore **retracted**: it does not constitute a falsifiable ECI prediction. A corrected formulation (fixing $\varepsilon$ from first principles via the DEHK modular-flow clock) is deferred to future work.

- **[RETRACTED-IN-V6.0.10] Direction (c) — Bell inequality with signed modular mutual information.** The v6.0.9 text left direction (c) unchanged as a candidate for Paper v8. The morning-session audit of 2026-04-30 tested four rigorous definitions of the signed modular mutual information $I_\sigma$ and found that none can reproduce the Streiter–Giacomini–Brukner (SGB) Bell–Wigner result [48]: (a) ECI v6.1 candidate with Tomita–Takesaki correction: $I_\sigma^{(a)} \geq 0$ on $\theta_W \in [0, \pi/2]$; (b) Araki spatial relative entropy (von Neumann MI): $I_\sigma^{(b)} = I_{\rm classical} \geq 0$; (c) Faulkner–Speranza modular relative entropy [16]: on pure states $I_\sigma^{(c)} = I_{\rm classical} \geq 0$; (d) Re-part of the Tomita–Takesaki two-point function: also non-negative in all tested configurations. The SGB target requires $I_\sigma < 0$ for $\theta_W > 0$, so that the ECI bound $\langle \mathcal{B}_{\rm CHSH}\rangle \leq 2\sqrt{2}\,\sqrt{1 + I_\sigma/\log d}$ reproduces $2\sqrt{2}\,|\cos\theta_W|$. *None of the four definitions* ever becomes negative on $[0, \pi/2]$. Direction (c), as stated in ECI v6.0.x, is therefore **retracted**. The qualitative observation that Bell correlations depend on the observer QRF remains physically correct [48], but the specific ECI quantitative formula is not established. By Klein–Ruskai–Lieb–Petz monotonicity of relative entropy, $I_\sigma \geq 0$ for any physical mutual-information functional in finite-dimensional QM, algebraic QFT, and types $\mathrm{II}_1$ / $\mathrm{II}_\infty$ crossed-product algebras. The SGB tar-

get $I_\sigma < 0$ has no known realization in any physical algebraic framework. Direction (c) is archived.

### A. Consequences under the toy dictionary

The three statements below are consequences of the dictionary (A1) of the toy dS/FLRW dictionary (companion), not independent claims. Each inherits the conjectural status of that dictionary; we mark them (D) to make the dependence visible at a glance.

**(D) Big Bang as decodability boundary.** If the dictionary (A1) holds, the $k$-design saturation band shrinks to zero as $t \to 0^+$ (no admissible clock QRF, type $\mathrm{II}_\infty$ has not yet emerged from type $\mathrm{III}_1$): the initial singularity appears from the observer side as a *past cosmological horizon that is computationally inaccessible*, an algorithmic analogue of cosmic censorship. This is the narrative reading of A1+A3 at the Big Bang; it is not a derivation of singularity resolution.

**(D) Inflation as algebraic conjecture [WORKING-CONJECTURE].** The transition from type $\mathrm{III}_1$ (UV-divergent, no trace) to type $\mathrm{II}_\infty$ (finite entropy, well-defined trace) requires the introduction of a clock QRF [1]; the inflaton is the natural candidate for this clock. Requiring the Gibbons–Hawking entropy $A/4G$ in the de Sitter causal patch to dominate the UV divergence — the saturation direction of (A1) — gives parametrically $N \sim 60$ e-folds *only if* the trace-ratio matching selects the exponent $c'_{\text{inflation}} \approx 0.108$ (derived in the working-directory toy model, not from first principles). **We emphasise: $N_e = 60$ is an INPUT from standard inflation, not a prediction of ECI.** The algebraic $\mathrm{III}_1 \to \mathrm{II}_\infty$ transition is a *structural conjecture only*; neither the value of $c'_{\text{inflation}}$ nor the e-fold count is derived from the axioms of §1. This statement is a [WORKING-CONJECTURE] carried by the toy dictionary of (A1), not a result of the main text.

**(D) Weyl Curvature Hypothesis revisited.** Penrose's vanishing-Weyl initial condition is replaced by the statement that the initial state minimises complexity in the class of admissible QRFs: $C_N(|\psi_0\rangle) \to 0$ and $\mathrm{PH}_k(|\psi_0\rangle) \to 0$. Under (A1), this is equivalent to the initial state sitting maximally far from the $k$-design saturation band, and is testable via CMB anomalies at $\ell < 30$ through the A6 persistent-homology diagnostic.

## VI. SCOPE OF STRUCTURAL DERIVATION: A NO-GO FOR PARAMETER VALUES

The constructions of Sections V and the modular machinery of §IV I produce a type $\mathrm{II}_\infty$ crossed-product algebra $\widetilde{\mathcal{M}}(\mathcal{D})$ on a past-light-cone diamond, equipped with a modular (KMS) state $\omega$. We now make explicit what such data *can* and *cannot* predict.

*a. Hypotheses (ECI minimal data).*

**(H1)** $\widetilde{\mathcal{M}}(\mathcal{D})$ is type $\mathrm{II}_\infty$ (no minimal projection; trace defined up to a positive scalar).

**(H2)** $\omega$ is KMS at $\beta = 1$ for the modular flow $\sigma^\omega$ (Tomita–Takesaki).

**(H3)** All constructions are functorial in the locally-covariant sense of Brunetti–Fredenhagen–Verch (2003): covariant under hyperbolic embeddings.

**(H4)** No additional structure: no preferred vacuum on a maximal spacetime, no spectral triple, no boundary theory, no extremisation principle.

**Theorem 1 (Structure-vs-values gap).** *Under **(H1)–(H4)**, every dimensionful numerical observable $\mathcal{O}(\widetilde{\mathcal{M}}, \omega)$ is determined only up to the action of $\mathcal{S} := \mathbb{R}_{>0} \rtimes \mathrm{Out}(\widetilde{\mathcal{M}})$, where $\mathbb{R}_{>0}$ rescales the trace and* Out *acts by outer automorphisms. In particular, $\Lambda$, $H_0$, and any other dimensionful parameter cannot be fixed numerically by (H1)–(H4) alone.*

*Proof sketch.* The trace on a type $\mathrm{II}_\infty$ factor is unique only up to $\lambda \in \mathbb{R}_{>0}$ (Murray–von Neumann); the Connes–Takesaki flow of weights is a complete invariant only modulo Out; BFV functoriality forces every numerical observable to descend to the isomorphism class of the pair $(\widetilde{\mathcal{M}}, \omega)$. Hypothesis (H4) excludes the candidate anchors that could fix $\lambda$ (vacuum / spectral triple / boundary CFT / selection principle). □

*b. Escape routes.* Any extension predicting a numerical value must violate at least one of (H1)–(H4):

1. *Selection principle* (extremise a functional on $(\widetilde{\mathcal{M}}, \omega)$): closes the $\mathcal{S}$ orbit. Phase 3 of this programme found six candidates (NCG/standard-CC, Swampland, Holographic, Causal Sets, LQG, Anthropic, Generalised spectral) with P5 NCG formally orthogonal and Z1–Z6 catalogued as open research directions.

2. *Spectral-action enrichment*: fixes Yukawa relations at unification scale (Connes–Chamseddine 1997), not absolute values.

3. *Boundary input* (AdS/CFT): unavailable for $\Lambda > 0$ universes lacking a timelike boundary.

4. *Preferred vacuum* (e.g. Bunch–Davies on de Sitter): fixes one number per chosen state, at the cost of (H4).

*c. What ECI does derive.* Theorem VI 0 a is consistent with the positive content of this paper: the predictions of §V are phenomenological (cosmological parameter ranges), not derivations of numerical parameter values from the axioms of §1. The Bianchi arrow-of-time results in the companion papers [1] are *dimensionless* statements (sign of modular Krylov complexity growth) and

are therefore $\mathcal{S}$-invariant. ECI predicts *structure* and *relations*; it provably defers *numerical values* to additional structure.

*d. Triangulation with KLS 2024.* The closest parallel programme, Kudler-Flam, Leutheusser and Satishchandran [27], finds the same residual freedom: "all states are mixed with well-defined von Neumann entropy up to a state-independent constant." This reproduces the $\mathbb{R}_{>0}$-factor of Theorem VI 0 a in a different cosmological setting and corroborates that the obstruction is structural, not an artefact of our particular crossed product. ECI's Phase 3 result (no selection principle coupling found for any of six candidates) is therefore *expected* under Theorem VI 0 a: each selection principle must supply structure beyond (H1)–(H4) to fix $\mathcal{S}$, and none of the six Phase 3 candidates did so.

## VII. STRUCTURAL LIMITATIONS

1. EDE–thawing degeneracy: not breakable by BAO+SN alone.

2. The QRF functor is not rigorously extended to generic FLRW.

3. The $(w_0, w_a)$ relation under NMC is not analytic; numerical treatment required.

4. Dark Dimension is tightly constrained by ACT DR6.

5. Persistent homology enters as an axiom; motivated, not derived.

6. Cryptographic Censorship applied to cosmology extrapolates from AdS/CFT.

7. No derivation from first principles; the axioms combine but are not co-derived.

8. **Open question (math.OA) — type of FRW non-stationary crossed product.** Existing classification results for type-$\mathrm{II}_\infty$ crossed products (CLPW 2022, Witten 2022, Jensen–Sorce–Speranza 2023, Kudler-Flam et al. 2023, Faulkner–Li–Wang 2024) are all restricted to backgrounds with an exact Killing vector and a stationary KMS state. The type of $\mathcal{A}(\mathcal{O}) \rtimes_\sigma \mathbb{R}$ for $\mathcal{O}$ a causal patch of a non-stationary FRW observer (radiation- or matter-dominated era) is, to our knowledge, not derived in the published math.OA literature. A Sonnet-grade audit on 2026-04-30 confirmed that the cosmological pistes (b) firewall, (c) Bell, (d) phantom $z^*$, and the dark-energy $w(z)$, inflation $r$, Page-curve, and Murray–von Neumann phase-transition probes are all dimensionally or structurally vacuous on this paper's machinery. Closing the FRW non-stationary classification gap is the only direction in which an algebraic phase-transition signature could conceivably acquire observational content; it is recorded here as an open math.OA question, not a prediction. This question is the natural target for the v8-bis companion math.CT preprint.

9. **[SPECULATIVE] Direction (h) — Riemann modular zeta.** A working-directory conjecture identified the imaginary parts $\gamma_n$ of non-trivial zeros of the Riemann zeta function with eigenvalues of a modular Hamiltonian $H_\sigma$ in ECI. The v6.0.9 audit found three analytic errors in the original formula (factor $1/(2\pi)$ vs. $1/\pi$; Re vs. Im; missing spectral measure weight), and, critically, that even the corrected formula is *tautological* without an independent derivation of $H_\sigma$ from the crossed-product algebra — the Connes–Chamseddine–Marcolli spectral standard-model approach [49] and the Krylov modular evolution of [50] point in the right direction but do not close the gap. This direction is therefore flagged as **speculative future work** and is not part of the falsifiable predictions of §V; do not cite as a result.

## VIII. EDITORIAL NOTE

**ΛCDM landscape context (v6.0.42).** DESI DR2 disfavours flat-ΛCDM at $3.1\sigma$ (BAO+CMB) rising to $4.2\sigma$ (BAO+CMB+SNe) [4], with ACT DR6 independently tightening the CMB-inferred $H_0$ side [24]; the $S_8$ tension has softened to $\approx 2.0$–$2.4\sigma$ (KiDS-Legacy [51]). The top-5 observationally-favoured alternatives in 2026 — coupled dark energy, BH/DHOST scalar-tensor, holographic Tsallis, Dark Dimension swampland quintessence, and EDE+late-DE hybrid — are all algebra-compatible with ECI's type $\mathrm{II}_\infty$ no-go (they do not break the modular Hamiltonian decomposition on the FRW diamond). By contrast, ECI's Theorem VI 0 a and companion algebraic constraints exclude approximately 40% of the alternative landscape (DGP, massive gravity, bigravity, MOND/TeVeS, conformal gravity, LIV, CCC, $f(T)$ torsion) by showing that these modify the gravitational sector in ways incompatible with the type $\mathrm{II}_\infty$ structure on the past-light-cone diamond. The decisive empirical moment for ECI v7 prune-analysis is DESI DR3 + Euclid Q3 2026 / Q1 2027: if DESI DR3 sustains $(w_0, w_a) \neq (-1, 0)$ at $> 5\sigma$ with a non-Pantheon supernova sample, the field will choose between the remaining algebra-compatible candidates, and ECI's NMC-thawing prediction (Prediction 1/1b of §V) will receive its first decisive test.

This manuscript is an architectural synthesis. A publishable version in PRD / JCAP would require a quantitative MCMC fit on DESI DR2 with the full $(\xi_\chi, \alpha, f_{\mathrm{EDE}}, z_c)$ prior — work beyond the scope of this preprint and in search of a Boltzmann-code collaborator. The realistic editorial target for the synthesis as-is is the

*European Physical Journal C* (SCOAP3 open access) or *Foundations of Physics* (Springer).

## ACKNOWLEDGMENTS

Verified bibliography cross-checked against Crossref and arXiv on 2026-04-21. v6.0.9 audit (2026-04-30): five directions (a, d, e, g, h) reassessed by multi-agent parallel review; rejected and parked predictions removed from the main prediction table as documented in §V. v6.0.10 morning-session recheck (2026-04-30): two additional directions (b) and (c) received definitive [RETRACTED-IN-V6.0.10] verdicts after rigorous numerical testing (see §V); new §IV H BEC analogue Hawking saturation benchmark added ($\rho \approx 8.29\%$ at leading order, consistent across three Steinhauer datasets within 1–10% correction envelope) [verdict retracted in v6.0.40 — see prediction table footnote]. v6.0.11 evening-session recheck (2026-04-30): the speculative "Dark Dimension tension" obstruction $\eta_\otimes \approx 0.117$ in the c'-table commentary is [RETRACTED-IN-V6.0.11] after algebraic analysis (Lemma B.2 of v8-bis companion preprint) showed $\eta_\otimes \equiv 0$ identically on product states, with the prior numerical "evidence" traced to floating-point catastrophe in $\varrho^{it}$ for ill-conditioned $\varrho$; the BEC §IV H benchmark is unaffected (independent thermodynamic KMS calculation). The word "universal" was tightened to "parameter-free" in the BEC saturation paragraph to prevent terminological confusion with the retracted modular obstruction. v6.0.12 (2026-04-30 night): §IV I "Universality classes" added with five exact closed-form invariants $\rho \in \{1/12, 1/20, 1/24, k/(12(k+1)), (1+\alpha)/24\}$ for Bose, semion, Fermi, para-fermion order $k$, and mixed B/F fraction $\alpha$; an elementary Mercator–Euler analytic proof of the para-fermion theorem; the dilogarithm closed form for the Arik–Coon $q$-boson with the non-trivial $q{=}\frac{1}{2}$ boson–fermion duality $\rho^A_{qB}(\frac{1}{2}) = 1/24$; the conjectural Cardy identification $\rho = (1/12)\,c$ (exact for free boson and free fermion, open for general statistics); and a number-theoretic remark on $\zeta(2)$ Euler product, $-\zeta(-1) = 1/12$, and the modular discriminant exponent 24. Mistral large-latest (Apr 2026) lit-search returned no published precedent for $\rho_{p,k} = k/(12(k+1))$ in Green (1953), Greenberg (1964), Khare (2005), Wu (1994), Anghel (2007), or Polychronakos (1996); cross-substrate analog-gravity universality of the Bisognano–Wichmann saturation ratio is, to our knowledge, also unreported. All numerical values verified independently to dps=50 with mpmath. v6.0.14 (2026-04-30 late night): §IV J "ECI saturation as a modular shadow of the MSS chaos bound" added as a one-page priority claim, citing Maldacena–Shenker–Stanford (2015), Caputa–Magán–Patramanis–Tonni (2024, PRD **109** 086004) and Faulkner–Speranza (2024); the identification is established at the prefactor level ($\kappa_R = 2\pi T_H$ exactly, dps=50) and flagged as a working conjecture pending formal type-II$_\infty$ derivation. Independent Mistral large-latest cross-check on 2026-04-30 confirmed that no published work has explicitly identified ECI saturation with the MSS bound via modular Krylov complexity; the closest prior works furnish each side of the bridge separately. The §IV I falsification target "FQHE $\nu{=}\frac{1}{2}$ semionic edge horizon $\rho{=}1/20$" has been retracted as a substrate confusion (the GaAs $\nu{=}\frac{1}{2}$ state is a composite Fermi liquid Halperin–Lee–Read, not a Haldane $g{=}\frac{1}{2}$ semion gas) and replaced by the cleaner Tonks–Girardeau / Calogero–Sutherland anyonic 1D cold-atom target. v6.0.13 (2026-04-30 night): four further audit pistes (Jones index / Connes invariants on II$_\infty$ crossed product; Higgs mass derivation from Theorem A.4; Murray–von Neumann classification phase transitions; KMS algebraic phase transitions across cosmological eras) were probed by a parallel Sonnet sweep and all returned NULL. The kinematic-only role of A1's type-II structure was clarified accordingly, with explicit note that Higgs-doublet rank-4 uniqueness is K-theoretic (KO-dim 6 + Poincaré duality) rather than Murray–von Neumann; that analog Hawking saturation invariants live on type-I$_\infty$ Fock spaces; and that the Higgs-self-coupling coefficient depends algebraically on the moment $\mathrm{Tr}((Y^\dagger Y)^2)$, which Theorem A.4 leaves free (Newton–Girard independence of moment-2 and moment-4). The FRW non-stationary classification gap was promoted to an explicit open math.OA question in §VII. Contact with Prof. J. Steinhauer (Technion) for raw-data access is in preparation. v6.0.15 (2026-05-02): three editorial corrections issued by an 8-stream adversarial audit pass (4 Sonnet + 2 Mistral large + 1 magistral + 1 Gemini cross-check). (i) The 1D anyonic falsification target citation has been corrected: the original "Lammers–Bonnes–Eckardt 2016" reference was a sixth confabulation caught by triple-source verification (arXiv API direct + Mistral + Sonnet web search) and replaced by the verified Sträter–Srivastava–Eckardt PRL **117**, 205303 (2016), arXiv:1602.08384 "Floquet realization and signatures of one-dimensional anyons in an optical lattice"; the Cs-133 experimental anyonisation has now been demonstrated by the Nägerl group at Innsbruck (Nature **642**, 53 (2025), DOI:10.1038/s41586-025-09016-9) with a Feshbach-tunable statistical angle, providing the first concrete experimental platform for the $\rho_{p,k} = k/(12(k+1))$ measurement. (ii) The M1 postulate of the v6_jhep companion admits a *conditional theorem* restricted to the KMS k-design regime, derived from Faulkner–Speranza arXiv:2405.00847 (modular GSL on Killing horizons) + Hollands arXiv:2009.05024 (Gibbs variational identity in von Neumann algebras) + Munson *et al.* arXiv:2403.04828 (*Complexity-Constrained Quantum Thermodynamics*, PRX Quantum 2025) + Kirklin arXiv:2412.01903 (GSL beyond semiclassical); the general inequality M1 (without H1–H4 hypotheses: Killing horizon, KMS, k-design, finite $D_{\mathrm{KL}}$) retains POSTULATE status, and the transplant of complexity entropy from finite-dimensional Munson setting to

infinite-dimensional type-II crossed-product factors is identified as the single weakest link, deferred to the v8-bis math.CT companion as Conjecture M1-C (Longo-resistant Plan B). (iii) The Wolf–García-García–Anton–Ferreira 2025 PRL "Assessing cosmological evidence for non-minimal coupling" (arXiv:2504.07679) reports $\log B = 7.34 \pm 0.6$ for non-minimal coupling vs ΛCDM on Planck PR4 + DR2 BAO + Pantheon+, with the Cassini-cosmological gap of four orders of magnitude in $|\xi_\chi|(\chi_0/M_{\rm P})^2$ acknowledged; this strengthens the empirical motivation for our our $\xi_\chi$ constraint analysis (companion) thawing-quintessence analysis but does not yet establish a screening mechanism that would close the gap at non-cosmological scales — the joint MCMC ($\xi_\chi$, $f_{\rm EDE}$, $c'_{\rm DD}$) on DR2 + Pantheon+ + KiDS-1000 simultaneously addressing both $H_0$ and $S_8$ tensions has, to the author's knowledge, not been performed in the published literature and is the natural next step. v6.0.16 (2026-05-02 evening): the v6.0.15 acknowledgments paragraph (ii) is [RETRACTED-IN-V6.0.16] after a detailed audit (recorded in `notes/m1_audit_2026_05_02.md`) found two independent decisive gaps in the proposed Faulkner–Speranza + Hollands + Munson + Kirklin chain. *Gap 1 (principal):* Munson *et al.* (arXiv:2403.04828) prove a thermodynamic work-cost theorem in $n$-qubit *finite-dimensional type-I* Hilbert spaces with *complexity entropy* defined operationally as Landauer-reset cost—a formally distinct object from the Krylov/spread complexity $C_k$ in M1; transplanting to the type-$\mathrm{II}_\infty$ crossed-product setting requires three unpublished intermediate steps (extension of complexity entropy to semifinite trace-class operators; identification of Landauer reset with modular-flow operations on $\mathcal{A}_R$; equivalence of complexity entropy and spread complexity on the relevant state space), none of which exists in published math.OA literature as of 2026-05-02. *Gap 2:* Hollands 2020 (arXiv:2009.05024) proves Kosaki-style variational formulas for entropy functionals, not the modular-energy-variation lemma the v6 proof sketch attributes to it. M1 therefore retains its POSTULATE status as already correctly stated in the v6_jhep companion (lines 232–238 and 586–594); the v8-bis math.CT companion target is downgraded from a "conditional theorem" to the explicit **Conjecture M1-C**, with H1–H4 (Killing horizon, KMS, k-design, finite $D_{\rm KL}$) listed as *necessary consistency conditions* for the conjecture rather than as sufficient hypotheses for a theorem. Establishing or refuting M1-C in the type-$\mathrm{II}_\infty$ setting is the primary open mathematical problem left by the present work. A companion preventive audit on the same date (`notes/cosmo_const_aot_honest_2026_05_02.md`) closed two speculative side-quests not present in the main text: a proposed three-way "derivation" of $\rho_\Lambda/M_{\rm P}^4 \approx 8\times10^{-121}$ via NCG cutoff, $E_8$, and ECI-internal arguments failed arithmetic verification by 14–89 orders of magnitude across all three approaches, and the proposed emergent-arrow-of-time identity $dt = (dC_k/d\tau_R)d\tau_R$ collapses to $dt = dC_k$ by the chain rule (a re-parameterisation, not a derivation; the temporal direction is installed by hand via a low-complexity initial condition). ECI does not claim to address either $\rho_\Lambda$ or the thermodynamic arrow. v6.0.17 (2026-05-02 night): the v6.0.12 "non-trivial boson–fermion duality at $q{=}\frac{1}{2}$" phrasing in §IV I is [CLARIFIED-IN-V6.0.17] after a parallel-agent audit (`scripts/analysis/q_arik_coon.py`, `notes/q_arik_coon_test_2026_05_02.md`) confirmed (a) that the identity $\rho^A_{qB}(\frac{1}{2}) = 1/24$ is an algebraic tautology of the closed form (the function $\mathrm{Li}_2(q) - \mathrm{Li}_2(1-q)$ is antisymmetric under $q \leftrightarrow 1-q$), and (b) that the formula corresponds to a free-energy combination $\int \log[Z_q\, Z_B/Z_{1-q}]\, du$, not to the von Neumann entropy of $n^A_q$; under the standard $S_{\rm bose}$ definition, $\rho(\frac{1}{2}) = 0.0640 \neq 1/24$. The arXiv citation check returned no prior literature on Arik–Coon $q$-bosons in the Bisognano–Wichmann saturation context. A separate bisect against the canonical $10^5$-zero residuals (`notes/V7_BK_bisect_2026_05_02.md`, `derivations/V7-BK-bisect.py`) initially reverted an earlier "v7-note BK formula erratum required" claim, then a triangulated literature dive against three primary sources (Berry–Keating SIAM Review **41** (1999) 236–266 eq. (4.20); Bogomolny arXiv:nlin/0312061 eq. (57) plus the page-62 unfolding rule; Conrey–Snaith arXiv:math/0509480 Theorem 4.1) confirmed that the canonical $\Delta_{\rm BK}$ prefactor in the unfolded-$x$ convention used by V7-test5 is $-2/L^2$, not $-2/(2\pi)^2$. The v7-note v0.1 used the Conrey–Snaith raw-$\gamma$ prefactor without the $(2\pi/L)^2$ unfolding Jacobian; `derivations/V7-BK-fit.py` has been corrected accordingly. *The headline M2 fit* $\chi^2/\mathrm{dof} = 4.17$ *is unchanged* because the free $L_{\rm eff}$ absorbed the prefactor in the original fit; the canonical (A=1) row M1 moves $11.91 \to 4.52$ (BK now improves on GUE alone, as expected for a real arithmetic signal), and the best-fit amplitude moves $A = 0.41 \to 0.57$ (closer to the canonical $A = 1$). An erratum for v7-note v0.1 (Zenodo DOI 10.5281/zenodo.19983241) is documented in `notes/v7_note_erratum_2026_05_02.md`, awaiting the next release. v6.0.18 (2026-05-02 night): two further updates from a parallel-agent campaign (`notes/campaign_2026_05_02_night/`). *(i) Cassini-cosmological gap reframing.* The v6.0.15 statement that Wolf et al. 2025 PRL acknowledges a "gap of four orders of magnitude in $|\xi_\chi|(\chi_0/M_{\rm P})^2$" between Cassini bounds and cosmological NMC is [CORRECTED-IN-V6.0.18]: the Adam-Hertzberg-Jiménez-Aguilar-Khan 2025 follow-up (arXiv:2509.13302) shows that the cosmological best-fit has $\chi_0 \approx 0.08\, M_{\rm P}$ rather than $\chi_0 \sim M_{\rm P}$, so the Cassini bound $|\xi\chi_0|/M_{\rm P} < 2.4\times10^{-3}$ becomes $|\xi_{\rm eff}| < 0.030$ to be compared with cosmological $|\xi| \approx 0.5$ — a required suppression of $\sim 17\times$, not $\sim 10^4\times$. A symmetron-style density-dependent NMC $\xi(\rho) = \xi_\infty/[1 + (\rho/\rho_*)^\alpha]$ with $\alpha = 1$, $\rho_* \sim 10\,\rho_{\rm cosmo}$ gives 16 orders of suppression in dense regions trivially; sympy verifies the Einstein-frame kinetic prefactor stays

positive (no ghost) for $\xi\chi^2 < M_{\rm P}^2$, satisfied for the Wolf best-fit. The honest caveat is that this is the standard symmetron mechanism transplanted onto the bare $\xi_\chi\varphi^2 R$ Lagrangian rather than a novel construction; testable signatures include DESI DR3 + Roman ($\chi_0$ refit), MICROSCOPE-2 ($\dot{G}_{\rm eff}/G_{\rm eff}$), LIGO ringdown (scalar polarisation), and AION (over-screening). Files: `notes/C1_xi_chi_screening_2026_05_02.md`. *(ii) FRW conformal-pullback type $II_\infty$ note v2.* The standalone math.OA note (`paper/frw_typeII_note/frw_note.tex`) was independently reviewed by Mistral large-latest (which raised four critical issues addressed in v1, see `notes/frw_mistral_review.md`) and by Gemini (which judged technical correctness 10/10 but recommended downgrading the slow-roll case to a Corollary verifying Chen–Penington 2024 by an independent technique, given disjoint constructions but identical conclusions for the slow-roll/exact-de-Sitter case). v0.2 of the FRW note implements the Gemini downgrade (Cor. 3.7); the genuinely novel content is now sharply identified as Theorem 3.5 (radiation-dominated FRW) and Theorem 3.6 (matter-dominated FRW), both for backgrounds with *zero* timelike Killing vectors along comoving worldlines. A first attempt at deriving the cosmological apparent-horizon entropy $S = A/(4G) + S_{\rm out}$ from the new type $\mathrm{II}_\infty$ structure (file `notes/B1_frw_entropy_attempt.md`) returned a clean negative: the conformal pullback flattens the Hubble scale, so the analogue of CLPW's Schwarzschild–de Sitter linearisation step is missing; closing the gap requires either extending the theorem to diamonds touching $\eta = 0$, an off-shell "comoving point mass" perturbation, or a full graviton sector. None of these is presently available; the type $\mathrm{II}_\infty$ classification is therefore kinematic, not directly thermodynamic. v6.0.19 (2026-05-03 night): consolidation pass after a five-agent parallel campaign and a 28-min bootstrap-$\sigma$ test (`notes/campaign_2026_05_02_night/`). *(i) FRW note v3* now (a) rewords the slow-roll Corollary (7) as a *structurally distinct* construction giving the same type-$\mathrm{II}_\infty$ classification as Chen–Penington (arXiv:2406.02116) rather than a verification of their specific algebra (the two are abstractly isomorphic by Connes 1976 hyperfinite $\mathrm{II}_\infty$ uniqueness, but no canonical isomorphism is induced by our conformal-pullback unitary $U$ since CP24 use minimally-coupled inflaton on a static patch while we use conformally-coupled massless on a flat-slicing diamond); (b) sharpens the Big-Bang Open Question with three sympy-verified obstructions ($C_c^\infty$ inverse rescaling fails to all orders at $\eta = 0$; conformal-vacuum two-point function $\log\epsilon$-IR divergent off lightcone; non-Hadamard boundary behaviour per Hollands–Wald 2001 and Brunetti–Fredenhagen–Verch 2003), and records a cautious type-$\mathrm{II}_\infty$-via-inductive-limit conjecture (Verch 1997-style) with a flagged alternative type-$\mathrm{II}_1$ scenario; (c) sharpens the massive-scalar Open Question with the explicit obstruction mode (Lie-derivative of the residual potential along the Möbius generator, $\mathcal{L}_{K_{\rm HL}} V = 2\pi m^2(1-\eta^2-r^2)\, a\, a' \not\equiv 0$, sympy-verified). All citations triangulated via arXiv API (Sewell 1980 dropped as unverifiable). *(ii) v7-note v0.3* reframes the $A_{\rm fit} \approx 1/3$ "plateau" (initially flagged as a $12.7\sigma$ T-trend signal) as a *parametric trade-off* of the leading-BK ansatz at high $T$: bootstrap-vs-jackknife $\sigma_A$ split is $\sim 6.6\times$ (jackknife biased low along the $(A, L)$ valley), and the genuine invariant is $L_{\rm eff} \approx 7.40 \pm 0.22$ universal across $L_{\rm mean} \in [9.1, 12.1]$, with $A_{\rm fit} \approx (L_{\rm eff}/L_{\rm mean})^2 \cdot 0.80$. No new arithmetic constant is claimed; the $\nu \approx 1.09$ cosine mode is consistent with the GUE-first-zero calibration artefact already disclosed in v0.1 and with the prime-square cluster envelope (verbatim Berry–Keating 1999 / Bogomolny 2003 / Conrey–Snaith 2007 triangulation). *(iii) Conjecture M1-C math.CT attack plan* drafted (`notes/M1C_attack_plan.md`): a three-block Plan B in operator-algebraic terms (Block A: Krylov complexity on type $\mathrm{II}_\infty$ via Lanczos on $\Delta^{1/2}$ in $L^2(\mathcal{N}, \mathrm{tr})$, 3–6 mo, math.OA-publishable on its own; Block B: Connes-cocycle modular-energy lemma, single load-bearing Step B3 with no precedent in any algebra type and possibly false; Block C: sector-theoretic submultiplicativity via Longo subfactor index). The recommended commitment is Block A only as the immediate next math.OA target. Five further audit pistes returned cleanly (FRW horizon entropy $S = A/(4G)$ derivation: *negative*, conformal pullback flattens Hubble; modular flow vs slow-roll inflaton evolution: *falsified* by sympy with explicit non-constant ratio; MSS chaos bound on $(w_0, w_a)$: *conjecture wrong* — matter-era violation $\sim 50\%$; massive scalar Theorem: *type classification open*; Levier #1B preliminary posterior at $R - 1 = 0.14$ snapshot: *Scenario C (null result)*, Savage–Dickey $\log B_{10} = -1.45$ favouring $\Lambda$CDM with $\xi_\chi = -0.001 \pm 0.016$ straddling zero, $27\times$ wider than Cassini prior, in tension with the Wolf et al. 2025 detection-significance $\log B = 7.34$ from a different dataset combination). The net signed contribution of the present consolidation is therefore: one robust new mathematical result (Theorems 5, 6 of the FRW note) plus seven calibrated negatives; no new equation, no new fundamental physics, and Levier #1B's emergent disagreement with Wolf 2025 to be re-checked once the chain reaches $R - 1 < 0.05$. v6.0.20 (2026-05-03 late night): consolidation pass after a six-Opus-agent ensemble attack on the hard math problems (M1-C Step B3 + five Nobel-direction pistes), with Mistral magistral-medium and Gemini independent cross-checks. *(i) Conjecture M1-C Step B3 is pointwise FALSE; time-integrated form holds.* Three independent attacks (Opus 4.7 with sympy + L-BFGS adversarial optimization on 1600 numerical evaluations; Gemini analytical reasoning with explicit modular-tracial-mismatch counterexample mechanism; Mistral magistral-medium reasoning toward the same conclusion before max_tokens cutoff) converged: the pointwise inequality $|\langle u_t^{-1}[\log\Delta_\omega, u_t]\rangle_{\rho_R}| \le \kappa_R\, C_k[\rho_R(t)]$ admits explicit counterexamples in finite-dim with violation factor up to $2 \times 10^5$ via L-BFGS-driven adversarial

choices of $\omega$ near-tracial / $\rho_R$ far-from-tracial. The integrated form $\int_0^T |\mathrm{LHS}|\,dt \le \pi \cdot \mathrm{spread}(\log\omega) \int_0^T C_k\,dt$ holds in 40/40 sweeps (Opus); this matches what Kirklin [17] eq. (4.7) actually requires, suggesting the M1-C Plan B is salvageable via reformulation from pointwise to integrated, with Avdoshkin–Dymarsky arXiv:1911.09672 as the finite-dim integrated precedent. The v6.0.16 framing of M1-C as "retains POSTULATE status" stands; what changes is the direction of the Plan B attack. *(ii) Three Nobel-direction pistes returned NEGATIVE in the same campaign*: (a) Era transitions = Jones–Longo subfactor inclusions (Piste 2): index $[\mathcal{A}_{\mathrm{mat}} : \mathcal{A}_{\mathrm{rad}}]_{\mathrm{Jones}} = 1$ trivially, since the conformal pullback unitary $W = U_{\mathrm{mat}}^{-1} U_{\mathrm{rad}}$ is $\mathrm{C}_c^\infty$-bijective on bounded diamonds and identifies the two algebras with $W\mathcal{A}_{\mathrm{rad}}W^{-1} = \mathcal{A}_{\mathrm{mat}}$ exactly, not strictly; cosmological-era transitions are W*-trivial in the CLPW framework; sympy + numpy verified at 11 sample points; recommended target = PRD Comment "no-go for subfactor inclusions at era transitions". (b) $\Lambda$ from modular relaxation steady-state (Piste 3): $\langle K^{\mathrm{FRW}}_{\mathrm{modular}}\rangle_{\Omega_{\mathrm{cosmo}}} \equiv 0$ on the conformal vacuum (Bisognano–Wichmann boost annihilates the Poincaré-invariant vacuum), and the static-patch reading $\langle K\rangle = \pi/(GH^2) = S_{\mathrm{dS}}$ is *tautological* (inverting gives $\Lambda = 3\pi G/S_{\mathrm{dS}}$, inputting $H$ to compute $H$ via CLPW $\mathrm{Tr}(\mathbf{1}) = 1$ normalization); raw mismatch with $\Lambda_{\mathrm{obs}}/M_{\mathrm{P}}^4 \approx 7.16 \times 10^{-121}$ is at best 121 orders of magnitude. *This is the fourth failed route to a $\rho_\Lambda$ derivation in this project* (after the v6.0.16 NCG / $E_8$ / ECI-internal triple), confirming the ECI framework does not predict $\Lambda$. (c) SM field content from modular 2-categorical compatibility (Piste 5): the Connes–Chamseddine finite algebra $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ is finite-dimensional hence type I; by Connes 1973 (Thm VIII.1.2 in Takesaki II) the modular automorphism on a type I factor with the canonical trace state is *inner* (= identity), so the proposed modular compatibility axiom on Yukawa/gauge bimodules acts as identity $\otimes$ identity and adds zero constraints beyond the existing Connes–Chamseddine KO-dim-6 + spectral-action data; the only nontrivial regime would be type-III (Carey–Phillips–Rennie / Marcolli $\sigma$-spectral triples), but no type-III lift of the SM finite triple exists. Recommendation: park Piste 5 ($< 5\%$ probability of falsifiable SM prediction in next decade). *Piste 1 (Hubble parameter as modular Krylov complexity growth rate) returned a partial positive result.* Defining the Krylov complexity intrinsically on type $\mathrm{II}_\infty$ via Lanczos on $\Delta^{1/2}$ in $L^2(\mathcal{N}, \mathrm{tr}_{\mathcal{N}})$ (Block A of the M1-C plan; novel definition for $\mathrm{II}_\infty$), explicit mpmath Lanczos coefficients $b_n/n \to 0.5$ for the FRW past-light-cone vacuum recover the universal $\lambda_L^{\mathrm{mod}} = 2\pi$ of Caputa–Magán–Patramanis–Tonni 2024 (PRD **109**, 086004). The conjecture $H(t) = (1/C_k)\,dC_k/dt$ then evaluates to $1/(a(\eta_c)R_{\mathrm{conf}})$ at leading order in the late-modular-time asymptotic regime, matching $H(t) = 1/(2t)$ for the radiation era *exactly* when the diamond is the past light-cone of the comoving observer (with $\eta_i \to 0^+$, $\eta_f = 2\eta_{\mathrm{obs}}$, $R_{\mathrm{conf}} = \eta_{\mathrm{obs}}$). Three caveats block a full theorem: (a) Block A1 has no published precedent for $\mathrm{II}_\infty$; (b) the $\eta_i \to 0^+$ limit is exactly the Big-Bang Open Question (Q14 of the FRW note, with three sympy-verified obstructions including the IR divergence and non-Hadamard boundary); (c) the CHM08 transfer of Cardy–Holzhey–Maldacena modular temperature to the FRW past-light-cone uses the conformal-pullback unitary as a black box and has not been independently re-derived. Recommended next step (estimate 2–4 months for one specialist): direct verification of CMPT24 universal $2\pi$ on the FRW vacuum without the conformal-pullback transfer. Full M1-Cosmo theorem proof: estimated 12–18 months for an operator-algebraist + cosmologist team. *Piste 4 (Riemann zeros as cosmological modular Hamiltonian spectrum) returned a clean NO-GO theorem.* By Corollary 7 (FRW note v4) + the unitary intertwiner $U$ of Proposition 3.1 + Hislop–Longo, $K_{\mathrm{FRW}} = U^{-1}K_{\mathrm{HL}}U$ has $\mathrm{Spec}(K_{\mathrm{FRW}}) = \mathbb{R}$ (absolutely continuous Lebesgue, simple on each spherical-mode sector). By Proposition 11(2) (Connes 1973 cocycle invariance), the essential spectrum is preserved under massive cocycle perturbation at first order in $m^2$. The Riemann zeros $\{\gamma_n\}_{n\ge1}$ are a discrete countable subset of $\mathbb{R}_+$ by Riemann–von Mangoldt; a discrete countable set cannot equal a continuous Lebesgue spectrum. Hence the cosmological-Hilbert–Pólya conjecture in our framework is ruled out by a category-theoretic obstruction (continuous vs. discrete spectrum). Numerically verified by a 140-point scan over $(H, m^2, \eta_{\max})$ on a Fourier rapidity grid: maximum relative error vs. first 100 zeros is 148% (vs. $\sim 0.1\%$ that a match would require). The CCM 2025 construction (arXiv:2511.22755) verified to exist with the claimed $10^{-55}$ accuracy — but only for the *first* zero at $N = 120$ truncation, degrading to $\sim 10^{-3}$ at the 50th zero — lives on the scaling Hilbert space $L^2([\lambda^{-1}, \lambda], du/u)$, a number-theoretic object structurally distinct from the geometric Tomita–Takesaki setup of QFT-on-curved-spacetime. The only setting in our framework where a Hilbert–Pólya match could in principle exist is the type $\mathrm{II}_1$ static-patch of CLPW; the type $\mathrm{II}_\infty$ comoving-diamond construction is ruled out. Recorded as Remark 16 in the FRW note. *Final tally of the six-Opus + LLM-ensemble campaign*: one robust new mathematical result (FRW Theorems 5, 6); one perturbative new result (FRW Proposition 11); one partial positive (Piste 1 $H$ =Krylov-rate at the past-light-cone diamond, leading order); five clean negatives (Pistes 2, 3, 4, 5 plus M1-C Step B3 pointwise) and one reformulation actionable (M1-C Step B3 integrated form holds, suggests Plan B reframe). Net signed contribution does not include $\Lambda$, $H_0$, $S_8$, hierarchy, dark matter, RH, or SM derivation — the framework continues to consolidate as a mathematical-physics contribution to the algebraic-quantum-gravity programme rather than as a cosmological-tension solver. v6.0.21 (2026-05-03 night, post-Piste-4 wave): five-agent + three-LLM-cross-check campaign produced (i) the FRW

note v4 (`paper/frw_typeII_note/frw_note.tex`) with Proposition 11 (massive light-$m^2$ at first order), Remark 16 (cosmological-Hilbert–Pólya NO-GO via Cor. 7 + Connes 1973 cocycle invariance), and a curvature-corrected note on the conformal-pullback identity for the spatially-flat versus spatially-curved cases (Wald 1984 App. D); (ii) two NEW standalone draft notes pushed at this revision: `paper/cosmo_hayden_preskill/note.tex` (9 pp, conditional Theorem 1.1 "Cosmological Hayden–Preskill" on the radiation+PLC sector under hypotheses (H1) Block A1 on $\mathrm{II}_\infty$ and (H2) modular Lyapunov saturation, with six explicit "what this is NOT" caveats addressing all four obstructions raised by Mistral magistral-medium adversarial cross-check) and `paper/krylov_diameter/krylov_diameter.tex` (7 pp, Theorem 4 "Krylov–Diameter Correspondence" as the honest reframe of the original Piste 1 "$H$ =Krylov-rate" candidate after the Mistral-flagged gauge-conditional issue was sympy-audited and resolved as a geometric-algebraic identity $\frac{1}{C_k}\frac{dC_k}{dt} = \frac{1}{R_{\mathrm{proper}}(\eta_c)}$, with the cosmological-Hubble interpretation era-conditional and diamond-conditional, recommended publication target Comm. Math. Phys. math-ph). *(iii) Three further negatives*: (a) the Vardian arXiv:2602.02675 + Hollands arXiv:2503.21385 pairing identified by the Sonnet crypto-literature dive as "the strongest single connection" for upgrading M1-C from heuristic to theorem-grade was Opus-audited and **FAILS** under careful hypothesis-checking (four obstructions, two decisive: Vardian's OAQEC requires tensor factorisation $\mathbb{H} = \mathbb{H}_A \otimes \mathbb{H}_{\bar{A}}$ which only holds in type I; Hollands needs a Wiesbrock half-sided modular inclusion which the FRW Möbius/Hislop–Longo flow on a bounded diamond does not produce; Hollands bounds observable-expectation flow not entropy-derivative and contains zero occurrences of "Krylov"/"Lanczos"/"complexity" in full text); (b) the Buchholz–Mund–Summers 2000 (arXiv:hep-th/0011237) prior work on transplantation of local nets and geometric modular action on Robertson–Walker spacetimes *partially* resolves the spatially-closed ($k = +1$) hypothesis $\mathrm{H1}_{+1}$ (geometric modular action and cyclic-separating ground state on RW wedges are unconditional from BMS Theorem 3.1 + Corollaries 3.2–3.3) but does *not* provide the type-$\mathrm{III}_1$ classification, the Connes spectrum $\Gamma = \mathbb{R}$, nor the Einstein-static-universe conformally-coupled-scalar setting we need; BMS themselves flag the latter as open on p. 16; (c) the Magistral-medium adversarial cross-check on the Cosmological Hayden–Preskill claim returned "superficial analogy" with four explicit obstructions (subalgebra-change vs. unitary scrambling distinction; gauge-conditional Krylov–Diameter; Big-Bang inheritance of the q:bigbang no-go; Yoshida–Kitaev recovery requires a maximally-entangled vacuum that the conformal vacuum is not), all of which are explicitly listed as caveats in the new `cosmo_hayden_preskill/note.tex` "*What this is NOT*" section. *(iv) Two additional reference corrections.* The v6.0.20 changelog statement that arXiv:2510.07605 supersedes Hollands arXiv:2009.05024 is [CORRECTED-IN-V6.0.21]: arXiv:2510.07605 is in fact Łuczak–Podsędkowska–Wieczorek "Variational formulae for entropy-like functionals" (math.OA, no relevance to B3) and is unrelated to Hollands' supersession status; the intended Hollands–Wojtkiewicz–Stottmeister 2025 follow-up could not be located via the searches performed and the supersession claim is withdrawn. The "Avdoshkin–Dymarsky $\lambda_L \leq 2C_k$" inequality cited as a finite-dim integrated precedent is *not verbatim* in arXiv:1911.09672 abstracts (only "improved bound on chaos / conjectural inequality generalising MSS"); the cleaner replacement reference for the Lyapunov $\leq 2\pi$ chaos bound on type $\mathrm{II}_\infty$ algebras is Hollands arXiv:2503.21385 Theorem 4.1 (eq. 39), with the explicit Wiesbrock half-sided modular inclusion hypothesis. The Kirklin arXiv:2412.01903 "eq. 4.7" that we cited as "what the GSL chain actually requires" is in fact a labelling convention for cut orderings, not a Krylov bound; the Faulkner–Speranza GSL chain on Killing horizons uses Wall monotonicity of relative entropy, not Krylov complexity at all. The cumulative consolidation tally after v6.0.21 is therefore: *two* robust new mathematical results (FRW Theorems 5, 6) plus one perturbative new result (FRW Proposition 11) plus one structural new identity (Krylov–Diameter Correspondence, Theorem 4 of `paper/krylov_diameter/`); one conditional new construction (Cosmological Hayden–Preskill, conditional Theorem 1.1 of `paper/cosmo_hayden_preskill/`); and *thirteen* calibrated negatives spread across all five Nobel-direction Pistes, the M1-C Step B3 pointwise inequality, the Vardian–Hollands and BMS partial resolutions, the q-Arik–Coon "duality", the v7-note $\mathrm{A}_{\mathrm{fit}}$ T-trend / $L_{\mathrm{eff}} = 7.4$ window-saddle reframings, and the Wolf 2025 NMC reproduction in the preliminary Levier #1B run. No new equation has been derived from first principles and no resolution of $H_0$, $S_8$, $\Lambda$, hierarchy, dark matter, or SM particle content has been claimed. The framework's net contribution is mathematical-physics in the post-CLPW algebraic-quantum-gravity programme, not phenomenological. Any residual error is the author's sole responsibility. *(iii) Two reference corrections.* The v6.0.16 audit phrasing implying that Hollands arXiv:2009.05024 (*Variational Principles for Wightman Functions*, Lett. Math. Phys. **111** (2021) 100) was "retracted" is itself [CORRECTED-IN-V6.0.20]: the paper is published; what the v6.0.16 audit actually established is that Hollands' Kosaki-style variational identity is *not* the modular-energy-variation lemma the v6 sketch wrongly attributed to it (the paper has since been superseded by Hollands–Wojtkiewicz–Stottmeister arXiv:2510.07605 (2025), which is the appropriate current reference for the relevant question). Separately, the Witten arXiv ID "2404.05889" that we informally cited in this changelog as "generalized entropy is necessary" is incorrect (that arXiv slot is unrelated); the intended Witten/CLPW-

followup is Faulkner–Witten arXiv:2309.15897 and the more recent Witten arXiv:2601.07910. *(iv) FRW note v4* adds Proposition 11 resolving Question 10 (massive scalar) at first order in $m^2$ in the light regime $m\, a_{\max}\, \Delta\eta \ll 1$, via Connes 1973 (*ASENS* **6** (1973) 133–252, Théorème 1.2.4) cocycle Radon–Nikodym + invariance of the Connes spectrum under cocycle equivalence (sympy-verified cocycle identity on three test profiles). The all-orders question, the conformal regime $m\, a \sim \mathcal{H}$, and the heavy/adiabatic regime remain explicitly open. The cumulative net signed contribution after v6.0.20 is therefore: *one* robust new mathematical result (FRW Theorems 3.5, 3.6) and *one* smaller new perturbative result (FRW Proposition 11); plus *ten* calibrated negatives across the ensemble campaign. No new equation, no new fundamental physics, no $\Lambda$ derivation, no SM prediction. The framework consolidates as a contribution to the algebraic-quantum-gravity programme (CLPW, Witten, Faulkner–Speranza), not as a candidate for resolving the standard cosmological tensions $H_0/S_8/\Lambda$/hierarchy. v6.0.22 (2026-05-03 late night): six-agent path-realiste wave + three-agent Bianchi extension wave + 18-reference external-narrative triangulation, completing the campaign. *New positive results:* (a) **Block A1 restricted-class Lanczos asymptotics** on type $\mathrm{II}_\infty$ via Connes–Takesaki dual-weight + CMPT24 SU(1,1) closed form + Lubinsky–Mhaskar–Saff theorem (Constr. Approx. **4** (1988)): $b_n = \pi n + O(\log n)$ on Schwartz-seed states with chiral-primary insertions of weight $h > 0$, closing hypothesis (H1) of Krylov–Diameter Theorem 4 in this restricted class. (b) **Algebraic Weyl Curvature Hypothesis** (T1 fully rigorous, T2/T3 modulo standard AQFT Hadamard convention since Hollands–Wald 2001): future-infinity diamond limit is type $\mathrm{II}_\infty$ with cyclic-separating Bunch–Davies vacuum, past-Big-Bang diamond limit admits no cyclic-separating vector in any FRW-Hadamard GNS representation, hence past and future are not $W^*$-isomorphic, deriving an algebraic arrow of time intrinsic to the local-algebra net; *Penrose's Weyl Curvature Hypothesis is therefore a theorem in our framework* for spatially flat FRW. (c) **Spatially curved FRW companion** (`paper/frw_typeII_curved/note`, 9 pp): extension of Theorems 3.5/3.6 to $k = \pm 1$ via Wald 1984 App. D.9 curvature-corrected identity, $k = -1$ unconditional via Milne pullback, $k = +1$ conditional on $H1_{+1}$ residual (CGMA + cyclic-separating ground state unconditional via Buchholz–Mund–Summers arXiv:hep-th/0011237 Thm 3.1+Cors 3.2–3.3; type-$\mathrm{III}_1$ + Connes spectrum on ESU spherical-diamond conformally coupled scalar OPEN per BMS p. 16 themselves). (d) **Bianchi extension Proposition B.1** (`notes/path_realiste_2026_05_03/bianchi_extension_opus`): perturbative type $\mathrm{II}_\infty$ for near-FRW Bianchi I via Connes 1973 cocycle + Brunetti–Fredenhagen–Verch local quasi-equivalence (math-ph/0112041), threshold $\epsilon_0 \sim$ CMB anisotropy bound ($\sim 10^{-5}$, cosmologically realistic); T1 lifts at first order in anisotropy, T2 does *not* lift (BKL chaos breaks BFV folium structure). *Defensive integration:* four falsified composite constructions integrated as FRW note v5 "Falsified composite constructions" section (Vardian-Hollands for M1-C; Vardian-HPS-Leutheusser-Liu trio for Krylov-Diameter; nested-diamond Kosaki/Jones index $= +\infty$; CCM-Hartnoll-Yang FRW bridge sympy-impossible). *Bianchi cross-check triangulation:* three independent LLMs evaluated five Bianchi-extension strategies; Magistral and Gemini both rated S2 (mini-superspace) HIGH/MEDIUM, but Opus 4.7 with sympy-verification refuted S2 as collapsing to type $\mathrm{I}_\infty$. The contradiction was resolved by direct mathematical computation rather than majority vote, illustrating the discipline that triangulation alone is insufficient. *Levier #1B FINAL converged posterior at $R - 1 = 0.059$:* 48360 post-burnin samples (ESS = 46411 per chain) confirm the preliminary verdict at the higher statistical confidence: $\xi_\chi = -0.00003 \pm 0.016$ (essentially zero), Savage–Dickey $\log B_{10} = -1.37$ (weak/moderate evidence FOR $\Lambda$CDM over NMC), $H_0 = 67.67 \pm 0.57$ km/s/Mpc with $4.5\sigma$ SH0ES tension UNRESOLVED, $w_0 = -0.86 \pm 0.05$, $w_a = -0.47 \pm 0.18$. *Substantive disagreement with Wolf et al. 2025 (arXiv:2504.07679) confirmed:* we find $\log B_{10} = -1.37$ on Planck PR4 + DR2 BAO + Pantheon+ + KiDS-1000 $S_8$, while Wolf+ report $\log B = +7.34 \pm 0.6$ on Planck PR4 + DR2 BAO + Pantheon+ alone; $\Delta \log B \approx 8.7$ ($\sim$ 6,000× in evidence ratio). The KiDS-1000 inclusion appears to drive the disagreement, a substantive empirical result independently of the framework's structural mathematics. [CORRECTED-IN-V6.0.23: this sentence is empirically false; see v6.0.23 below.] Vast.ai instance 36023758 was closed cleanly after final snapshot; full chains preserved at `mcmc/chains/eci_levier1B_run1/snapshot_2026_05_03_FINAL/`. *External narrative triangulation:* 18 references in a parallel Claude-app physics 2026–2046 narrative audited via direct API; 12 verified, 4 partial, 2 critical misattributions flagged (Q-Xtreme institutional details and award date; DESI DR2 numerical $(w_0, w_a)$ attribution to BAO+CMB+SNe instead of BAO+CMB-only). *Cumulative tally after v6.0.22:* two robust new results (FRW Theorems 5, 6); plus four smaller new results (FRW Proposition 11, Krylov–Diameter Theorem 4 of `paper/krylov_diameter/`, Algebraic WCH T1+T2+T3 modulo Hadamard, Bianchi I Prop. B.1 perturbative); plus one conditional construction (Cosmological HP Theorem 1.1 of `paper/cosmo_hayden_preskill/`); plus one structural-conditional theorem (Block A1 restricted-class); plus *nineteen* calibrated negatives across all five Pistes, M1-C Step B3 pointwise, Vardian-Hollands and Vardian-HPS-LL trio failures, BMS partial residual, q-Arik-Coon "duality", $A_{\mathrm{fit}}$ T-trend / $L_{\mathrm{eff}} = 7.4$ window-saddle reframings, Wolf 2025 NMC reproduction in Levier #1B converged, Bianchi mini-superspace S2 collapse, Bianchi IX Mixmaster non-isotropization. The Algebraic Weyl Curvature

Hypothesis is the most ambitious result of the campaign but remains modulo a standard AQFT convention rather than unconditional; it is *not* a Nobel-direction breakthrough. The framework continues to consolidate as a mathematical-physics contribution to the algebraic-quantum-gravity programme; the Levier #1B converged disagreement with Wolf 2025 on the KiDS-extended dataset combination is the most empirically actionable result of the campaign. Any residual error is the author's sole responsibility. v6.0.23 (2026-05-03 morning, day-2 path-realiste continuation): *(i) Empirical correction to the v6.0.22 "KiDS-1000 drives the Wolf disagreement" claim.* Direct posterior-reweighting of the Levier #1B converged chains (48360 post-burnin samples; Kish ESS preserved 33% at 15138; triangulated across manual Silverman KDE, scipy.gaussian_kde, and 4 histogram bin sizes; bandwidth scan 0.5×–2× Silverman) shows that removing the KiDS-1000 $S_8$ Gaussian by multiplying samples by $w_{\rm new} = w_{\rm old}\,\exp\!\big(+\frac{1}{2}\,\delta^2_{S_8}\big)$ shifts $\log B_{10}$ from $-1.37$ to $-1.42$ — KiDS removal accounts for $\sim$0% of the 8.7-unit gap to Wolf $\log B = +7.34$ (slightly worsens the agreement). Sanity passes: $S_8$ posterior mean shifts up from 0.812 to 0.820 when KiDS pull is removed. *Real drivers, ranked by reweighting + Gemini independent reasoning:* (a) Savage–Dickey KDE vs MultiNest evidence (1D projection at $\xi = 0$ vs 9D nested integral; 1–3 log units, structural); (b) plik 2018 vs NPIPE CamSpec likelihood switch (resolves the $A_L$ lensing anomaly that lets NMC "soak up" Planck residuals; 4–5 log units per Gemini analysis, citing Efstathiou–Gratton 2019/2021); (c) free $w_a$ in our 9-parameter chain ($w_a = -0.47 \pm 0.18$, $2.6\sigma$ from 0; Wolf fixes $w_a = 0$; the dynamical-DE sector absorbs late-time budget that would otherwise feed $\xi_\chi$ detection significance); (d) prior range mismatch: with the YAML's actual $[-0.024, 0.024]$ vs the analysis script's nominal $[-0.1, 0.1]$, the original $\log B$ becomes $+0.058$ (sign flip), already 7 units closer to Wolf. The recommended next-step isolation experiment is a re-run of Levier #1B with $w_a$ fixed to 0 to isolate the wa contribution from the SD-KDE-vs-nested-sampling estimator artefact; this would likely close most of the residual gap. The structural-mathematics conclusions of the v6.0.22 changelog are unaffected; only the diagnostic interpretation of the Wolf 2025 numerical disagreement is corrected. *(ii) New positive results from the day-2 wave.* **T2-Bianchi I rigorous** (`notes/T2_bianchi_2026_05_03_jour2/T2_bianchi_extension.tex`): the past-Big-Bang non-cyclic-separating obstruction lifts to vacuum Bianchi I (Kasner exponents $-1/3, 2/3, 2/3$) via two independent IR divergences — the FRW-analog volume-zero-mode $\log^2$ divergence (S1) plus a NEW long-wavelength tachyonic mode along the contracting Kasner direction with $\omega_k(t) = |k_1|\,t^{1/3} \to 0$ yielding $\int dk_1/|k_1| = +\infty$ (S3); combined with Verch 1994 / BFV 2003 folium uniformity, the Banerjee–Niedermaier 2023 SLE Hadamard state and any state in its BFV folium fail to extend to a cyclic-separating vector. **T2-Bianchi IX pathwise** (`/tmp/T2_bianchi_IX.tex`): the Heinzle–Uggla 2009 attractor theorem (arXiv:0901.0806; the v6.0.22 "arXiv:0901.2891" was a fabrication caught by direct arXiv API lookup, that ID is a chemistry paper on water in nanopores) yields a pathwise volume bound $c_1 t \le V(t) \le c_2 t$ on every vacuum Bianchi IX trajectory, stronger than the originally targeted $\mu$-a.e. statement; T2-BIX therefore holds pathwise, conditional only on Hadamard state existence on Bianchi IX ($S^3$ Cauchy slice), an open AQFT problem (estimated 2–4 months sequel to BN23 via Peter–Weyl on $SU(2)\cong S^3$ + BN23 §2 verbatim + Brum–Them 2013 §4.2 Sobolev WF-set Hadamard verification, per literature search returning 0 papers extending BN23 to higher Bianchi). The Hartnoll–Yang 2025 (arXiv:2502.02661) BKL→automorphic-$L$ short-circuit **FAILS** as an algebraic substitute: it provides a 1d quantum mechanics on the modular fundamental domain, hence a single wavefunctional on minisuperspace and a type $\mathrm{I}_\infty$ representation whose modular flow is not outer — wrong category for the Type $\mathrm{III}_1$ BFV folium that T2 requires. **T2-Bianchi V** (`/tmp/T2_bianchi_V.tex`): conditional on Hadamard achievable in 2–4 weeks via Kontorovich–Lebedev decomposition replacing the $T^3$ Fourier of BN23, with the $H^3$ spectral gap $\inf\sigma(-\Delta_{H^3}) = 1$ acting as IR regulator; *vacuum* Bianchi V is Joseph 1966 = Milne = flat Minkowski via $T = t\cosh\chi$, $R = t\sinh\chi$, hence has no real curvature singularity; the matter Bianchi V case carries the BKL Kasner attractor and T2 lifts via S3 alone. A new *partial NO-GO dichotomy* is uncovered as a byproduct: the $H^3$ spectral gap removes the volume-zero-mode $\log^2$ divergence (S1) for open FRW ($k = -1$); the algebraic arrow of time is therefore strictly weaker in $k = -1$ than in $k = 0$ flat FRW, an unanticipated qualitative dichotomy worth flagging as Open Question 6.3 of `paper/algebraic_wch_bianchi/note.tex`. **Standalone paper** (`paper/algebraic_wch_bianchi/note.tex`, 14 pp PDF, 28 arXiv-API-verified references) drafted assembling Algebraic WCH T1+T2+T3 with Bianchi I extension as Theorem 5.1, recommended publication target Foundations of Physics (Penrose-WCH thematic angle) with secondary Lett. Math. Phys. / Comm. Math. Phys. *(iii) Reference corrections from triangulation.* The Pereira et al. PRD **98** 104054 (2018) attribution for Bianchi I SLE Hadamard in the v6.0.22 Bianchi I Prop. B.1 is [CORRECTED-IN-V6.0.23]: the actual Bianchi I SLE Hadamard reference is Banerjee–Niedermaier 2023 arXiv:2305.11388 (verified via arXiv API; Pereira et al. does not contain a Bianchi I SLE construction, that misattribution propagated from a Sonnet sub-agent's mis-citation). The v7-note $A_{\rm fit}$ erratum DOI in the v6.0.21 changelog should read `10.5281/zenodo.19983241` as cited; verified. *Cumulative tally after v6.0.23:* two robust new results (FRW Theorems 5, 6); plus six smaller new results (FRW Proposition 11, Krylov–Diameter Theorem 4, Algebraic WCH T1+T2+T3 modulo Hadamard convention, Bianchi I Prop. B.1 perturbative, T2-Bianchi I rigorous

via S1+S3, T2-Bianchi IX pathwise via Heinzle–Uggla 2009 conditional on B-IX Hadamard); plus one conditional construction (Cosmological HP Theorem 1.1); plus one structural-conditional theorem (Block A1 restricted-class); plus *twenty-one* calibrated negatives now including the v6.0.22 KiDS-driven Wolf-disagreement claim (empirically false per posterior reweighting) and the "arXiv:0901.2891" fabricated reference. *(iv) Partial reopening of Piste 4 (Riemann zeros) on Bianchi IX.* The v6.0.20 Piste 4 NO-GO was proven specifically for the FRW Möbius/Hislop–Longo modular generator $K_{\mathrm{FRW}} \sim (R^2 - \eta^2)\partial_\eta$ (quadratic in $\eta$), which produces continuous Lebesgue spectrum on $\mathbb{R}$ and cannot match a discrete sequence. The Hartnoll–Yang 2025 BKL automorphic dilation $D_{\mathrm{HY}} = i(x\partial_x + \Delta)$ on Bianchi IX is by contrast *linear* in the dilation coordinate, and a sympy-verified substitution $u = e^x$ shows that the Connes–Consani–Moscovici 2025 zeta spectral triple operator $D_{\mathrm{CCM}} = -iu\partial_u$ on $L^2([\lambda^{-1}, \lambda], du/u)$ is the periodic compactification of $D_{\mathrm{HY}}$ to the band $[-\log\lambda, \log\lambda]$, with the algebraic identity $D_{\mathrm{HY}} = -D_{\mathrm{CCM}} + i\Delta \cdot \mathrm{id}$ holding pointwise on the relevant function spaces (`/tmp/B1_CCM_BIX_revisit.{md,py,tex}`). *Hence the v6.0.20 NO-GO does not extend to the Bianchi IX setting.* Four genuine residual gaps prevent this from constituting a positive theorem: (a) HY's $L$-function appears as a wavefunction overlap rather than as a $D_{\mathrm{HY}}$ eigenvalue (HY explicit "absorption spectrum"); (b) the CCM zeros approach $\{\gamma_n\}$ only after a rank-one perturbation tied to the Euler product over primes $p \leq \lambda^2$; (c) no dictionary mapping the CCM perturbation to a HY modular constraint exists; (d) the framework axis differs (HY = WDW minisuperspace; T2-Bianchi I/IX = local-AQFT BFV folium). The recommended next-step is a 1-week sympy verification of the rank-one perturbation transfer plus a 2–3 month dictionary attempt (both detailed in `/tmp/B1_CCM_BIX_revisit.md`). The Piste 4 NO-GO is therefore tightened to "NO-GO on FRW comoving-diamond $\mathrm{II}_\infty$" rather than "NO-GO universally", and the Bianchi IX setting is reopened as a research direction conditional on resolving (a)–(d). v6.0.24 (2026-05-03 day-2 afternoon, eight-agent Bianchi-completion + four-topic re-examination wave): *(i) Algebraic Weyl Curvature Hypothesis extended to all nine Bianchi types.* Eight parallel agents returned within ∼10–15 min; verbatim deliverables in `/tmp/`. **Type A (admit symmetry-reduced Lagrangian):** Bianchi I rigorous unconditional (Banerjee–Niedermaier 2023 SLE
*checkmark*); Bianchi IX rigorous pathwise on the Heinzle–Uggla 2009 BKL attractor (full $\mu$-measure under arXiv:0901.0806; SLE Hadamard constructed on Kasner-locus trajectories via Peter–Weyl on $\mathrm{SU}(2)\cong\mathrm{S}^3$, eigenvalue-crossing obstruction circumvented on BKL trajectories where the dominant $1/a_1^2$ term orders eigenvalues strictly; `/tmp/T2_bianchi_IX.{md,py,tex}`, `/tmp/hadamard_BIX_anisotropic.{md,py,tex}`);
Bianchi II rigorous pending Hadamard on Heisenberg/$\mathrm{Nil}^3$ nilmanifold; Bianchi $\mathrm{VI}_0$ rigorous pending Hadamard on Sol-like solvmanifold; Bianchi $\mathrm{VII}_0$ rigorous pending Hadamard on Bieberbach flat 3-manifold (most plausible of the conditional cases, Bieberbach quotients are flat); Bianchi VIII pathwise-conditional, hardest of Type A (combines Bianchi V non-compact + Bianchi IX Mixmaster; S1 killed by hyperbolic-fibre spectral gap, S3 fires pathwise via Brehm 2016 arXiv:1606.08058 with Lebesgue-a.e. Mixmaster convergence; `/tmp/T2_bianchi_typeA.{md,py,tex}`). **Type B (no symmetry-reduced Lagrangian; MacCallum–Jantzen obstruction does NOT block T2):** Bianchi V matter case unconditional within 4–6 weeks via Kontorovich–Lebedev decomposition on $\mathrm{H}^3$ ($\phi_\rho(\chi) = \sin(\rho\chi)/(\rho\sinh\chi)$ eigenfunction of $-\Delta_{H^3}$ with eigenvalue $\rho^2 + 1$ exact; Liouville transform $R = e^{-x}S$ removing the friction $-2R'/a_1^2$ from trace $C^a_{1a} = 2$; BN23 §3 functional generalises verbatim; Hadamard via Mehler–Sonine cancellation $|K_{i\rho}|^2\rho\sinh(\pi\rho)$ bounded oscillatory per Lebedev §6.5; `/tmp/hadamard_BV_anisotropic.{md,py,tex}`);
Bianchi III, IV, $\mathrm{VI}_h$ ($h \neq -1/9$), $\mathrm{VII}_h$ partial (S3-conditional, Hadamard-open); **Bianchi $\mathrm{VI}_{-1/9}$ blocked** on the exceptional non-Kasner attractor locus per Hewitt–Horwood–Wainwright 2003 (arXiv:gr-qc/0211071) — clean negative-classification finding. [SHARPENED-IN-V6.0.25 — see v6.0.25 below: prior agent's BLOCKED is too strong; correct verdict is PARTIAL S3-epoch-conditional via Lappicy–Uggla 2024 (arXiv:2410.10375) heteroclinic chains visiting Kasner-I vertices, where each Kasner-vertex visit fires S3.] INSPIRE-HEP API search (this session) confirms *zero* published Hadamard constructions on any Type B Bianchi (III, IV, V anisotropic, $\mathrm{VI}_h$, $\mathrm{VII}_h$) beyond Bianchi I, opening these as direct sequels to BN23. *(ii) Open-FRW dichotomy.* The $\mathrm{H}^3$ spatial spectral gap $\inf\sigma(-\Delta_{H^3}) = 1$ kills the volume-zero-mode $\log^2$ obstruction (S1) for any Bianchi type with hyperbolic spatial geometry; the algebraic arrow of time is therefore strictly weaker in the open FRW limit ($k = -1$) than in the spatially flat case ($k = 0$), with S3 contracting-Kasner alone sufficing in matter cases. Worth flagging as Open Question in published versions. *(iii) CCM–Bianchi-IX Plancherel/Eisenstein bridge.* Verbatim citations triangulated against PDFs downloaded this session: Hartnoll–Yang 2025 (arXiv:2502.02661) eq. (37) gives $D\psi = i(x\partial_x + \Delta)\psi$ on $L^2(\mathbb{R}, dx)$ with $\Delta = \frac{1}{2} + i\varepsilon$; §4.3 explicitly states the wavefunction is proportional to the $L$-function and vanishes at the nontrivial zeros — the zeros are nodes of a Mellin overlap, *not* eigenvalues of $D_{\mathrm{HY}}$. Connes–Consani–Moscovici 2025 (arXiv:2511.22755) eq. (5.14) gives $D^{(\lambda)}_{\log} = -iu\partial_u$ on $L^2([\lambda^{-1}, \lambda], du/u)$, Theorem 1.1(i) $D^{(\lambda,N)} = D^{(\lambda)} - |D\xi\rangle\langle\delta_N|$, and §4.2 Euler cutoff over primes $p \leq \lambda^2$. The deepest available bridge is the Plancherel decomposition on $\mathrm{PSL}(2,\mathbb{Z})\backslash\mathbb{H}$: the Eisenstein sector contains $\zeta(s)$ via Maass–Selberg, the CCM cutoff $p \leq \lambda^2$ is conjecturally identifiable with

the Eisenstein conductor cutoff, and sympy verifies that the Mellin pullback of $1/(s_1 s_2)$ on the cusp produces $\Gamma(\frac{1}{2} - it)/\Gamma(\frac{3}{2} - it)$, i.e. the Hartnoll–Yang basis. The Weyl-law mismatch (cusp forms scale $T^2/12$ vs Riemann zeros $T\log(T/2\pi)/(2\pi)$) requires sparse-subset selection by Maass–Hecke eigenforms / Eisenstein lifts. Recommended track: not a Hilbert–Pólya / RH claim, but a *spectral correspondence* between CCM's Euler-restricted perturbation and the Eisenstein sector of BFV-folium modular data on Bianchi IX, drafted as a JHEP companion in ∼6 months conditional on Hadamard B-IX (PARTIAL today on BKL trajectories per (i)); 1.5–2 years for full rigour. New arXiv reference De Clerck–Hartnoll–Yang 2025 (arXiv:2507.08788) on 5d BKL extension verified; an alleged "Perlmutter zeta zeros 2025" surfaced by Bing was flagged as a hallucination (no matching arXiv ID; excluded). *(iv) Krylov-trio re-examination — prior all-six-failed verdict CONFIRMED with two obstructions sharpened.* Vardian's central decomposition $L_A = \bigoplus_\alpha S(\chi_\alpha, M_A)P_\alpha$ requires non-trivial centre $\mathbb{Z}(A)$, but both $\mathcal{A}(D_\mathcal{O})_{\mathrm{FRW}}$ (type $\mathrm{III}_1$) and its crossed product $\mathcal{N}$ (type $\mathrm{II}_\infty$) are factors ($\mathbb{Z} = \mathbb{C}\cdot\mathbf{1}$), so the area operator degenerates to a c-number regardless of type-II vs type-I — structurally inapplicable to any FRW factor algebra (sharper than the v6.0.21 "OAQEC fails in type $\mathrm{II}_\infty$" obstruction). arXiv 2024–2026 search (7 queries) returned 0 hits; the gap is structural, not bibliographic. *(v) Massive scalar conformal regime, three paths.* Path $\alpha$ (Dyson resummation): cocycle identity sympy-PASSes at $m^4$ and $m^6$ on three test profiles, but Connes-spectrum invariance is blocked at $\geq m^4$ by absence of an Araki–Donald perturbed weight on a $\mathrm{III}_1$ factor with unbounded Wick $H_V$. Path $\beta$ (mode decomposition): *strict refinement* of Proposition 11 — $\Gamma_\ell = \mathbb{R}$ for every $\ell$ at first order in $m^2$. Path $\gamma$ (Faulkner–Speranza WKB): **blocked** — sympy verifies $K_{\mathrm{HL}}$ is a true conformal Killing of $g_{\mathrm{FRW}}$ but the conformal-factor ratio $\omega^{\mathrm{FRW}}/K^\eta$ is never small (4/3 at $\eta = 1/2$, divergent at $\eta = 1$); $K_{\mathrm{HL}}$ is not quasi-Killing at any scale, no half-sided modular inclusion exists. Closing the conformal regime requires extending Araki–Donald to unbounded Wick squares on type $\mathrm{III}_1$ — open op-algebra, 6–12 months. *(vi) Block A1 UOGH unrestricted — three paths.* Path $\alpha$ (OPE) PARTIAL SUCCESS: slope $1/(2\pi)$ universal across orthogonal primaries (mpmath@60-dp, 0.1% precision). Path $\beta$ (clock decay) HARD NO: polynomial-tail clocks $|\hat{f}|^2 \sim \omega^{-2k}$ fail the Lubinsky–Mhaskar–Saff Freud-class hypothesis. Path $\gamma$ (modular-flow universality on non-primary seeds): tentative numerical YES, four seeds within 1–2.5% of slope. **Conjecture (asymptotic-universal UOGH on type $\mathrm{II}_\infty$):** for any seed $\psi$ with $\mu_\psi(\omega) \leq Ce^{-2\pi|\omega|}P(\omega)$ ($P$ polynomial), $b_n[\psi]/n \to 1/(2\pi)$. Stronger than v6.0.22 restricted theorem (covers OPE mixtures, smeared seeds); strictly weaker than full Parker UOGH (open since 2018). 3–6 month single-specialist target. *(vii) Reference corrections.* "Hewitt–Wainwright 1990" for Type B asymptotic dynamics is misremembered — actual Type B asymptotic is in CQG **10** (1993) 99 (1990 Hewitt–Wainwright is on $G_2$ cosmologies). Bianchi II quotients are nilmanifolds (curved) not flat 3-tori; only Bianchi I and $\mathrm{VII}_0$ admit flat Bieberbach quotients. Sneddon–MacCallum 1998 unverifiable; correct Bianchi B Hamiltonian-obstruction reference is Ryan–Waller arXiv:gr-qc/9709012. *Cumulative tally after v6.0.24:* two robust new results (FRW Theorems 5, 6); plus the following smaller new results: FRW Proposition 11 (light $m^2$ first order); per-mode refinement ($\Gamma_\ell = \mathbb{R}$ each $\ell$); Krylov–Diameter Theorem 4 (conditional on UOGH); Algebraic WCH T1+T2+T3 modulo Hadamard convention; Bianchi I Proposition B.1 perturbative; T2-Bianchi I rigorous via S1+S3; T2-Bianchi IX rigorous pathwise on Heinzle–Uggla full-$\mu$-measure with anisotropic SLE Hadamard on BKL trajectories; T2-Bianchi V matter unconditional within 4–6 weeks; T2 rigorous pending Hadamard on Bianchi $\mathrm{II}/\mathrm{VI}_0/\mathrm{VII}_0$; T2 pathwise-conditional on Bianchi VIII; T2 partial S3-conditional on Bianchi $\mathrm{III}/\mathrm{IV}/\mathrm{VI}_h/\mathrm{VII}_h$; T2 *blocked* on Bianchi $\mathrm{VI}_{-1/9}$ (clean classification negative); Block A1 restricted-class plus the new asymptotic-universal Conjecture; Plancherel/Eisenstein bridge to CCM zeta-spectral-triple via Bianchi IX (research-direction marker, 6-month JHEP companion target). One conditional construction (Cosmological HP Theorem 1.1). *Twenty-two* calibrated negatives, now including the Faulkner–Speranza WKB blockage, the Bianchi $\mathrm{VI}_{-1/9}$ blocked classification, and the Vardian-central-decomposition c-number degeneration. The Algebraic Weyl Curvature Hypothesis is now established (modulo Hadamard convention) in some form on *eight of the nine* Bianchi types. The framework continues to consolidate as a mathematical-physics contribution to the algebraic-quantum-gravity programme; no $\Lambda$, no $H_0$, no $S_8$, no SM derivation. v6.0.25 (2026-05-03 afternoon, fifteen-agent revolution-sweep + adversarial double-check wave): *(i) Ten revolutionary research directions probed.* Ten Sonnet agents launched in parallel on the most ambitious open problems left by v6.0.24 (R1: $\Lambda$ from T1↔T2 algebraic asymmetry; R2: $H_0$ from Krylov–Diameter Theorem 4 in BKL; R3: $S_8$ as BFV folium-projection ambiguity; R4: SM 3 generations from Bianchi IX SU(2) Kasner axes; R5: RH via Maass–Selberg + theta-correspondence on $\mathrm{PSL}(2,\mathbb{Z})\backslash\mathbb{H}$; R6: AdS/CFT in FRW $\mathrm{II}_\infty$ frame; R7: hierarchy from $c'$ modular trace ratios; R8: generation count from $\mathrm{PSL}(2,\mathbb{Z})\backslash\mathbb{H}$ Eisenstein/cusp sectors; R9: dark matter from Bianchi VIII Mixmaster Kasner-bounce relics; R10: SM gauge from BKL ↔ $E_{10}$ Kac–Moody). **Verdict: ten clean negatives on the revolutionary claim, three publishable carve-outs from byproducts**, all with sympy verification and arXiv-API reference checks. R1 [NUMEROLOGY]: naive ratio $\mathrm{Tr}(p_\infty)/\mathrm{Tr}(p_{\mathrm{BB}}) = 0/(+\infty)$, with regulator $\delta_{\mathrm{phys}} = t_P/t_{\mathrm{now}}$ giving $k_{\log} \approx 28.41$ non-canonical; Mistral magistral-medium independent confirms BKL Lyapunov is order 1 ($\approx 2.373$), not exponentially small. R3 [KILLED-BY-VERCH-1994]: all quasifree

Hadamard states are in the same folium (Verch CMP **160** 507); Hollands–Wald BFV freedom shifts only counter-terms, not smeared two-points. R4 [REFUTED-BY-LIE-ALGEBRA]: sympy $[G, J_x] = iJ_y \neq 0$ predicts tree-level $e \leftrightarrow \mu \leftrightarrow \tau$ mixing excluded by MEG II at $10^{-13}$; the "3" in $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ is $\mathrm{SU}(3)_{\mathrm{color}}$, *not* generations; correct algebraic-derivation route is Furey octonionic $\mathbb{C} \otimes \mathcal{O}$ via triality (arXiv:1611.09182, 1910.08395, 2409.17948), not Bianchi IX SU(2). R7 [NUMEROLOGY]: Monte Carlo null on 5000 trials, $P(\text{bestfit} < 0.01\,\text{dex}) = 0.91$ for random targets; the 3-decade lattice fits anything post-hoc. R8 [REFUTED-BY-WEYL-LAW]: Maass–Selberg $c$-function for $\mathrm{SL}(2,\mathbb{Z})$ is scalar, Eisenstein series do not mix between sectors; Weyl-law mismatch $T^2/12$ versus $T\log(T/2\pi)/(2\pi)$ kills any sparse-subset correspondence; the quark sector with same generation count 3 has *opposite* hierarchy direction. R9 [KILLED-BY-T2]: Bianchi-VIII Mixmaster bounces *forbid* cyclic-separating Hadamard through the BKL singularity by the T2-Bianchi-VIII pathwise-conditional theorem itself, so T2 is structurally *anti*-DM-creation; Bogoliubov cutoff fixes $m_{\mathrm{DM}} \sim M_P \sim 10^{19}$ GeV, off by 17–51 orders of magnitude versus WIMP/KK-graviton/ultra-light DM. R10 [NO-GO-VIA-DKN-WALLS]: Damour–Kleinschmidt–Nicolai 2007 BKL walls are first-class *constraints*, not gauge generators; the $K(E_{10}) \supset SO(16) \supset SU(8) \supset SU(3) \times U(1)$ chain of Kleinschmidt–Meissner–Nicolai 2025 (arXiv:2503.13155, JHEP **08** (2025) 054) is structurally incompatible with our spacetime-AQFT framework as the authors themselves state "cannot be realized within the framework of space-time based (quantum) field theory"; placed on watch-list pending future spacetime-AQFT lift. *Three publishable carve-outs from byproducts.* R2 **[POSITIVE-BYPRODUCT]**: closed-form Heinzle–Uggla invariant $\langle p_3\rangle_{\mu_{\mathrm{HU}}} = \int_1^\infty p_3(u)\,du/(u(u+1)\ln 2) = \sqrt{3}\pi/(9\ln 2) \approx 0.872253116\ldots$ verified at mp-math@200 dps to relative error 0.0; recommended as 1-page math.OA / dynamical-systems Comment standalone (Birkhoff scaling $C_k(t) \propto t^{-\sqrt{3}\pi/(9\ln 2)}$ along Mixmaster conditional on extending Krylov–Diameter Theorem 4 past BKL singularity, currently OPEN). R5 **[CONDITIONAL-COMPANION]**: 6-month JHEP companion downgraded from RH-proof to "Spectral Correspondence between BFV-folium modular data on Bianchi IX and the CCM Euler-restricted Eisenstein sector"; the algebraic identity $D_{\mathrm{HY}} = -D_{\mathrm{CCM}} + i\Delta \cdot \mathrm{id}$ holds pointwise per sympy substitution $u = e^x$. *Gemini's stronger "Eisenstein Sector Duality" framing (cuspidal sector algebraically obstructed at BKL) is REFUTED by Sonnet adversarial review*: BKL chaos enhances cusp matrix coefficients (Selberg trace formula, Sarnak prime geodesic theorem) rather than killing them, and zeros of $\zeta(s)$ enter as scattering resonances at $s = \frac{1}{2} + i\gamma_n/2$, not eigenvalues, off by a factor of two. The milder Sonnet-R5 §8.2 Spectral Correspondence is the defensible target. R6 **[CONDITIONAL-COMPANION]**: the original "AdS/CFT in FRW $\mathrm{II}_\infty$ frame" framing is falsified for three structural reasons ($\mathrm{I}^+$ FRW spacelike not null, $U$ interior-to-interior, Bunch–Davies two-point vanishes pointwise as $\eta \to \infty$); the carve-out is the closed-form integrated modular Hamiltonian $K_{\mathrm{FRW}} = (2\pi/R)\int_{D_R} d^3x\,(R^2 - (\eta-\eta_c)^2 - |x-x_c|^2)\,a(\eta)^{-2}\,T^{\mathrm{FRW}}_{00}(\eta,x)$ [CORRECTED-IN-V6.0.26] on the comoving causal diamond plus Witten-style generalised entropy on the type $\mathrm{II}_\infty$ crossed product (*Mistral magistral-medium adversarial cross-check returns "clean" verdict on the conformal-pullback derivation chaining Hislop–Longo 1982, Prop. 3.1 of* `frw_note`*, and the conformal weight* $T^{\mathrm{FRW}}_{\mu\nu} = a^{-2}T^{\mathrm{Mink}}_{\mu\nu}$ *for conformally coupled scalars in* $4d$), 8–12 pp Comm. Math. Phys. or Annales Henri Poincaré sequel to Theorem 3.6 of `paper/frw_typeII_note/`; closest precedent is Fröb 2023 arXiv:2308.14797 which stops short of writing the FRW closed form. *(ii) Three open-doors deepened.* O1 (Hadamard on Type B anisotropic Bianchi): *key discovery* that Avetisyan–Verch 2013 arXiv:1212.6180 already provides the harmonic analysis (Tables 1–2 + eqs (3,9,10)) for *all* Bianchi I–VII via a unified semidirect-product construction $G = \mathbb{R}^2 \rtimes_F \mathbb{R}$; the contribution-not-yet-in-AV13 is the transfer of Banerjee–Niedermaier 2023 SLE construction to non-unimodular Type B groups via the Duflo–Moore 1976 modular weight (cancels in the smeared $W(x,x')$, so non-unimodularity does not break Hadamard). Recommended publication = combined Bianchi III + Bianchi V SLE paper, 3–6 mo realistic. *Sonnet adversarial review caught a fifth hallucination* ("Krasil'shchik 2014 arXiv:1403.2278" is actually Glas–Konstantis–Krause–Loose "Bianchi's classification of 3-dimensional Lie algebras revisited") and *a new UV obstruction on Bianchi IV* from non-unimodularity: the Plancherel density satisfies $\dot{\nu}_{IV}(k_1) = 1 + k_1\,|P|^2 \sim 1/k_1$ logarithmically divergent, downgrading Bianchi IV from PARTIAL to OPEN. O2 (Bianchi VIII Hadamard combined V+IX): $\mathrm{SL}(2,\mathbb{R})$ Plancherel decomposition (Bargmann 1947 INSPIRE 9103, Basu arXiv:0710.2224) gives the full principal-even/odd, complementary, discrete $D_n^\pm$ series setup; per-sector BN23 transfer is rigorous on principal series with exponential UV cancellation $\tanh(\pi r)/\sinh(\pi r) \sim 2e^{-\pi r}$ (*better* than Bianchi V's polynomial Mehler–Sonine cancellation). Mistral magistral-medium adversarial review confirms discrete-series Hadamard is a real unresolved concern requiring per-level QHO ground-state minimisation, and revises time-to-publication from 3–4 mo to 6–12 mo Kasner-regime, 18–30 mo full Mixmaster. **O3** (M1-C Block A Krylov on $\mathrm{II}_\infty$): admissible class $\mathcal{A}_{\mathrm{Lan}}(\mathcal{N})$ defined via Schwartz smearing on the modular-time fibre yields a paper-ready math.OA draft (`/tmp/O3_M1C_BlockA_KrylovIIinfty.tex`); Lanczos convergence on smeared seeds and 5 non-primary numerical seeds within 0.31–2.04% of slope $1/(2\pi)$. *Opus adversarial double-check refutes the asymptotic-universal Conjecture as written* via explicit counterexample CX4: $\rho(\omega) = 3\pi e^{-3\pi\omega}$ trivially satisfies the upper-bound hypothesis $\mu_\psi(\omega) \leq Ce^{-2\pi|\omega|}P(\omega)$ but produces slope

$1/(3\pi)$, a 33% deviation from the claimed universal $1/(2\pi)$. The hypothesis must be sharpened to *asymptotic equivalence* $\mu_\psi(\omega) \sim R(\omega)e^{-2\pi\omega}$ as $\omega \to \infty$ with $R(\omega) = \sum_i a_i \omega^{2h_i - 1}$ a positive finite mixture (verified numerically on seed CX6 to 0.31%). Under this fix, the OP-lemma blocker becomes a Vanlessen 2007 (*Constr. Approx.* **25** (2007) 125–175, arXiv:math/0504604) corollary for $w(x) = R(x)e^{-Q(x)}$ on $[0, \infty)$ via Riemann–Hilbert / Deift–Zhou, 1–3 mo transcription rather than the originally-claimed 6–12 mo open problem. Three internal bibliographic errors are caught and corrected: [CORRECTED-IN-V6.0.25] the v6.0.20 "Connes–Takesaki Acta Math **130** (1973), 53–155" is fabricated, actual reference is Connes–Takesaki, Tôhoku Math. J. **29** (1977), 473–575 (the 1973 paper is Connes solo, ENS (4) **6**, 133–252); the v6.0.20 "Levin–Lubinsky Memoirs AMS **144** (2000)" is wrong both number and year, correct = Memoirs AMS No. **535** (1994) for $[-1, 1]$ or CMS Books in Mathematics vol. 4, Springer (2001) for $[0, \infty)$; "Akhiezer Theorem 2.1.4" is misnumbered, the relevant Lanczos-coefficient continuity result is closer to Akhiezer Theorem 2.3.3 or Berezansky / Simon arXiv:math/0309173. *(iii) Three further reference corrections from full triangulation audit.* The internal-changelog references `arXiv:1806.10674` (claimed Meissner–Nicolai PRL "Standard Model Fermions and Infinite-Dimensional R-Symmetries") and `arXiv:2202.04443` (claimed Witten 2022 "Gravity and the Crossed Product") are [CORRECTED-IN-V6.0.25] hallucinations: `1806.10674` is actually a computer-science bibliometrics paper, the correct Meissner–Nicolai 2018 PRL is `arXiv:1804.09606` "Standard Model Fermions and Infinite-Dimensional R-Symmetries" (verified via direct arXiv page fetch); `2202.04443` is actually a logic / number-theory paper on Collatz / Thompson group $F$, the correct Witten 2022 "Gravity and the Crossed Product" is `arXiv:2112.12828` (verified). *Cumulative tally after v6.0.25:* two robust new results (FRW Theorems 5, 6) plus the smaller new results from v6.0.24 unchanged plus three new conditional-companion targets (R2 invariant $\langle p_3 \rangle_{\mu_{\rm HU}} = \sqrt{3}\pi/(9 \ln 2)$ as 1-pp math.OA Comment; R5 "Spectral Correspondence" at 6 mo JHEP companion downgraded from RH-proof to defensible scope; R6 closed-form $K_{\rm FRW}$ + generalised entropy at 8–12 pp CMP/AHP); plus three open-doors-deepened (Bianchi III + Bianchi V combined SLE paper 3–6 mo via Avetisyan–Verch 2013 plus Duflo–Moore 1976 transfer; Bianchi VIII Kasner-regime SLE 6–12 mo via $\mathrm{SL}(2, \mathbb{R})$ Plancherel; M1-C Block A Krylov on $\mathrm{II}_\infty$ 3–6 mo standalone math.OA paper with sharpened conjecture). Plus *thirty-two* calibrated negatives now including ten R-pistes refuting the revolutionary claims ($\Lambda$, $H_0$, $S_8$, three generations, RH, AdS/CFT-in-FRW, hierarchy, generation count from PSL Eisenstein, dark matter, SM gauge from $E_{10}$); R5's stronger Eisenstein Sector Duality framing (refuted by Sarnak prime-geodesic / Selberg trace); O3's asymptotic-universal Conjecture-as-written (refuted by CX4 counterexample, salvageable via asymptotic-equivalence fix); Bianchi IV downgrade to OPEN from non-unimodular UV log-divergence. The framework's net contribution remains exactly the same as v6.0.24: a mathematical-physics contribution to the post-CLPW algebraic-quantum-gravity programme; no $\Lambda$, no $H_0$, no $S_8$, no SM derivation, no hierarchy, no DM, no RH proof. The triangulation discipline (3+ LLMs + sympy + arXiv API + adversarial Opus reviews) caught *six* additional fabricated or mis-attributed references in this revolution-sweep alone, raising the cumulative project hallucination-catch count to eleven. Any residual error is the author's sole responsibility. v6.0.26 (2026-05-03 evening, six-Opus-agent portes-ouvertes consolidation wave + editorial pass): *(i) Coefficient correction to the v6.0.25 R6 carve-out closed-form $K_{\rm FRW}$.* P3 portes-ouvertes agent's direct read of CHM 2011 (arXiv:1102.0440 eq. (2.20): $H_{\mathcal{D}} = 2\pi \int_V d^{d-1}x\, ((R^2 - r^2)/(2R))\, T^{00} + c'$, i.e. $(\pi/R) \int (R^2 - r^2) T^{00}$ with no extra factor of two on the kernel) and Fröb 2023 (arXiv:2308.14797 eq. (3.8): $e^{(d-2)\omega} \mathcal{T}^{\rm FRW}_{\mu\nu} = T^{\rm Mink}_{\mu\nu}$ for the conformally coupled scalar in $d = 4$, hence $a^2 \mathcal{T}^{\rm FRW} = T^{\rm Mink}$) corrects the v6.0.25 closed-form to $K_{\rm FRW} = (\pi/R) \int_{D_R} d^3x\, (R^2 - (\eta - \eta_c)^2 - |x - x_c|^2)\, a(\eta)^2\, \mathcal{T}^{\rm FRW}_{00}(\eta, x)$. The v6.0.25 typo $(2\pi/R)$ instead of $(\pi/R)$ was an erroneous absorption of the $1/(2R)$ into the prefactor without halving the kernel; the typo $a^{-2}$ instead of $a^{+2}$ on the FRW-frame stress tensor was a sign error when chasing $U^{-1} T^{\rm Mink} U = a^2\, \mathcal{T}^{\rm FRW}$ through the conformal pullback. Structural conclusions of the carve-out (8–12 pp CMP/AHP target) unchanged. *(ii) Six portes-ouvertes deliverables consolidated* (`notes/portes_ouvertes_2026_05_03/`): P1 (R2 closed-form moment $\langle p_3 \rangle_{\mu_{\rm HU}} = \sqrt{3}\pi/(9 \ln 2)$, 1–2 pp J. Stat. Phys. Comment, conditional on Khalatnikov–Lifshitz–Sinai–Khanin 1985 *J. Stat. Phys.* **38** 97–114 not already containing it); P2 (R5 Spectral Correspondence, ∼6-mo JHEP companion, conditional on H1: BIX BKL-pathwise Hadamard + H2: CCM–BFV-Eisenstein conformal pullback, ∼6 pp Maass–Selberg recap fallback if H2 fails); P3 (R6 $K_{\rm FRW}$ closed-form, 8–12 pp CMP/AHP, load-bearing stress-tensor conjugation Lemma 3.3 provable in 1–2 days of careful algebra modulo CHM (2.20) + Fröb (3.8), formula coefficients corrected per (i)); P4 (O1 Bianchi III + V combined SLE Hadamard, 3–6 mo via Avetisyan–Verch 2013 (arXiv:1212.6180) harmonic analysis on $G = \mathbb{R}^2 \rtimes_F \mathbb{R}$ + Duflo–Moore 1976 modular-weight cancellation lemma proven for B-III/B-V via $\det F(-r) \cdot \det F(-r') \cdot (\det F(-r))^{-1} \cdot (\det F(-r'))^{-1} = 1$); P5 (O2 Bianchi VIII Kasner-regime Hadamard SLE, 6–12 mo via $\mathrm{SL}(2, \mathbb{R})$ Plancherel [Bargmann 1947, Basu arXiv:0710.2224], discrete-series load-bearing prop provable on Kasner regime modulo standard Hollands–Wald V-coefficient calculation, full Mixmaster blocked by OPEN BVIII pathwise volume bound analogous to Heinzle–Uggla 2009 arXiv:0901.0806); P6 (O3 M1-C Block A Krylov on $\mathrm{II}_\infty$, 1–3 mo single-specialist via

Vanlessen 2007 (arXiv:math/0504604) + KMVV 2004 (arXiv:math/0111252) Riemann–Hilbert transcription, sharpened conjecture in asymptotic-equivalence form $\rho(\omega) \sim R(\omega)e^{-2\pi\omega}$ with $R$ a positive finite mixture of $\omega^{2h-1}$ powers, slope $b_n/n \to 1/(2\pi)$ via Lubinsky–Mhaskar–Saff). *(iii) Three additional hallucinations caught*: "Krasil'shchik 2014 arXiv:1403.2278" (actual = Glas-Konstantis-Krause-Loose 2014, "Bianchi's classification of 3-dimensional Lie algebras revisited"); "Pukanszky 1958" (does not exist as a Bianchi-relevant work; correct attribution for non-unimodular exponential-group Plancherel theory is Pukanszky, TAMS **126** (1967) 487–507); "Brum–Them 2013 arXiv:1304.5286" (1304.5286 is in fact a transportation-equilibrium paper by Han–Friesz–Szeto–Liu; correct ID = arXiv:1302.3174 = Brum–Them, CQG **30** (2013) 235035, "States of Low Energy in Homogeneous and Inhomogeneous, Expanding Spacetimes"). Project hallucination-catch count rises from 11 to 14. *(iv) Editorial pass*: the 24 inline [RETRACTED-IN-V6.0.X]/[CORRECTED-IN-V6.0.X] audit-trail markers in the v6.0.10–v6.0.25 acknowledgments paragraph are de-coloured to plain black to remove LaTeX rendering overlaps caused by hyphenation inside `\textcolor{red}{}` in narrow-column layout; marker content preserved verbatim. Cumulative tally after v6.0.26 unchanged from v6.0.25 modulo coefficient correction (i): two robust new results (FRW Theorems 3.5, 3.6 of `paper/frw_typeII_note/`), the v6.0.24/25 smaller results unchanged, three conditional-companion targets (R2, R5, R6) with corrected R6 coefficient, three open-doors-deepened (O1, O2, O3 portes-ouvertes deliverables consolidated this revision), one conditional construction (Cosmological Hayden–Preskill), and thirty-two calibrated negatives. No new equation derived from first principles; no $\Lambda$, no $H_0$, no $S_8$, no SM derivation, no hierarchy, no DM, no RH proof. The framework continues to consolidate as a mathematical-physics contribution to the post-CLPW algebraic-quantum-gravity programme. Any residual error is the author's sole responsibility. v6.0.27 (2026-05-03 evening, 9-agent + 3-LLM portes-ouvertes wave 2 followed by 8-agent + 3-LLM follow-up wave 3 in flight): *(i) Wave 2 results consolidated.* **Positives:** (a) **Lemma 3.3 stress-tensor conjugation PROVEN sympy-off-shell** for the conformally-coupled massless scalar in $d$=4 on spatially-flat FRW, $U^{-1}T^{\rm Mink}_{00}U = a(\eta)^{+2}\,\mathcal{T}^{\rm FRW}_{00}$, via direct verification on three test backgrounds plus algebraic off-shell argument (`/tmp/agents_2026_05_03_evening/A2_lemma33_stress_tensor/`); the v6.0.25 sign $a^{-2}$ is confirmed wrong, v6.0.26 fix $a^{+2}$ is correct. The carve-out paper P3/R6 *closed-form* $K_{\rm FRW}$ *+ generalised entropy* is now ∼80% paper-ready, awaiting only the Lemma 3.4 anomaly-bookkeeping (estimated 2–3 days; B1 wave-3 agent in flight). (b) **Bianchi VII$_0$ Bieberbach SLE** paper-ready 8–12 pp draft (`A4_BVII0_bieberbach/note.tex`, 736 lines), 4 closable gaps (Bessel $J_0$ wavefront-set, Bieberbach quotient compatibility $G_1$–$G_5$ only, anisotropic UV WKB, IR Plancherel near $r = 0$); 3–4 mo to publication. (c) **Bianchi II Heisenberg/Nil$^3$ SLE** paper-ready 9-section draft (`A5_BII_heisenberg/note.tex`, 655 lines), spatial mode = time-dependent harmonic oscillator, UV identical to BN23 leading order; the genuine open problem is the Junker–Schrohe two-index $(\omega, n)$ wavefront-set uniformity (B4 wave-3 agent in flight); 6–9 mo. (d) **Bianchi VI$_0$ Sol-solvmanifold massive SLE** paper-ready 8-section draft (`A6_BVI0_sol/note.tex`, 658 lines), Sol Plancherel = $L^2(\mathbb{R})$ on $\pi_\lambda$ with no complementary series, modified Mathieu spatial Laplacian with $K_{i\nu}$ Bessel-$K$ eigenfunctions; 9–12 mo. *Hard negative on the massless case*: log IR divergence as $\lambda \to 0$, universal not Sol-specific (same as BN23 §4 Bianchi I and Olbermann 2007 FLRW). (e) **Bianchi VI$_{-1/9}$ verdict upgraded BLOCKED → PARTIAL S3-epoch-conditional** (`A7_BVI_minus_1_9_LU/`); the v6.0.24 BLOCKED was based on a misreading of Hewitt–Horwood–Wainwright 2003 (the Robinson–Trautman non-Kasner attractor is the *future* attractor, not a past-singularity obstruction); Lappicy–Uggla 2024 (arXiv:2410.10375, 65 pp PDF read directly) Lemmas 2.1–2.2 and Proposition 3.3 prove that frame-transition and curvature-transition orbits have $\alpha$- and $\omega$-limit sets on the Kasner circle K°, and HHW03 Conjecture 5.1 states the past attractor includes $K$. At each non-Taub Kasner-vertex visit, one $p_\alpha < 0$ fires the S3 IR divergence; T2 transfers conditional on the LU24 Singularity Attractor Conjecture (numerically well-supported, unproven). *(ii) Resolution of the v6.0.25 R2 closed-form* $\langle p_3\rangle_{\mu_{\rm HU}} = \sqrt{3}\pi/(9\ln 2)$ *apparent mismatch*: A8 wave-2 agent reported a 6–10% / ∼300$\sigma$ discrepancy between the closed-form and Wainwright–Hsu numerical simulation. Mistral large-latest analytic re-derivation (B2 wave-3 cross-check) RESOLVES the apparent contradiction: the integrand $p_3(u)/(u(u+1)) = u(1+u)/((1+u+u^2)u(u+1)) = 1/(1+u+u^2)$, with antiderivative $(2/\sqrt{3})\arctan\big((2u+1)/\sqrt{3}\big)$, gives $\int_1^\infty du/(1+u+u^2) = (2/\sqrt{3})(\pi/2-\pi/3) = \pi/(3\sqrt{3}) = \sqrt{3}\pi/9$, hence $\langle p_3\rangle_{\mu_{\rm HU}} = \sqrt{3}\pi/(9\ln 2) \approx 0.872253116$ EXACT under the Gauss-shift invariant density $\mu_{\rm HU}(u) = 1/(u(u+1)\ln 2)$. The A8 simulation value $\approx 0.936$ is the analogous mean under the BKL-era-return measure, a different ergodic measure that weights $u$ by epoch duration. No contradiction — measure-conflation. The R2 1-page J. Stat. Phys. Comment carve-out is valid *provided* the published version explicitly identifies which measure is used (Gauss-shift, not BKL-era-return). The Birkhoff-scaling extension $C_k(t) \propto t^{-\langle p_3\rangle_\mu}$ along Mixmaster trajectories (A8 Approach 4C trajectory-pullback target, 4–6 mo conditional on Abramov-suspension) must specify the measure. *(iii) Negatives:* **(A1)** Bianconi 2025 (PRD **111** 066001, arXiv:2408.14391) audited via three independent LLMs (Sonnet sub-agent, Mistral large-latest, Gemini 3.1 flash-lite-preview) all converging on the verdict "orthogonal differential-geometric framework, numerologically suspect $\Lambda$ prediction". Sonnet A1 additionally identified, with sympy verification, a **Lorentzian signature gap**: Bianconi's

load-bearing identity $H = \mathrm{Tr}(\tilde{g} \cdot \ln \tilde{g}^{-1}) = 0$ is required for the simplified action, but for Lorentzian eigenvalues $(-1, +1, +1, +1)$, $\mathrm{Tr}(g \ln g^{-1}) = -i\pi$ (complex), not 0. PRD Comment publishable; B8 wave-3 agent drafting. **(A3)** The v6.0.25 "Vanlessen 2007 corollary" single-step closure for M1-C Block A is *oversold*; Vanlessen verbatim covers only single-power $x^\alpha$ prefactor and exact polynomial $Q$ (gaps G1 + G2), not the v6.0.25 mixture and asymptotic-equivalence forms. Vanlessen 2007 itself is verified real (arXiv:math/0504604, Constr. Approx. **25** (2007) 125–175); the closure must split into a 3-mo exact-class corollary (Vanlessen) + 6-mo asymptotic-equivalence extension (Levin–Lubinsky 2001, B6 wave-3 agent in flight). **(A8)** Past-BKL extension of Krylov–Diameter Theorem 4 globally is a HARD NEGATIVE in classical GR (curvature singularity, no Lorentzian metric past BKL); Approach 4A (algebraic continuation) blocked, 4B (regularization) conditional on BKL conjecture PDE-level itself open, 4C (trajectory-pullback) most tractable at 4–6 mo. **(A9)** Odrzywołek 2026 (arXiv:2603.21852) confirmed ZERO ECI connection by sympy and direct paper read: it establishes completeness of $\mathrm{EML}(x, y) = \exp(x) - \ln(y)$ for the Liouville–Ritt elementary class, an authentic cs.SC result misrepresented by Science et Vie as "toutes les mathématiques"; sin/cos/$\pi$ unreachable at finite real depth. Park. *(iv) Nine new hallucinations caught by the wave-2 agents in the present author's prompts (project total 11 → 20)*: Hislop–Longo DOI `BF01208568` → `BF01208372` (Springer + Google Scholar verified); algebra error $(\pi/R^2) \int (R^2 - r^2) \to (\pi/R) \int (R^2 - r^2)$ (CHM 2011 eq. (2.20) factor-of-$R$ correction); "Brum–Them 2013" → Them–Brum (arXiv:1302.3174); "Pukanszky 1967 TAMS **126** 487–507" → J. Funct. Anal. **1** (1967) 255–280 DOI:10.1016/0022-1236(67)90015-8; conformal coupling $\xi = 1/8 \to \xi = 1/6$ in $d$=4; CMPT24 arXiv `2402.02035` → `2306.14732` (the previous ID is a solar-energy paper); "Simon *Trace ideals* arXiv:math/0309173" → no arXiv preprint (AMS S&M vol. 120; the previous ID is a withdrawn combinatorics paper by Pretorius–Swanepoel); "Krasil'shchik 2014 arXiv:1403.2278" → Glas–Konstantis–Krause–Loose 2014 (Bianchi Lie-algebras revisit); BFV03 arXiv `math-ph/0112228` → `math-ph/0112041`. *(v) Wave 3 launched (8 sub-agents + Mistral large + Gemini 3.1 flash-lite-preview)*: B1 Lemma 3.4 anomaly-bookkeeping (Opus); B2 Mixmaster exponent re-derivation (Opus, Mistral cross-check completed and confirms closed form under Gauss-shift); B3 $\mathrm{BVII}_0$ Gap 1 (Bessel-$J_0$ WF-set) + Gap 4 (IR convergence near $r = 0$) (Sonnet); B4 BII Junker–Schrohe two-index $(\omega, n)$ parametrix (Sonnet); B5 $\mathrm{BVI}_0$ massive Lemmas C+D (Mathieu-spectral lower bound + $K_{i\nu}$ second-order WKB) (Sonnet); B6 Levin–Lubinsky G1+G2 closure (Opus); B7 G3 spectral-rescaling $2\pi^2$-factor convention reconciliation between v6.0.22 $b_n = \pi n$ and v6.0.25 slope $1/(2\pi)$ (Sonnet); B8 Bianconi PRD Comment draft (Sonnet). *(vi) Cumulative tally after v6.0.27*: two robust new mathematical results (FRW Theorems 5, 6); plus seven smaller new results (FRW Prop. 11, Krylov–Diameter Theorem 4 with explicit measure $\mu_{\mathrm{HU}}$ now identified, Algebraic WCH T1+T2+T3 modulo Hadamard, Bianchi I Prop. B.1 perturbative, T2-Bianchi I rigorous via S1+S3, T2-Bianchi IX rigorous pathwise on Heinzle–Uggla, Lemma 3.3 stress-tensor conjugation NEW); plus T2-Bianchi V matter unconditional within 4–6 weeks; plus T2 rigorous-pending-Hadamard on Bianchi $\mathrm{II}/\mathrm{VI}_0/\mathrm{VII}_0$ with closure plans drafted (B3, B4, B5 wave-3 in flight); plus T2 partial S3-conditional on Bianchi $\mathrm{III}/\mathrm{IV}/\mathrm{VI}_h/\mathrm{VII}_h$; plus T2 **upgraded** from BLOCKED to PARTIAL on Bianchi $\mathrm{VI}_{-1/9}$ (NEW v6.0.27 via LU24); plus T2 pathwise-conditional on Bianchi VIII; plus Block A1 restricted-class; plus four conditional-companion targets (R2 Comment with $\mu_{\mathrm{HU}}$ identification, R5 Spectral Correspondence, R6 $K_{\mathrm{FRW}}$ + generalised entropy with Lemma 3.3 NEW, M1-C Block A asymptotic-equivalence Vanlessen+Levin–Lubinsky split); plus Bianconi-2025 PRD Comment forthcoming (3-LLM convergent verdict + Lorentzian signature gap NEW). One conditional construction (Cosmological HP). *Thirty-seven* calibrated negatives now including A1 Bianconi orthogonal-plus-Lorentzian-signature-gap, A3 Vanlessen single-step oversold, A6 $\mathrm{BVI}_0$ massless log IR universal, A8 past-BKL global-extension hard-negative, and A9 Odrzywołek zero-connection. Net signed contribution unchanged in spirit: mathematical-physics in the post-CLPW algebraic-quantum-gravity programme; no $\Lambda$, no $H_0$, no $S_8$, no SM derivation, no hierarchy, no DM, no RH proof. Triangulation discipline (3+ LLMs + sympy + arXiv API + adversarial Opus reviews) caught nine additional hallucinations in the present author's wave-2 prompts, raising the cumulative project hallucination-catch count to twenty. Any residual error is the author's sole responsibility. v6.0.28 (2026-05-03 night, wave-3 follow-up consolidation + 3 new paper drafts published locally + Bianchi C+C-extension landscape survey in flight): *(i) Wave 3 results* (8 sub-agents B1–B8 + Mistral + Gemini double-checks). **Positives:** (a) **Lemma 3.4 (anomaly bookkeeping)** drafted and sympy-verified (`paper/k_frw_generalised_entropy/lemma34.tex`); $\Delta_{\mathrm{anom}}[D_R] = (8\pi^2/15)\, a(\eta_c)^2\, R^4\, \langle T_{00}^{\mathrm{FRW}} \rangle_{\mathrm{anom}}$ in c-number sector preserves type $\mathrm{II}_\infty$ factoriality; radiation-era $\langle T^\mu_\mu \rangle = 1/(240\pi^2\eta^8) \neq 0$ (ICRT contributors $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = 24/\eta^8$, $R_{\mu\nu}R^{\mu\nu} = 12/\eta^8$, $R = 0$, sympy-derived); BD-1978 PRSA transcription needed for full publication (∼8–11 days). (b) **Mixmaster $\sqrt{3}\pi/(9 \ln 2)$ closed form CONFIRMED** via three independent paths: Mistral analytic, Opus B2 with explicit five-line completing-the-square proof ($\int_1^\infty du/(1 + u + u^2) = \pi/(3\sqrt{3})$, then $\boldsymbol{\nabla} \cdot \ln 2$), and SYMPY@200dps verified to absolute error 0. The A8 wave-2 "300$\sigma$ rejection" was traced to an off-by-one bug at `mixmaster_numerics.py:121` (`u_era_start = 1.0 + 1.0/x` should be `1.0/x`); A8's $\langle p_3 \rangle \approx 0.93578$ is the integral under the wrong measure $du/((u - 1)u \ln 2)$

on $(2,\infty)$, also sympy-verified. The R2 1-page J. Stat. Phys. Comment carve-out is now *ready for submission* as `paper/mixmaster_invariant/note.tex`. (c) **Bianchi VII$_0$ Bieberbach Gap 1+4 closed unconditionally** (`B3` agent): the Bessel-$J_0$ kernel $K_r(x,x') = J_0(r\,\rho(x,x'))$ is real-analytic on $\mathbb{R}^3\times\mathbb{R}^3$ via the evenness of $J_0$ (Taylor series in $\rho^2$, which is itself a trigonometric polynomial), so $\mathrm{WF}(K_r)=\emptyset$ for fixed $r>0$ — the feared obstruction does not exist. IR Plancherel convergence at $r\to 0$ trivial via $|\beta_r|^2 = O(r^2)$; for compact Bieberbach quotients $G_1$–$G_5$ the spectral gap $r \geq r_1 > 0$ makes IR completely trivial. Time-to-publication stays ∼4–6 mo (parametrix matching Gap 3 still open, ∼1–2 weeks; UV anisotropic WKB Gap 3', ∼3–6 weeks). (d) **Bianchi VI$_0$ Sol massive Lemmas C+D substantially closed** (`B5` agent): Lemma C (uniform mode-frequency positivity $\omega^2_{n,\lambda}(t) \geq m^2 + R_{\min}/6 > 0$) closed unconditionally via Mathieu spectral lower bound $V_{\min} = 2\lambda^2\sqrt{\alpha\beta}$ (NB: *quadratic* in $\lambda$, not linear as the v6.0.24 sketch said); Lemma D (adiabatic order $\geq 2$, $|\beta|^2 = O(\lambda^{-4})$) substantially closed via $K_{i\nu}$ second-order WKB plus Reed–Simon Thm XIII.71. Time-to-pub massive case 9–12 mo → **3–5 mo**; primary bottleneck Obstruction E (Radzikowski wavefront-set on Sol sectors). Massless case re-confirmed as universal log IR hard-negative (BN23 §4 + Olbermann 2007). (e) **Bianchi II Heisenberg/Nil$^3$ Junker–Schrohe two-index $(\omega,n)$ uniformity G1 near-rigorous** (`B4` agent): single-$\varepsilon$ scaling fails as my prompt stated (cross-term $(2n{+}1)|\omega|/(a_1a_2)$ scales $O(\varepsilon^{-3})$ while $\omega^2/a_3^2$ scales $O(\varepsilon^{-2})$); correct joint parameter is $\Lambda_{\mathrm{eff}} = \omega^2 + n$. With this, $|\beta_{n,\omega}(f)|^2 = O((\omega^2+n)^{-N})$ uniform in $(n,\omega)$ proven modulo two sub-gaps: G1.1 (∼2–3 weeks Olver two-parameter uniformity) and G1.4 (∼4–6 weeks gating: pseudodifferential calculus on Nil$^3$ via Fischer–Ruzhansky 2016). Time-to-pub 6–9 mo → **4–6 mo**. (f) **Convention reconciliation G3 closed** (`B7` agent + sympy@100dps on SU(1,1) chiral primary $h$=0.5, 1, 1.5, 2): the v6.0.22 result $b_n = \pi n$ (slope $\pi$, modular MSS $\lambda_L = 2\pi$) and the v6.0.25/A3 Vanlessen transcription $b_n/n \to 1/(2\pi)$ (slope $1/(2\pi)$, $\lambda_L = 1/\pi$) are *both correct* but use *different operators*: v6.0.22 + `krylov_diameter.tex` use Convention C1 (modular Hamiltonian $H_{\mathrm{mod}} = -\log\Delta$), Vanlessen uses Convention C2 (half-modular $\Delta^{1/2}$). Conversion factor $2\pi^2$ (NOT $2\pi$ from linear rescaling, NOT $4\pi^2$ from $\Delta$-vs-$\Delta^{1/2}$ confusion) arises from the nonlinear spectral reparametrization $E = -2\log\omega$. v6.0.25's labeling of the Vanlessen transcription as a "corollary of CMPT" was the gap, now closed. (g) **Bianconi 2025 PRD Comment drafted** (`B8` agent): 4-page revtex4 draft `paper/bianconi_comment/comment.tex`, identifies Lorentzian signature gap $\tilde{H} = \mathrm{Tr}(\tilde{g}\cdot\ln\tilde{g}^{-1}) = \pm i\pi$ for $\mathrm{diag}(-1,+1,+1,+1)$ (sympy + mpmath@200dps), three-LLM convergent verdict (Sonnet+Mistral+Gemini): orthogonal differential-geometric framework, numerologically suspect $\Lambda$, and the $\tilde{H}=0$ step holds only under implicit Wick rotation. *(ii) New paper drafts in repo* (committed to GitHub at d375b7a + ea980ae): `paper/bianconi_comment/` (4 pp revtex4, ∼1 week to submission); `paper/mixmaster_invariant/` (3 pp J. Stat. Phys. note, ∼1 week to submission); `paper/k_frw_generalised_entropy/` (5 pp CMP/AHP draft, ∼2–3 weeks to submission after BD-1978 transcription). *(iii) Six additional hallucinations caught in the present author's wave-3 prompts (project total 20 → 26)*: Junker–Schrohe arXiv `math-ph/0107024` → `math-ph/0109010` (the previous ID is Jovanovic on Euler–Poincaré–Suslov, completely unrelated); Avetisyan–Verch journal "CMP (2013)" → "CQG **30** (2013) 155006"; Hollands–Wald 2005 review arXiv `gr-qc/0404074` → `1401.2026` (the previous ID is the conservation-of-stress-tensor paper, not the review); CLPW arXiv `2206.10780` mistyped as "type II$_\infty$" → correct: type II$_1$ (de Sitter static patch); the type II$_\infty$ source is Witten arXiv `2112.12828`; KLKSS 1985 author list "Khalatnikov–Lifshitz–Sinai–Khanin" → "Khalatnikov–Lifshitz–Khanin–Shchur–Sinai" (Shchur was missing); $V_{\min} \geq 2\lambda/(a_1a_2) \to V_{\min} = 2\lambda^2\sqrt{\alpha\beta} = 2\lambda^2/(a_1a_2)$ (quadratic, not linear in $\lambda$ — distinction critical for IR behaviour). *(iv) Cross-LLM double-checks of wave-3 critical findings*: `Mistral large-latest` confirmed B2's Mixmaster bug claim independently via the same closed-form; `Gemini 3.1 flash-lite-preview` attempted to refute B3's $\mathrm{WF}(K_r)=\emptyset$ but missed the evenness of $J_0$ (which converts $\rho$-dependence into $\rho^2$-dependence, eliminating the apparent square-root singularity); `Mistral` on B1 Lemma 3.4 made one arithmetic error (claimed $R_{\mu\nu}R^{\mu\nu} = 12/\eta^4$, sympy direct gives $12/\eta^8$) and one convention difference (kernel integral $8\pi R^5/15$ without the Hislop–Longo factor of $1/(2R)$ vs. B1's $4\pi R^4/15$ with it); independent sympy on $R_{\mu\nu}R^{\mu\nu}$ confirms B1, refutes Mistral. The triangulation discipline continues to surface real issues and false positives in roughly equal measure. *(v) Cumulative tally after v6.0.28*: same "two robust + seven smaller new results" baseline; new R6 carve-out paper draft (`paper/k_frw_generalised_entropy/note.tex`) with Lemmas 3.3 + 3.4; new R2 Comment (`paper/mixmaster_invariant/note.tex`) with five-line elementary proof; new Bianconi PRD Comment (`paper/bianconi_comment/comment.tex`); G3 convention reconciled. The ten-paper publication pipeline (`notes/PUBLICATION_PLAN_v6_0_27.md`) lays out the next 12 mo of releases. Net signed contribution unchanged: post-CLPW algebraic-quantum-gravity programme; no $\Lambda$, $H_0$, $S_8$, hierarchy, DM, RH, SM derivation. Twenty-six hallucinations attrapées dans les prompts du présent auteur, raised cumulative project hallucination-catch count from 20 to 26 in this single wave; the triangulation discipline (sympy + arXiv API + 3-LLM-cross-check + adversarial Opus reviews) remains the operational standard. Any residual error is the author's sole responsibility. v6.0.29 (2026-05-03 night, paper-pipeline materialization

wave): *(i) 7 paper drafts now in repo* (committed at e504515, c977d6c, and following): `paper/m1c_block_a/` (M1-C Block A theorem, 7 pp, B6 deliverables with corrected LL2005–2006–2007 trilogy citation chain plus MNT 1985 continuity argument; numerics @200dps gives slope 0.42% off target $1/(2\pi)$); `paper/bvii0_sle_bieberbach/` (Bianchi $\mathrm{VII}_0$ Bieberbach SLE Hadamard, 10 pp, A4 base + B3 Gap 1+4 closure via $J_0$ evenness $\Rightarrow \mathrm{WF}(K_r) = \emptyset$ unconditionally; time-to-pub 4–6 mo); `paper/bii_sle_heisenberg/` (Bianchi II Heisenberg/$\mathrm{Nil}^3$ SLE Hadamard, 9 pp, A5 base + B4 Junker–Schrohe two-index parametrix with correct joint parameter $\Lambda_{\mathrm{eff}} = \omega^2 + n$ NOT single-$\varepsilon$ scaling, G1.4 Fischer–Ruzhansky pseudodifferential calculus on $\mathrm{Nil}^3$ remains gating; time-to-pub 4–6 mo); `paper/bvi0_sle_sol/` (Bianchi $\mathrm{VI}_0$ Sol-solvmanifold massive SLE Hadamard, 9 pp, A6 base + B5 Lemmas C+D substantially closed via Mathieu spectral lower bound $V_{\min} = 2\lambda^2\sqrt{\alpha\beta}$ and $K_{i\nu}$ second-order WKB; time-to-pub massive case 3–5 mo down from 9–12 mo; massless case re-confirmed as universal log IR hard-negative); `paper/t2_bvi_minus_1_9/` (T2-Bianchi $\mathrm{VI}_{-1/9}$ S3-epoch-conditional sharpening, A7 deliverable verbatim, 4–6 pp Lett. Math. Phys. target). Plus the three already-published-in-repo Tier 1/2 carve-outs: `paper/bianconi_comment/` (4 pp PRD Comment, author = Kévin Remondière, ready for editorial polish + submission), `paper/mixmaster_invariant/` (3 pp J. Stat. Phys. note, author = Kévin Remondière, ready for submission), `paper/k_frw_generalised_entropy/` (5 pp CMP/AHP carve-out, ∼80% paper-ready, BD-1978 transcription pending ∼8–11 days). *(ii) Phase 2 calculation re-verification independent reruns*: SYMPY re-passes for all four core derivations: Bianconi Lorentzian $\tilde{H} = -i\pi$ (sympy 1.12 + mpmath 200 dps); Mixmaster $\sqrt{3}\pi/(9\ln 2) = 0.87225311598275358480948438656331485954 44$ exact at 200 dps with the bonus closed forms $\langle p_1\rangle$ and $\langle p_2\rangle$ summing to 1; Lemma 3.3 stress-tensor conjugation off-shell PASS on three test backgrounds; Lemma 3.4 anomaly contribution PASS for radiation-era $\langle T^\mu_\mu\rangle = 1/(240\pi^2\eta^8)$ and matter-era. arXiv API live-verified all 13 critical references (Fröb 2308.14797, Witten 2112.12828, CLPW 2206.10780, CHM 1102.0440, Heinzle–Uggla 0901.0806, Banerjee–Niedermaier 2305.11388, CMPT24 2306.14732, Vanlessen math/0504604, Avetisyan–Verch 1212.6180, Bianconi 2408.14391, Junker–Schrohe math-ph/0109010, BFV math-ph/0112041, Hollands–Wald review 1401.2026); all titles and authors match. *(iii) One additional title-correction caught* (28th hallucination): the Fröb 2023 paper title is *"Modular Hamiltonian for de Sitter diamonds"* (JHEP **12** (2023) 074), not the "Conformal scattering and modular flow on cosmological backgrounds" bibliographic entry of the v6.0.28 K_FRW carve-out — corrected at commit c977d6c. The paper does cover conformally-flat spacetimes generally (de Sitter is a corollary), so the carve-out's reliance on Fröb's intertwiner remains valid. *(iv) Cumulative hallucination-catch count: 27 → 28*; the discipline (3+ LLM cross-check + sympy + arXiv API + adversarial Opus reviews) continues to surface real issues including paper-body title typos that survived multiple internal reviews. *(v) Pipeline status after v6.0.29*: Tier 1 (Bianconi PRD Comment, R2 Mixmaster Comment, M1-C Block A theorem) ready for editorial polish + submission within 1–4 weeks; Tier 2 (K_FRW + entropy carve-out) needs BD-1978 transcription (8–11 days) + shape-residual quantification; Tier 3-4 (Bianchi $\mathrm{VII}_0$, II, $\mathrm{VI}_0$ paper drafts) residual gaps identified and time-bounded at 3–6 mo each. Net 7 paper drafts in repo plus eci.tex master (32 pp) covering the v6.0.27–29 mathematical-physics consolidation, with explicit non-coupling to cosmological tensions ($\mathrm{H}_0$, $\Lambda$, $S_8$, generations, RH, hierarchy, DM, SM gauge — all NO-GO per v6.0.20–25 R-pistes). Any residual error is the author's sole responsibility. v6.0.30 (2026-05-03 night, 8-agent parallel-lecture-ABC wave + Mistral + Gemini cross-checks): *(i) Lecture (a) follow-up papers materialized in repo (4 new)*: `paper/bv_matter_sle/` (Bianchi V matter unconditional SLE Hadamard, 11 pp; D2 deliverable; Kontorovich–Lebedev decomposition on $H^3$ with spectral gap $\inf\sigma(-\Delta_{H^3}) = 1$ killing volume-zero-mode IR; sympy 5 modes Wronskian drift $< 4\times 10^{-9}$; $V_{\mathrm{eff}} = a''/a = 2/\tau^2 \neq 0$ *correction caught* from v6.0.24 sketch (was claimed $V_{\mathrm{eff}} = 0$ wrongly), but the spectral gap absorbs the potential and the Hadamard argument is unaffected — 30th hallucination); `paper/r5_spectral_correspondence/` (R5 Spectral Correspondence skeleton, 9 pp; D3 deliverable; *D3 RETRACTION of v6.0.23 algebraic identity* sympy-confirmed: $D_{\mathrm{HY}} \neq -D_{\mathrm{CCM}} + i\Delta\cdot\mathrm{id}$ pointwise under $u = e^x$ — the difference is $i(x-1)\partial_x$ vanishing only at $x = 1$; what holds is *spectral type equivalence* both unitarily equivalent to multiplication by $t$ on $L^2(\mathbb{R}, dt)$ via Mellin/Plancherel — 29th hallucination); `paper/k_frw_generalised_entropy/` updated with D1 **lemma34_v2.tex** sympy-rigorous coefficients: B1 ansatz $\langle T_{00}\rangle_{\mathrm{anom}} = R_{\mu\nu}R^{\mu\nu}/(4\cdot 2880\pi^2)$ was *dimensionally inconsistent* (missing $1/a^2$ redshift); D1 corrects via 1st-order linear ODE $\rho' + 4H\rho = -HA$ from trace+conservation, integrating factor $a^4$, $C_0 = 0$ Bunch-Davies; *coefficient retraction*: B1 (radiation $+R^4/(1800\eta_c^6)$, matter $+R^4/(150\eta_c^8)$) → D1 (radiation $+R^4/(1800\eta_c^4)$, matter $-R^4/(90\eta_c^4)$, both $\eta_c^{-4}$ uniform, matter sign flipped negative driven by $\Box R = -216/\eta^{12}$); 28 sympy assertions PASS + 2 mpmath@200dps cross-checks PASS, no FAIL. *(ii) Lecture (b) scope and limitations* (E1 deliverable, paper/scope_and_limitations.tex + paper/SCOPE.md): explicit 'What ECI claims' / 'What ECI does NOT claim' tables; the 5 Pistes + 10 R-pistes NO-GO catalogue collectively documented as '17 tested and retired candidate bridges — a calibration success'; surgical overclaim audit produced corrections for Bianconi PRD Comment (add 2 sentences that ECI itself has no $\Lambda$ derivation), WCH T3 (add Remark that algebraic past↔future non-isomorphism

does NOT derive arrow of time independently of the Past Hypothesis), Levier 1B citations (always flag null $\xi_\chi + \log B_{10} = -1.37 + H_0$ UNRESOLVED at $4.5\sigma$), and M1-C demote Theorem → Conjecture 4.2 in eci.tex. *(iii) Lecture (c) selection-principle / second-framework coupling* (4 new papers + verdict): `paper/connes_embedding_FRW/` (F2 deliverable, 7–8 pp; *Theorem established*: FRW Theorem 3.5/3.6 type $\mathrm{II}_\infty$ factor is hyperfinite = $\mathbb{R}_{0,1}$ (unique AFD $\mathrm{II}_\infty$); 5-lemma chain: A type $\mathrm{III}_1$ (BFV+HL or Verch1994), A AFD (BDF1987 transfer or Verch+Fewster2015+D'Antoni-Hollands2006), Connes-Takesaki 1977 + Connes 1976 → $\mathbb{R}_{0,1}$, separability automatic for Bunch-Davies GNS; Mistral large-latest cross-check VALIDATES proof chain with refined hypothesis-checking; sharpens R3 [KILLED-BY-VERCH-1994] and R8 [REFUTED-BY-WEYL-LAW]); `paper/bec_steinhauer_krylov/` (F1 deliverable, 10 pp; *first ECI experimental falsifier outside MCMC*: BEC sonic horizon Krylov saturation prediction $\lambda_L^{\mathrm{sonic}} = 2\pi k_B T_H/\hbar = 288\ \mathrm{s}^{-1}$ for Steinhauer 2019 $T_H = 0.35$ nK, $t_K = 3.47$ ms, accessible via $g^{(2)}(\tau)$ decay rate measurement, not done in Steinhauer 2019; *caveat*: KD Theorem 4 conjecture $R_{\mathrm{proper}} = \xi$ is order-of-magnitude only, predicts $T_H \sim 1.2$ nK vs measured 0.35 nK, factor of 3.5 off, NOT a precision cross-check); `paper/wedge_reconstruction_FRW/` (F3 deliverable, 10 pp; *2 binding obstructions identified*: O1 $\partial D_R$ not conformal boundary so reconstruction is modular-flow Cauchy prescription *not* AdS/CFT-style holography; O4 $\partial D_R$ not Killing horizon of $g_{\mathrm{FRW}}$, QFT Markov property required by Faulkner-Lewkowycz unproven; conditional Theorem 3 with sympy modular flow $\sigma_s^{\mathrm{FRW}}[\phi](\eta, x) = [a(\eta_s)/a(\eta)]\phi(\eta_s, x_s)$ Möbius+rescaling, 4 PASS); `paper/bi_sequel/` + `paper/ncg_eci_orthogonality/` (F4 deliverable, 10 pp + 5 pp; *BIPP Theorem*: $\mathcal{A}_{\mathrm{BI}}(D_R)$ type $\mathrm{III}_1$ via BFV, crossed product type $\mathrm{II}_\infty$, $K_{\mathrm{BI}}$ Plancherel reduces to $K_{\mathrm{FRW}}$ in isotropic limit; *Lemma 3.3 conformal pull-back NO-GO* sympy-confirmed for general anisotropic Bianchi I, BIPP bypasses via Tomita-Takesaki direct on BN-state; AND *NCG-ECI orthogonality theorem*: $A_F \otimes A_{\mathrm{FRW}}(D_R)$ is type $\mathrm{II}_\infty$, modular flow on $A_F$ trivial since type I, S_gen decomposes additively NO cross-term, NCG provides ZERO selection principle for ECI — the second-framework lecture-(c) coupling via NCG Connes-Chamseddine is closed CLEAN). *(iv) Cross-LLM double-checks of wave findings*: Mistral large-latest VALIDATED F2 Connes embedding proof chain with refined citations; Gemini 3.1 flash-lite-preview on D1 anomaly was REFUTED by direct sympy ($R_{\mu\nu}R^{\mu\nu} = 12/\eta^8$ for radiation, NOT $6/\eta^4$ as Gemini claimed) but its (c) prefactor $8\pi^2/15$ and (d) shape $\mathrm{O}(R^2)$ confirmed; Mistral on B2 mixmaster CONFIRMED A8 off-by-one bug. *(v) Eight new hallucinations caught from the present author's wave prompts (project total 28 → 37)*: $V_{\mathrm{eff}} = 0$ v6.0.24 sketch → $V_{\mathrm{eff}} = 2/\tau^2$; $D_{\mathrm{HY}}$ pointwise identity → spectral type equivalence; Stanford from MSS arXiv:1503.01409 (had "Stankovics"); Schroer 1989 unsourceable → BDF 1987 + Fewster 2015 + D'Antoni-Hollands 2006 chain; Marcolli-van Suijlekom arXiv:1405.7860 → 1301.3480 (1405.7860 is Clarkson-Umeh-Maartens-Durrer cosmology); Birrell-Davies $\lambda\phi^4$ DOI 10.1098/rspa.1978.0223 fabricated (correct $\lambda\phi^4$ paper is Phys. Rev. D 22 (1980) 322); Bunch-Davies 1978 conflated with Bunch-Davies 1977 ('Stress tensor and conformal anomalies' is BD 1977 PRSA A 356:569, NOT 1978; the 1978 paper is 'QFT in dS Space' PRSA A 360:117); Faulkner-Lewkowycz arXiv:1704.05732 → 1704.05464 (1704.05732 is Cooley-Kang-Koch hypergraph combinatorics); Mukohyama-Speranza arXiv:2402.10362 → unfindable (2402.10362 is Hejazi et al. quantum simulation, no M-S stretched-horizon paper at that ID); plus the v6.0.10 BEC "$\rho \approx 8.29\%$ saturation" RETRACTION CANDIDATE confirmed by F1: 4 arXiv API queries find no Steinhauer-group 2024-2026 BEC saturation paper, the term "saturation envelope" does not appear in any analogue gravity literature found. *(vi) Cumulative project paper inventory after v6.0.30*: 11 paper drafts in repo (eci.pdf master 32 pp + bianconi_comment 4pp + mixmaster_invariant 3pp + k_frw_generalised_entropy 5pp + m1c_block_a 7pp + bvii0_sle_bieberbach 10pp + bii_sle_heisenberg 9pp + bvi0_sle_sol 9pp + bv_matter_sle 11pp + r5_spectral_correspondence 9pp + connes_embedding_FRW 5pp + bec_steinhauer_krylov 10pp + wedge_reconstruction_FRW 10pp + bi_sequel 5pp + ncg_eci_orthogonality 4pp); plus markdown notes paper/t2_bvi_minus_1_9 (T2 sharpening A7) + paper/scope_and_limitations.tex + paper/SCOPE.md; total 13 LaTeX papers + 2 scope documents covering the v6.0.27–30 mathematical-physics consolidation. The framework is now *paper-pipeline-pipelined* with Tier 1 (1-2 weeks editorial polish + submission) Bianconi PRD Comment, R2 Mixmaster Comment, M1-C Block A theorem, Connes embedding hyperfiniteness; Tier 2 (2-3 weeks) K_FRW + entropy carve-out (D1 corrections applied); Tier 3 (3-6 mo) BV matter SLE + Bianchi $\mathrm{VII}_0/\mathrm{II}/\mathrm{VI}_0$ SLE pipeline + R5 Spectral Correspondence + Wedge Reconstruction conditional; Tier 4 (6-12 mo) BIPP Bianchi-I theorem + various extensions. NCG-ECI orthogonality is a clean negative observation publishable in 2-3 mo. The lecture-(c) selection-principle search must look elsewhere than NCG. *Cumulative tally*: two robust new mathematical results (FRW Theorems 5, 6); plus the smaller new results documented in v6.0.21–29 unchanged plus FRW Theorem 3.5/3.6 = $\mathrm{R}_{0,1}$ identification (NEW v6.0.30 via F2); plus BIPP Bianchi-I theorem (NEW v6.0.30 via F4); plus Bianchi V matter unconditional SLE (NEW v6.0.30 via D2); plus R5 Spectral type equivalence (NEW v6.0.30 via D3, downgraded from pointwise identity per retraction); plus FRW wedge reconstruction conditional theorem (NEW v6.0.30 via F3, conditional on H1+H2). *Thirty-seven calibrated negatives* unchanged plus the v6.0.10 BEC saturation retraction candidate (now

38). Net signed contribution unchanged: post-CLPW algebraic-quantum-gravity programme; no $\Lambda$, $H_0$, $S_8$, hierarchy, DM, RH, SM derivation; the framework consolidates as a focused mathematical-physics contribution with explicit scope-and-limitations documentation, an experimental falsifier proposed (BEC analogue gravity), and one second-framework selection-principle direction (NCG) closed clean. Any residual error is the author's sole responsibility. v6.0.41 (2026-05-03 evening, Z7 cross-paper audit + Phase 3 selection-principle search closed) [superseded by v6.0.42, see below]: *(i) Phase 3 closure (definitive structural negative).* Six independent selection-principle frameworks tested as candidate "second framework" to supply phenomenology compatible with ECI's algebraic $\mathrm{II}_\infty$ structure: NCG Connes–Chamseddine (F4 closed clean: $A_F \otimes A_{\mathrm{FRW}} = \mathrm{II}_\infty$ with trivial modular flow on type-I $A_F$), Swampland (Z1: orthogonal + domain-sharpening only), Holographic cosmology / dS-CFT / DSSYK (Z2: orthogonal, FRW→dS embedding only), Causal Sets (Z3: $\delta\Lambda \sim \ell_P^2 V^{-1/2}$ fluctuation matches $\Lambda_{\mathrm{obs}}$ but requires anthropic step), LQG / spin foam (Z4: $\mathrm{I}_n \otimes \mathrm{II}_\infty$ orthogonal; LQC bounce a non-trivial coupling *REFUTED* by Mistral large-latest cross-check via Fewster-Verch 2013 `arXiv:1304.1775`: polymer measure mutually singular with Fock measure, so bounce $a(\eta) > 0$ does NOT extend to a Hadamard quasifree state on the diamond), Anthropic principle / multiverse (Z5: low-priority philosophical; KLS 2024 `arXiv:2406.01669` "Algebraic Observational Cosmology" independently validates ECI's type-$\mathrm{II}_\infty$ programme on inflationary FLRW asymptotically dS in past — prior art to acknowledge), Generalized spectral action via type-III $\sigma$-spectral triples (Z6: Brown 1981 contractibility of $U(N)$ on type-III blocks SM gauge derivation; necessary but not sufficient). *Conclusion: P5 (standard NCG / Connes–Chamseddine spectral action): formally orthogonal theorem proved (companion paper ncg_eci_orthogonality). Z1 swampland, Z2 holographic, Z3 causal sets, Z4 LQG/spin-foam, Z5 anthropic, Z6 generalised spectral: exploratory work-plan notes (Phase 3); no coupling theorem yet; recorded as open research directions. Phase 3 closed P5 formally; Z1–Z6 catalogued as open exploratory directions. ECI is algebraic-quantum-gravity self-consistent, not a TOE; the search for a complementary phenomenology-supplying framework remains open after this first exploration. (ii) Z7 equation cross-paper audit (Opus max-effort, 281 numbered displays + 25 boxed centrals)*: framework dimensionally consistent, 11 sympy/mpmath@200dps cross-checks (9 PASS + 1 FAIL Issue #1 + 1 partial Issue #2 + 1 prose-table Issue #3, all three fixed in this revision). Issue #1: `paper/bi_sequel/note.tex` stress-tensor weight $T^{\mu\nu}_{\mathrm{FRW}} = a^{-(d+2)} \cdot \ldots$ wrong (raising indices on lemma33 lower-index $a^{d-2}$ via $g^{\mathrm{FRW},\mu\alpha} = a^{-2}\eta^{\mu\alpha}$ gives $a^{d-6} = a^{-2}$ for $d = 4$, NOT $a^{-6}$); corrected. Issue #2: `paper/k_frw_generalised_entropy/lemma33.tex` prose "off-shell, identical zero" wrong (sympy + direct test give on-shell only with residual $(1/3)\Phi a'(\Phi a' + 2\Phi' a)$); restated as "on-shell vanishing modulo FRW field equation". Issue #3: this paper's §BEC table verdicts "consistent" (3 rows) replaced with "prediction awaiting test" + footnote retracting v6.0.10 envelope claim. *(iii) Opus OCR proper visual audit (149 pages all individually inspected)* caught residual issues that Sonnet v1 missed: **ETHICS BLOCKER** in BEC paper §8.2 (named Steinhauer/Muñoz de Nova/Parentani/Liberati as candidate co-authors with affiliations + recruitment timing in public preprint) replaced by neutral group-engagement framing; `[Author]+[Institution]` placeholder typeset on `bv_matter_sle` cover (v1 sweep missed) and "Agents A5, B4, G5" codename on `bii_sle_heisenberg` cover, both replaced; "[ECI programme draft]" placeholders on `wedge_reconstruction_FRW`, `m1c_block_a`, `ncg_eci_orthogonality` replaced with author info; 3 LaTeX double-superscript/subscript bugs fixed (`\lambdaL^{KD}`, `\Nil^3` when `\Nil=\mathrm{Nil}^3`, `M_\DR`). *(iv) KLS 2024 prior-art acknowledgment* `arXiv:2406.01669` added in `paper/connes_embedding_FRW/` introduction and bibliography, and in `paper/bi_sequel/` abstract: KLS provide independent type-$\mathrm{II}_\infty$ construction for FLRW *asymptotically dS in past* (inflationary regime, inflaton zero-mode); ECI's Theorems 3.5/3.6 cover the disjoint setting of post-inflationary radiation/matter eras via conformal pullback of Hislop–Longo on conformally-flat FRW. *(v) Cumulative hallucination-catch count: 46 → 54 → 56 (v6.0.42 adds +2: Brown 1981 Trans.AMS 266 p.91 (Z6 sub-agent fabrication, never propagated to repo, fix logged); Sorkin 1987 IJMPD DOI 10.1142/S0218271887000174 (CrossRef 404, flagged by Z3 sub-agent, never propagated to repo))* (Phase 3 wave caught 8 additional bibliographic mis-attributions in present author's prompts: Bousso `hep-th/0610132` = Boltzmann brain not $\Lambda$ via causal patch; Sorkin 1987 IJMPD impossible (IJMPD founded 1992); DEHK acronym conflated with swampland but is in fact the De Vuyst-Eccles-Höhn-Kirklin QRF papers; Susskind `2110.04054` = robotics paper not DSSYK; Carey-Phillips-Rennie `1003.4735` = condensed matter not CPRS, correct = `math/0411019`; Devastato Lizzi Martinetti 2014 = actually Devastato+Martinetti only; Ashtekar-Lewandowski `gr-qc/9412049` = Conradi 1994 Kantowski-Sachs; EPRL `0708.1595` = Freidel-Krasnov, correct EPRL = `0711.0146`). Z7 verified *no new* hallucinations beyond these were introduced in any of the 15 audited papers. *(vi) Cumulative pipeline state*: 15 papers + eci master + paper/frw_typeII_note (16th paper, 18 pp parent of Theorems 3.5/3.6 not previously included in formal audit, now added). All compile clean: `eci 34pp, bv 13, bii 10, bvi0 9, wedge 10, bec 13, m1c 12, ncg 4, bi_sequel 8, k_frw 5, frw 18, etc.` GitHub release v6.0.40 has 15 PDFs as assets; Zenodo record 20014341 DOI `10.5281/zenodo.20014341` has 15 PDFs published. *(vii) Z7 refactoring recommendations not adopted at*

*v6.0.41* (deferred to a future major revision to avoid breakage): `paper/_common/preliminaries.tex` and `paper/_common/sympy_geometry.py` shared modules (15-20% page-count reduction); KD Theorem 4 disambiguation between m1c_block_a Theorem 4.1 (math.OA modular Krylov slope) and BEC §3.2 phenomenological $c_s/\xi$ conjecture (DONE in this revision, see BEC paper §3.2); Lemma 3.4 promotion to standalone short note (deferred: would split the K_FRW carve-out paper). Net signed contribution unchanged: post-CLPW algebraic-quantum-gravity programme, no $\Lambda$, $H_0$, $S_8$, hierarchy, DM, RH, SM derivation, with explicit scope-and-limitations documentation, KLS 2024 acknowledged as independent prior art for the inflationary disjoint setting, and Phase 3 closed P5 formally; Z1–Z6 catalogued as open exploratory directions. Any residual error is the author's sole responsibility. v6.0.42 (2026-05-03, W1–W5 Phase-4 audit corrections): *(i) Phase-3 orthogonality claim tightened*: P5 NCG (standard CC) is formally orthogonal (theorem in ncg_eci_orthogonality); Z1–Z6 are open exploratory directions with no coupling theorem yet, not closed orthogonality proofs. *(ii) BEC changelog narrative retracted*: v6.0.10 "consistent across three Steinhauer datasets within 1–10% correction envelope" appended with verdict retraction tag; see prediction table footnote. *(iii) Hallucination counter 54 → 56*: +2 new catches — Brown 1981 Trans.AMS 266 p.91 (Z6 sub-agent fabrication, never propagated to repo; fix logged in A1 action); Sorkin 1987 IJMPD DOI 10.1142/S0218271887000174 (CrossRef 404, flagged by Z3 sub-agent, never propagated to repo). *(iv) bi_sequel raising convention clarified*: parenthetical at eq. Lemma33-FRW now specifies $\eta$-raising (not $g_{\mathrm{FRW}}$-raising) so the weight $a^{-2}$ is consistent. *(v) DESI DR2 and alternative-cosmology context*: § Discussion updated with DESI DR2 BAO+CMB $3.1\sigma$ ($\to 4.2\sigma$ with SNe), top-5 alternatives algebra-compatible with ECI type-$\mathrm{II}_\infty$ no-go, ECI prunes ∼40% of alternative landscape, decisive moment DESI DR3 + Euclid Q3 2026/Q1 2027. *(vi) ECI Yukawa non-prediction noted*: ECI does not predict Yukawa couplings, consistent with universal pattern across 13 reviewed frameworks; only $\mathrm{m_{top}}$ via Eichhorn–Held asymptotic safety arXiv:1707.01107 ever genuinely predicted before measurement; modular flavour SL(2,$\mathbb{Z}$) (Feruglio arXiv:1706.08749) noted as structural overlap with ECI's Tomita–Takesaki modular flow, flagged for v7 R&D. *(vii) New bibliography entries*: KiDS-Legacy arXiv:2503.19441 (Wright et al.); Eichhorn–Held arXiv:1707.01107; Feruglio arXiv:1706.08749; Constantin–Leung–Lukas–Nutricati arXiv:2507.03076 (heterotic flavour fit). *(viii) §"Scope of structural derivation" inserted before Bianchi attractor section*: no-go Theorem 1 (Structure-vs-values gap) formally records that dimensionful observables are determined only up to the $\mathcal{S} = \mathbb{R}_{>0} \rtimes \mathrm{Out}(\widetilde{\mathcal{M}})$ orbit; three escape routes listed. v6.0.28-addendum (B6 wave-3 returned LATE): the v6.0.25 "Levin–Lubinsky 2001 (CMS Books in Math vol. 4)" citation IS REAL but does NOT cover the half-line $[0,\infty)$ Laguerre-type weights needed for M1-C Block A; LL2001 covers real-line symmetric weights $e^{-Q(x)}$ on $\mathbb{R}$ only. The correct citation chain is the trilogy LL2005 (J. Approx. Th. **134** 199–256), LL2006 II (J. Approx. Th. **139** 107–143), LL2007 (J. Approx. Th. **144** 260–281). LL2007 covers "rapidly decreasing" weights $Q'(x) \to \infty$ which excludes linear-$Q$ verbatim; the borderline-linear regime is closed by Szegő 1939 (classical Laguerre) plus Maté–Nevai–Totik 1985 (Constr. Approx. **1** 231–248) continuity. Numerical verification on mixture $R(\omega) = 0.3\omega^{-1/2} + 0.5\omega + 0.2\omega^3$, $Q(\omega) = 2\pi\omega + 0.1\sin\omega$ at 200 dps: Lanczos slope 0.15983 vs target $1/(2\pi) \approx 0.15915$, relative error 0.42% (`paper/m1c_block_a/` pending). 27th hallucination caught (LL2001 misapplied + ISBN typo 978-0-387-98941-5 should be 978-0-387-98941-9). v6.0.43 (2026-05-04 dawn patch): Opus direct-OCR audit of all 16 PDFs caught three single-pass `pdflatex` BLOCKERS (bii_sle_heisenberg, r5_spectral_correspondence, ncg_eci_orthogonality, all rendering ?? for ∼95 cite calls due to missing `.bbl` regeneration) and 11 CONCERNS resolved via full `latexmk` re-run + `xr` cross-paper setup + 8 codename-leak strips across 7 sub-papers; bibliography unchanged, scientific content unchanged, hallucination counter unchanged at 56. v6.0.44 (2026-05-04 forenoon, multi-pool agent campaign): 20 sub-agents in 4 pools (P1 modular flavour SL(2,$\mathbb{Z}$), P2 algebraic-envelope E6 escape route, P3 coupled-DE phenomenology, P4 top-5 W1 prune-analysis) audited v6.0.43 against arXiv API + CrossRef and investigated v7 R&D directions; counter unchanged at 56 (no new repo hallucinations); 16 reference-level catches in agent briefs / external task statements only (none propagated to `eci.tex` or `eci.bib`): KSTTT correct = arXiv:1906.10341 not 1908.07457 (=Nomura-Okada-Popov scotogenic); Devastato et al. correct = arXiv:2002.11700 not the previously fabricated PRD 103 035003; Karwal–Kamionkowski correct = arXiv:1608.01309 not 1601.05005 (=elastomer paper); Dark Dimension correct = arXiv:2205.12293 not 2203.10757; KiDS-Legacy $S_8 = 0.815^{+0.016}_{-0.021}$ at $0.73\sigma$ from Planck (now consistent, not $3\sigma$); Bedroya $c' = \frac{1}{3}c_{\mathrm{DDC}}$ identified as a fabricated relation: $c$ and $c'$ are independent in arXiv:2507.03090; $\mathrm{m_{KK}}$ DM mass scale is keV not $10^{-3}$ eV; the speculation $[M:N] = (1+\sqrt{3})^2 = 3 + 2\sqrt{3}$ for the E6 subfactor is wrong, correct E6 indices are $2+\sqrt{3}$ (string algebra) or $3+\sqrt{3}$ (GHJ A11⊂E6); Connes math/9811068 abstract verified to contain no SL(2,$\mathbb{Z}$) action on $K_R$; CCM arXiv:2511.22755 verified to contain no SL(2,$\mathbb{Z}$) action. `compute/C4_joint_mcmc/preregistration/design_doc_v1.md` drafted as pre-registration for the joint 10-model MCMC on Planck PR4 + DESI DR2 + Pantheon+ + KiDS-1000 + ACT DR6 lensing, locked priors, PolyChord $n_{\mathrm{live}}$=500 (1000 for EDE), Jeffreys decision thresholds, ECI-specific falsification window $\log B_{\mathrm{ECI},\Lambda\mathrm{CDM}} \in [-3,+1]$ (Bayes-comparable, not falsified) pre-registered. v7-R&D status: (a) primary external falsification risk = Wolf–García-García-Anton-Ferreira NMC quintessence

(PRL **135** 081001, `2504.07679`, $\log B = 7.34 \pm 0.6$ vs $\Lambda$CDM, $\xi = 2.31$, $G_{\rm eff}(0)/G_N = 1.77 \pm 0.20$ at $4.3\sigma$); LISA standard sirens at Q4 2026 launch + 4 yr will test $\Xi(z)$ at few-% precision and discriminate against ECI's H4 hypothesis directly. (b) algebraic-envelope E6 escape route from Theorem 1: closest existing anchor is the Weyl-group rigidity theorem of arXiv:2508.08194 (Out of $L(\Lambda \rtimes G/H)$ is finite for semisimple $G$); the specific extension to ECI's type $\mathrm{II}_\infty$ $\widetilde{\mathcal{M}}(\mathcal{D})$ is an original-result gap. (c) the $(\infty, 2)$-functor U conjecture of the v7 manifesto is REFUTED in current form: Krylov $2\pi$ saturation [Parker et al. PRX **9** 041017] is incompatible with rational E6-modular-tensor-category target [Rabinovici et al. JHEP 03 (2022) 211]; rational CFTs exhibit Krylov localisation $b_n \to b_\infty$, not the linear $b_n \sim \alpha n$ growth required for chaos-bound saturation; the v7 manifesto must either weaken C1 to a spectral condition compatible with rational MTCs or replace the target with a non-semisimple/non-rational algebraic structure. (d) most promising structural overlap candidate for ECI's Tomita–Takesaki modular flow with Feruglio modular flavour $\mathrm{SL}(2,\mathbb{Z})$ is the non-holomorphic Maass-form Yukawa-coupling extension (Qu–Ding JHEP 08 (2024) 136 polyharmonic framework, $\Delta_k Y(\tau) = \lambda Y$); two genuine obstructions remain (chirality of SM Yukawas; continuous $\mathbb{R}$ modular flow vs discrete $\Gamma_N$ finite quotient). v6.0.45 (2026-05-04 afternoon, retraction patch): two cross-audits launched on the v6.0.44 multi-pool campaign output (Opus vision audit of `/tmp/eci_nobel_path.md` and a Pan-Yang verification agent on the cited source for the morning's $\Delta w_a^{\rm NMC} = -2\xi_\chi \Omega_{\phi 0}$ derivation) caught FIVE corrections that supersede the v6.0.44 audit log claim of "counter unchanged at 56". (i) The on-PC sympy script `w0wa_nmc.py` from the morning Task B cited "Pan-Yang JCAP 2018 arXiv:1804.05064 eq. 18" as the source for $\Delta w_a^{\rm NMC} = -2\xi_\chi \Omega_{\phi 0}$; direct arXiv API hit reveals arXiv:1804.05064 is Henden-Puchwein-Shen-Sijacki, "The FABLE simulations: A feedback model for galaxies, groups and clusters" MNRAS 2018 — a galaxy-cluster hydrodynamics paper, NOT a Pan-Yang NMC quintessence paper. There is no Pan-Yang JCAP 2018 paper on $\xi R \chi^2$ NMC at all; genuine 2018 Pan-Yang papers (1804.08455, JCAP 09 (2018) 019) concern interacting dark energy, not NMC. The structural form $\Delta w_a \propto \xi \Omega_\phi$ is plausible at leading order in dS ($R \approx 12 H^2 \Omega_\phi$) but the coefficient $-2$ has no justified source. $\Delta w_a^{\rm NMC}$ **retagged from [PROVED 2026-05-04, NEW HERE] to [WORKING-CONJECTURE — pending sympy first-principles derivation from $\mathcal{L}_\chi$ NMC action]**; the deferred status of eci.tex line 153 is reaffirmed; the 35× discrimination claim against Coupled-DE Amendola is held conditional on the formula being correct. (ii) The Scherrer-Sen 2008 baseline citation drift: arXiv:0712.2083 was wrong; correct ID is arXiv:0712.3450 (PRD 77, 083515). The pure-quintessence baseline $w_0 \approx -1 + \lambda^2/3$ in the morning's script also missed the $\Omega_{\phi 0}$ factor: leading order should be $w_0 \approx -1 + (\lambda^2/3)\Omega_{\phi 0}$. (iii) In the v7-R&D internal note `notes/eci_v7_aspiration/eci_nobel_path_v1_PATCHED.md`, the §2 framework table was fabricated vs eci.tex line 632; corrected to the actual list (DGP, massive gravity, bigravity, MOND/TeVeS, conformal gravity, LIV, CCC, $f(T)$ torsion). (iv) arXiv:0905.0462 (mis-cited as Lurie cobordism hypothesis in the v7-R&D note) is in fact Lurie's "$(\infty, 2)$-Categories and Goodwillie Calculus I"; the cobordism hypothesis paper is arXiv:0905.0465. Corrected. (v) arXiv:2503.19898 was attributed to BH/DHOST $\alpha_M \sim 3\sigma$ in the v7-R&D note; the actual paper is Pan-Ye NMC at $\sim 3\sigma$ from DESI DR2. Re-attributed to NMC row. Three inflations also flagged in v6.0.45: (a) the v7-R&D note's claim "R1+R3 prunes 65–75% of the alternative landscape" is unjustified and retracted (the empirical prune remains $\sim$ 40%, $\geq$ 80% is conditional on observational falsification); (b) the v7-R&D note's claim that ECI's H4 $\to \Xi = 1$ singularises a Nobel-narrative for an LISA $\Xi(z = 1)$ detection is OVERSTATED (H4 $\to \Xi = 1$ is shared with $\Lambda$CDM, Coupled-DE, Dark Dimension, Tsallis HDE; an LISA $\Xi = 1$ detection rules out Wolf-NMC and DHOST but does not select ECI specifically); (c) the Rabinovici 2112.12128 attribution in P2_4 generalises beyond the paper's actual scope (integrable XXZ, not "rational CFTs" globally) — the U-conjecture refutation argument survives qualitatively (rational MTC discrete spectrum is structurally incompatible with linear $b_n \sim \alpha n$) but the cited reference does not directly establish this for rational CFTs. **Hallucination counter** 56 → 57: the propagated Pan-Yang 1804.05064 fabrication is the 57th catch (it propagated to the v6.0.44 audit log paragraph and to the on-PC sympy script, both corrected here). New v7-R&D internal notes added: `notes/eci_v7_aspiration/eci_nobel_path_v1_PATCHED.md` (with retraction notice header), `notes/eci_v7_aspiration/eci_nobel_audit_2026_05_04.md`, and `notes/eci_v7_aspiration/pan_yang_verification_2026_05` Strategic verdicts of the Nobel-trajectory synthesis (no Nobel for ECI alone 2026–2030; Foundational/Breakthrough Prize trajectory; LISA + Innsbruck as external discriminator hooks; v7 manifesto MCC/CCF retired; C4 joint MCMC \$500–750 Q3 2026 = action #1) are NOT contested by the audit and remain valid. v6.0.46 (2026-05-04 late afternoon, six-agent Sonnet exploration sweep): six parallel agents (A1 NMC slow-roll first-principles sympy; A2 Hecke closure on polyharmonic Maass forms; A3 latest 2026-Q2 cosmology lit scan; A4 Wolf-NMC vs ECI-$\xi_\chi$ structural relation; A5 Innsbruck Cs-133 $\rho_{p,k=2}$=1/18 protocol design; A6 Vast.ai Profile L pre-booking checklist) returned six concrete deliverables now archived under `notes/eci_v7_aspiration/A{1...6}_*_2026_05_04.md`. Net findings: (I) A1 closed the v6.0.45 working-conjecture: a proper first-principles slow-roll expansion of the NMC Klein–Gordon $\ddot{\chi} + 3H\dot{\chi} + V'(\chi) = \xi_\chi R \chi$ at leading order in $\varepsilon_V$ and $\xi_\chi$ gives $\Delta w_a^{\rm NMC} =$

$-(2/3)\,\xi_\chi\lambda^2\Omega_{\phi,0}$, with the $\lambda^2$ factor (mandatory since $\dot\chi/H = \lambda M_P\Omega_\phi$ on the attractor) missing from the morning's hallucinated coefficient $-2\xi_\chi\Omega_{\phi,0}$. The corrected value at $\xi_\chi$=0.05, $\lambda$=1, $\Omega_{\phi,0}$=0.7 is $\Delta w_a^{\rm NMC} \approx -0.023$ (not $-0.07$), giving only $0.23\sigma$ discrimination at DESI DR3 $\sigma(w_a)\approx$0.10 instead of the previously claimed $0.7\sigma$. The discrimination ratio over Coupled-DE Amendola at $\beta^2$=$10^{-3}$ is $\sim$ 12×, not 35×. This deepens the v6.0.45 brutally-honest verdict that the ECI-NMC $(w_0, w_a)$-plane signal is sub-$5\sigma$ at DR3 precision. The full derivation is in `derivations/w0-wa-nmc/derivation.py` and the LaTeX block at line 153 supersedes the prior deferred-status pointer. (II) A2 confirmed Hecke closure for harmonic and polyharmonic Maass forms at scalar level: T($p$) eigenvalues 1+$p$ verified by exact Fraction arithmetic for primes $p \nmid N$ at levels $N \in \{2,3,4\}$, and the level-11 cusp-newform $\eta(\tau)^2\eta(11\tau)^2$ benchmark passes. By Bruinier–Ono–Rhoades (arXiv:0802.0963 Thm. 1.1) T($p$) preserves $H_k(\Gamma(N))$, and an inductive filtration argument extends this to polyharmonic kernels $\Delta_k^r Y = 0$ used in Qu–Ding JHEP 08 (2024) 136 (arXiv:2406.02527; **CrossRef DOI 10.1007/jhep08(2024)136 verified** 2026-05-04, contradicting A2's interim "unverified" flag — the JHEP attribution is correct, the arXiv journal-ref metadata is missing). **Gap G4 of P1_4 is now WORKING-CONFIRMED at scalar level** for the Maass-form ↔ KMS overlap programme (axis (d) of v7-R&D); the vector-valued $A_4/S_4$ multiplet structure remains partially open. (III) A3 returned 15 verified 2026-Q2 references; the highest-priority new entry is **Karam–Sánchez López–Terente Díaz** arXiv:2604.16226 demonstrating that the PPN Cassini constraint on the Wolf-NMC parameter $\xi$ is *substantially weakened* in the **Palatini** formulation via reinforced Yukawa suppression. This means Wolf-NMC at $\xi = 2.31$ (log B= 7.34) might evade the $G_{\rm eff}/G_N$=1.77 local-gravity tension if reformulated in Palatini, which *nuances* ECI's claimed Cassini-clean structural advantage (A4) but does not negate it: the Wolf paper uses metric formulation, the Palatini bridge is hypothetical for the same field. Other A3 entries: 2604.02204 (Wang J-Q et al. NMC sign-changing interaction favoured over ΛCDM), 2604.13535 (Bella–Poulin–Vagnozzi–Knox multi-axion EDE allowing $f_{\rm EDE} < 0.014$ and easing the H2 ECI constraint), 2604.04556 (Moshayedi BV–BFV → Reshetikhin–Turaev (3-2-1)-TQFT supporting partial H3 functoriality), 2602.02675 (Vardian modular-Krylov OAQEC supporting H1), 2604.01422 (de Medeiros Varzielas–Paiva $S_4'$ + Kähler quark sector with Hecke closure unproven — ECI gap G4 vector-valued multiplet level remains open), 2604.00805 (Euclid Collab. photo-$z$ calibration 0.004(1+$z$) for DR1 3×2pt), and 4 more. (IV) A4 confirmed ECI-$\xi_\chi$ at $-0.00003 \pm 0.016$ from Levier #1B (R-1=0.059, 48 360 samples, Cassini wall ON) is mutually exclusive with Wolf $\xi = 2.31$; ECI does NOT inherit the Cassini gap; **Cand C LISA** $\Xi(z{=}1)$ **is a CONDITIONAL GO** as a survival test (necessary not sufficient): $\Xi$=1 confirms ECI's H4 algebra-compatible position but does not singularise it (shared with ΛCDM, Coupled-DE, Dark Dim, Tsallis HDE), while $\Xi\neq1$ at $> 3\sigma$ falsifies ECI in its present form. Recommendation for v6.0.46+: frame ECI as the "Cassini-compliant, screened-zero limit" of the NMC parameter space, complementary to but distinct from Wolf's large-$\xi$ regime, with explicit caveat that the Karam–Palatini bridge (arXiv:2604.16226) may shift this distinction. (V) A5 produced a complete 10-step Innsbruck Cs-133 protocol design for measuring $\rho_{p,k=2}$=1/18: observable is mode-resolved quasi-particle occupation via Bragg spectroscopy + $g^{(2)}$ Steinhauer-style correlator extraction; physical $k$=2 realisation via three-body loss quench in 1D Cs tube near Feshbach resonance $B \approx 787$ G; discrimination from Bose (1/12) at $> 10\sigma$, from Fermi (1/24) at $5.6\sigma$ with $J$=400 runs, from parafermion-3 (1/16) at $3\sigma$ with $J\sim$2000 runs; Wu Haldane $g$=2/3 degeneracy broken at low frequency by ECI-Gentile hard saturation $\langle n\rangle \leq 2$ vs Wu $1/u$ divergence; alternative platforms ranked Rb-87 box (Paris/MIT) > K-40 (Munich MPQ) > Sr-87/Yb-171 optical clocks; timeline $3\sigma$ feasible Q2 2028, $5\sigma$ Q4 2028–Q1 2029. (VI) A6 verified Vast.ai pricing live (0.73–2.00/h spot per A100 80 GB PCIe; 4×A100 total 4–8/h, the design-doc \$5/h was a working estimate not a confirmed spot price); cosmopower-jax API confirmed (`pip install cosmopower-jax`, distinct from `cosmopower_cobaya` TF wrapper); AxiCLASS repo `PoulinV/AxiCLASS` confirmed; hi_class_public requires `make` then `make classy` sequence; KiDS-Legacy $S_8$=0.815 patch in `eci_kids_s8.py` flagged as a mandatory pre-registration step (still pending: the value in the .py file is 0.815 since v6.0.45, but the C4 design-doc text references both 0.759 and 0.815 in different paragraphs — to harmonise). **Verdict: NOT ready to book Profile L**; book 1× Profile S RTX 4090 (∼\$6 total) first to close 5 blockers (AxiCLASS+hi_class compile, cosmopower-jax M5+M9 emulator training, PolyChord MPI on Vast.ai LXC, full data install), then tag `c4-preregistration-v1`, then verify live Profile L spot price, then book. Hallucination counter **unchanged at 57**: the Qu–Ding JHEP 08 (2024) 136 attribution is correct (CrossRef verified, contradicting A2's interim flag), and no new propagated repo hallucination was caught in this sweep. Three new bib entries added in this commit: `Karam2026` (arXiv:2604.16226), `Wang2026NMC` (arXiv:2604.02204), `Bella2026EDE` (arXiv:2604.13535). Final author responsibility statement applies. v6.0.47 (2026-05-04 evening, four-agent Sonnet exploration sweep + LLM-triangulation cross-check + C6 v3 MCMC bugfix): four parallel Sonnet agents (B1 Bianchi V Hadamard WKB lemma closure attack; B2 Jing–Alestas–Kuroyanagi [52] GW constraint vs ECI; B3 $S_4'$ quark-doublet Hecke closure; B4 Wolf-NMC $\xi = 2.31$ Kasner ghost rigorous refutation) returned four concrete deliverables now archived under `notes/eci_v7_aspiration/B{1...4}_*/`. Net findings:

**(B1) Bianchi V WKB closure plan identified — PARTIALLY CLOSED.** The single missing temporal SLE–WKB lemma $|T_\rho^{\rm SLE}(t) - T_\rho^{\rm WKB}(t)| \le C(\delta,T)\rho^{-1}$ admits a two-step closure: STEP A applies Dunster 2024 [53] (uniform Liouville–Green expansions of $K_{i\rho}(z)$ as $\rho \to \infty$ with explicit error bounds in unbounded complex domains, the critical new mathematical input unavailable before late 2024) directly to give $|K_{i\rho}(u) - K_{i\rho}^{\rm WKB}(u)| \le C(u)\rho^{-1}$; STEP B adapts BN23 [54] §4 Gronwall verbatim to the Bianchi V frequency $\omega(\rho,t)^2 = (\rho^2+1)/a_1(t)^2 + R(t)/6$ with $\omega \sim \rho/a_1$ at large $\rho$. Assembly via Brum–Them 2013 Sobolev wavefront makes `thm:Hadamard-BV` unconditional and `thm:T2BV` (FRW open-limit) unconditional. Estimated 4–6 weeks of mathematical work; no post-2020 paper treats Bianchi V Hadamard states (literature gap confirmed by arXiv search). **(B2) Jing 2605.00735 GW constraint verified NON-applicable to ECI.** Jing et al. [52] constrains *quintessential $\alpha$-attractor inflation* where the same scalar drives both inflation and late-time dark energy; the GW signal arises from a kination phase between inflation and reheating that blue-tilts the GW background and adds to $\Delta N_{\rm eff}$, capping $n_s \le 0.967$ against the ACT+Planck preference $n_s \approx 0.974 \pm 0.003$ and giving $\ln B = -12.47$. ECI's $\chi$ field is a pure late-time DE scalar, generates no kination GW background, and produces no $\Delta N_{\rm eff}$ contribution: the surface similarity (both involve exponential factors in the potential) is notational rather than physical. ECI's working point ($\xi_\chi \approx 0$, $\lambda \approx 1$, $\chi_0 \approx M_P/10$) is outside the scope of Jing's constraint, and this is favourable: no parameter revisions needed. **(B3) $S_4'$ quark-doublet Hecke closure CLOSED at the theoretical level via Schur's lemma.** On $X(4)$ (Fermat quartic, genus 3): $\dim M_2(\Gamma(4)) = 8 = \dim S_2(3) + \dim E_2(5)$; the cuspidal $S_2(\Gamma(4))$ transforms as the $S_4'$ *triplet* (Fermat differentials), the Eisenstein $E_2(\Gamma(4))$ contains the $S_4'$ *doublet* $Y_2^{(2)} = (Y_a, Y_b)$. Schur's lemma applied to $T(p) \in {\rm End}_{S_4'}(M_2(\Gamma'(4)))$ on the irreducible $V_2$ representation forces $M(p) = \lambda(p)\cdot I_2$ for all $p \nmid 4$, unconditionally. Predicted character-twisted eigenvalue $\lambda(p) = \chi_4(p) + p$ (giving $\lambda(3)$=2, $\lambda(5)$=6, $\lambda(7)$=6, $\lambda(11)$=10, $\lambda(13)$=14) requires Week 2 numerical verification with the true CG-doublet basis from [55] Appendix B (the natural $(\theta_3^2, \theta_4^2)$ proxy fails because $T(3)\theta_3^2$ has nonzero coefficient $a(9) + 3a(1) = 16$ outside span$\{\theta_3^2, \theta_4^2\}$, a basis identification error rather than a closure failure). Together with the C4-morning $A_4$-triplet result, the v7 R&D Gap G4 is now closed at both the $A_4$ triplet and $S_4'$ doublet levels — a substantive new theorem. References [55–57] confirmed via arXiv API on 2026-05-04. **(B4) Wolf-NMC Kasner ghost argument is NOT a refutation.** Sympy verifies that on any Kasner solution ($\sum p_i = 1$, $\sum p_i^2 = 1$) the Ricci scalar $R \equiv 0$ identically, so the $\xi R\chi^2$ coupling vanishes on Kasner and Wolf's NMC reduces to minimal coupling there: the proposed "NMC-specific" ghost is the universal logarithmic divergence of any massless scalar near a singularity, present even at $\xi = 0$. Klein–Gordon on Kasner yields $\chi(t) = C_1 + C_2 \ln t$; the alleged ghost crossing $t_* = t_0 \exp[(-\chi_{\rm crit} - \chi_0)/C_2]$ for typical slow-roll quintessence ($C_2 \sim 10^{-2}$ to $10^{-1}\,M_P$) gives $t_* \sim 10^{-80}$ to $10^{-52}$ s, far below the Planck time $t_{\rm Pl} \sim 5 \times 10^{-44}$ s where classical ghost concepts cease to apply. Vainshtein screening is irrelevant since NMC has no derivative self-interactions. The Wolf et al. original paper already establishes $M_{\rm eff}^2 > 0$ throughout observable FRW history with $\Delta\phi/M_P \sim 0.1 \ll \phi_{\rm crit} = M_P/\sqrt{2.31} = 0.658\,M_P$. Verdict: $\xi_\chi = 2.31$ remains viable; the Kasner-ghost argument dissolves. **(C6 v3 MCMC bugfix on PC gamer.)** The morning's $C_6$ NMC+DESI+Pantheon++KiDS-Legacy MCMC failed at runtime with `CAMBUnknownArgumentError: output` because the YAML `theory.camb.extra_args.output: 'mPk'` line is `classy`-(CLASS)-syntax, not CAMB syntax: in Cobaya/CAMB the matter power spectrum and $\sigma_8$ derived parameter are auto-triggered by listing $\sigma_8$ under `params` rather than via an explicit `output` key. With the line removed the model builds cleanly, returns $\log p_{\rm post} = -711.55$ at the reference point with derived $\Omega_m = 0.3148$, $\sigma_8 = 0.8112$ (giving $S_8 = 0.831$, in $0.86\sigma$ tension with the KiDS-Legacy 2025 prior $0.815 \pm 0.0185$, perfectly physical), and the chain is now running on six P-cores at ∼30 steps/rank/min with first-order autoselect covmat. **LLM-triangulation methodology check.** The companion Gemini-CLI + Mistral-API cross-check on three substantive questions (Karam-Palatini Yukawa suppression factor; Jing $\alpha$-attractor bounds; $S_4'$ doublet $q$-expansion) caught a clean Mistral fabrication: on the Jing question, default-routing Mistral (presumed `mistral-large-latest`) invented numerical bounds $\alpha < 0.05$ and $\lambda < 0.3$ that disagree with the actual paper's posterior mean $\alpha = 3.908$ $(-2.40/+0.98)$ read directly by the B2 agent from the arXiv source. Gemini-CLI free-tier returned empty responses on all three questions because the shared free-tier Gemini Pro quota was exhausted at 100%; verification of remaining free models (Flash 56% / Flash-Lite 97%) and Mistral catalogue (incl. `magistral-medium-2509` reasoning model) is logged in this commit's `notes/` entry. **Hallucination counter** 57 → 58: the 58th catch is the Mistral Q2 fabrication-on-demand pattern; counter increments only for substantive hallucinations propagated to a deliverable, here caught *in cross-check* before any propagation to the manuscript. Forward-looking infrastructure note: the cosmopower-jax POC plus a JAX/blackjax-NUTS GPU stack on a single Vast.ai RTX 4090 (Profile S, ∼$5–10 budget) is projected to compress the full six-lever MCMC suite from ∼1 week of CPU wall-clock to ∼1–2 h GPU, a ∼200× aggregate speedup combining ∼70 000× theory-emulator gain with ∼50× NUTS-vs-Metropolis sampler-efficiency gain.

## DATA AVAILABILITY

All derivation scripts, MCMC chains, and posterior-analysis artefacts underlying the numerical claims of this paper are available in the repository `https://github.com/AIdevsmartdata/crossed-cosmos` under `derivations/` and `mcmc/`, and archived at CERN Zenodo (DOI 10.5281/zenodo.19686398). DESI DR2, ACT DR6, and Pantheon+ data are obtained from their respective public releases as cited in the text.

## CONFLICTS OF INTEREST

The author declares no competing financial or non-financial interests.


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## Appendix A: Speculative: A3 toy dictionary for Cryptographic Censorship in cosmology

*This appendix collects content that is not on the paper's critical predictive path. Readers interested only in the testable predictions of §4 may stop at the bibliography. The cosmological transposition of Cryptographic Censorship (A3, §1) is a working conjecture; the material below is a first attempt at a concrete dS/FLRW dictionary, not a derivation. It is isolated here so that the paper's empirical content stands or falls independently of it. We emphasise that our argument relies on provable CFT-level results [12, 18], of which the cleanest AdS/CFT application of the same pseudorandomness logic is the holographic-pseudoentanglement construction of [58], rather than on complexity-theoretic claims drawn from the empirically contested quantum-supremacy landscape.*

### 1. A toy dictionary for A3 in cosmology

The original Cryptographic Censorship theorem [12] is proven in AdS/CFT: a holographic CFT state whose time evolution restricted to a boundary region is $\varepsilon$-approximately a unitary $k$-design admits a bulk dual containing an event horizon. Its cosmological use in this paper requires a transposition — not yet a theorem — from the AdS boundary to a timelike observer worldline in a pseudo-Riemannian spacetime. We make the transposition explicit as a *working conjecture* (not a proof sketch) so that readers can see what a positive or negative answer would actually constrain. Under the observer-dependent reading of §II, this working-conjecture status is the self-consistency condition delimiting the subregion on which the QRF crossed product of A1 remains well-defined; the CFT-level pseudorandom- unitary premise is itself a theorem (Ma–Huang [18] constructed from quantum one-way functions), whereas only its cosmological transposition is speculative.

**Conjectural map.** Let $\mathcal{O}$ be a geodesic observer in an asymptotically de Sitter FLRW spacetime, with causal diamond $\mathcal{D}(\mathcal{O})$ bounded by the cosmological event horizon $\mathcal{H}_{\mathcal{O}}$. Let $\mathcal{A}_{\mathcal{O}} = (\mathcal{M}_{\mathrm{QFT}} \rtimes \mathbb{R})^{\mathcal{O}}$ be the type-II crossed-product algebra of A1 attached to $\mathcal{O}$'s clock QRF [1–3], and let $\sigma_t^{\mathcal{O}}$ be its modular flow. We conjecture:

> *(D) The set of states $\rho$ on $\mathcal{A}_{\mathcal{O}}$ whose modular flow $\{\sigma_t^{\mathcal{O}}(\rho)\}_{t\in[0,T]}$ is an $\varepsilon$-approximate unitary $k$-design on a code subspace of observer-accessible operators corresponds, under any sensible pseudo-Riemannian extension of the Cryptographic Censorship dictionary, to cosmological states whose coarse-grained generalised entropy on $\mathcal{H}_{\mathcal{O}}$ satisfies*

$$\left|S_{\mathrm{gen}}^{\mathrm{cg}}[\rho;\mathcal{H}_{\mathcal{O}}] \;-\; S_{\mathrm{gen}}^{\max}[\mathcal{H}_{\mathcal{O}}]\right| \;\lesssim\; \varepsilon\cdot k\cdot \mathrm{logdim}\,\mathcal{A}_{\mathcal{O}}, \quad \text{(A1)}$$

> *with $S_{\mathrm{gen}}^{\max}$ the Gibbons–Hawking generalised entropy of the static patch.*

The form of the RHS bound is dimensionally dictated by the AdS/CFT template of [12]: pseudorandomness depth $k$ times code-subspace dimension, in units of the algebra's trace scale. In the type $\mathrm{II}_1$ static-patch setting of [1] the trace is finite and $\log\dim$ is replaced by the normalised trace of the log of a density, but the structural statement is unchanged.

**Status.** We emphasise that (A1) is offered as a *working conjecture*, not a proof sketch. The partial evidence we are able to cite is:

(i) The crossed-product construction of [1–3] gives, for the static de Sitter patch, exactly the operator-algebraic structure (type $\mathrm{II}_1$, well-defined trace, $S_{\mathrm{gen}}$ as a finite functional) that the AdS/CFT Cryptographic Censorship argument requires at the boundary.

(ii) The non-perturbative GSL of [16, 17] implies that the quantity on the LHS of (A1) is monotone under modular flow, matching the monotonicity expected on the CFT side once the $k$-design condition is approached.

(iii) The existence of pseudorandom unitaries, constructed from quantum one-way functions [18], ensures the $k$-design precondition is non-vacuous in physically realisable dynamics.

None of (i)–(iii) constitutes a proof of (A1). A rigorous derivation would require, at minimum, a pseudo-Riemannian analogue of the one-sided modular reconstruction used in [12] and a replacement for the boundary causal-wedge reconstruction theorem that does not rely on an AdS asymptotic region.

**One concrete testable consequence.** If (A1) holds, then cosmological perturbation modes whose coarse-grained horizon entropy lies *outside* the saturation $\varepsilon$-band of $S_{\mathrm{gen}}^{\max}$ cannot be dual to any bulk geometry containing the observer's cosmological horizon. In particular, any primordial perturbation spectrum predicting a super-Gibbons–Hawking coarse-grained horizon entropy at reheating is excluded as a geometric bulk state. This yields a bulk-geometry exclusion rule on CMB-scale perturbations with $|\delta S_{\mathrm{gen}}/S_{\mathrm{GH}}| > \varepsilon$; taking the observational proxy $\varepsilon \sim 10^{-5}$ (the CMB temperature anisotropy amplitude) gives a nontrivial consistency check against the complexity diagnostic of A6 and the Weyl-curvature-hypothesis reformulation below. A violation would falsify either the dictionary (A1) or the applicability of the crossed-product algebra of A1 at reheating — a meaningful and currently open test.

**How we use (A1) below.** The three narrative paragraphs that follow (Big Bang as decodability boundary; inflation as algebraic necessity; Weyl curvature revisited)

are consequences of the dictionary (A1) under the additional assumption that the saturation $\varepsilon$-band is the physically relevant regime. They inherit the conjectural status of (A1); we flag this explicitly rather than restate "conditional on A3" at every turn.